TANGENT STABILITY ON TORIC FANO BLOWUPS AND BUNDLES

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Abstract. For Fano blowups of products of projective spaces along invariant codimension-two centres indexed by a tree, using distinct coordinate hyperplanes at each factor, anticanonical tangent stability is equivalent to one divisor-degree inequality at each vertex. The exceptional contributions are mixed intersections on the blowup centres. Stability holds for every tree of maximum degree at most three whose nonleaf factor dimensions equal their valences, with arbitrary positive leaf dimensions. For smooth toric Fano varieties that are locally trivial toric bundles with fibre a product of degree-six del Pezzo surfaces and base a product of projective spaces, the largest Picard number compatible with tangent stability in dimension $n \geq 4$ is $\lfloor 5n/4 \rfloor + 1$, allowing arbitrary integral twists. Every fixed tree of maximum degree at most three with an edge yields bundles attaining this bound after sufficiently long independent edge subdivisions. Their number, up to isomorphism of toric fans, is unbounded as the dimension grows through multiples of four. The proofs use relative tangent bundles, ranks of subspaces spanned by rays, and integral comparisons. The bundle existence proofs also use finite exact rational calculations with uniform error estimates and induction.

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1. Introduction

Let X be a smooth toric Fano variety of dimension n. We study slope stability of its tangent bundle with respect to $H = -K_X$. The tangent bundle of a nontrivial product is not stable. Toric bundles and invariant blowups of products can have stable tangent bundles, although either construction can also admit a proper tangent subsheaf of excessive slope. We give stability criteria for two classes of these varieties, with unbounded dimension and Picard number.

For an invariant prime divisor D_ρ, write

$$\delta_\rho = \frac{nD_\rho H^{n-1}}{H^n}.$$

These positive weights sum to n. The toric slope criterion states that the tangent bundle is stable if and only if

$$\sum_{u_\rho \in V} \delta_\rho < \dim V$$

for every nonzero proper subspace spanned by primitive ray generators [Kly90, HNS22]. For the blowups considered here, we reduce these inequalities to one divisor-degree inequality at each vertex of a tree and express its terms as mixed intersections on the blowup centres. For the toric bundles, we determine the largest Picard number compatible with stability, allowing arbitrary integral twists. We construct bundles attaining this maximum from every fixed tree of maximum degree three, after sufficiently long independent subdivisions of its edges.

Blowups of products of projective spaces. Let G be a finite simple graph and choose integers $b_v \geq 1$. Start with $\prod_v \mathbb{P}^{b_v}$ and assign distinct coordinate hyperplanes at each factor to its incident edges. For each edge $\{v, w\}$, blow up the codimension-two centre defined by its two assigned hyperplanes. These star subdivisions commute. The resulting variety $Y(G, b)$ is Fano exactly when $b_v \geq d_v := \deg_G(v)$ for every v, and then

$$\dim Y(G,b) = \sum_v b_v, \qquad \rho(Y(G,b)) = |V(G)| + |E(G)|.$$

At each vertex there is an unassigned hyperplane. Its inverse image has divisor class $D_v = f_v^* \mathcal{O}_{\mathbb{P}^{b_v}}(1)$, where f_v is the original factor projection composed with the blowdown. Set $A_v = nD_v H^{n-1}/H^n$.

Theorem 1.1. *If G is a tree with at least one edge and $b_v \geq d_v$ at every vertex, then $T_{Y(G,b)}$ is anticanonically stable if and only if $A_v < 1$ for every v. It is semistable if and only if $A_v \leq 1$ for every v.*

Theorem 3.4 reduces the slope inequalities for subspaces meeting any number of factors to the stated vertex inequalities. At a nonleaf vertex with $b_v = d_v$, the distinguished ray defines a rank-one foliation with first Chern class D_v. Its degree satisfies the identity

$$(b_v + 1)A_v = b_v + \sum_{e \ni v} J_{v,e}.$$

Each $J_{v,e}$ is a positive first moment of the corresponding exceptional facet. Section 4 expresses this moment as a mixed intersection on the blowup centre. The factor projection specifies an ordering of the two normal directions. Moreover,

$$J_{v,e} + J_{w,e} = \frac{nE_e H^{n-1}}{H^n}, \qquad \sum_{e \ni v} J_{v,e} = \frac{nf_v^*(-K_{\mathbb{P}^{b_v}})H^{n-1}}{H^n} - b_v.$$

The stability inequality for this foliation therefore depends on the sum of the exceptional contributions at the vertex. The mixed-intersection formula also applies to compatible blowup morphisms whose incidence graph is not a tree.

Call a vertex *saturated* when $b_v = d_v$.

Theorem 1.2. *Let G be a tree with at least one edge and maximum degree at most three. If every nonleaf vertex is saturated and the leaf dimensions are arbitrary positive integers, then $T_{Y(G,b)}$ is anticanonically stable. A saturated degree-four vertex adjacent to four saturated degree-four vertices forces instability, independently of the farther branches of a Fano tree.*

The assertions are proved in Theorems 6.1 and 6.4. The stable case allows arbitrary distances between branching vertices and imposes no symmetry assumption. A saturated tree with M edges has dimension $2M$ and Picard number $2M + 1$. Thus maximum degree three is the largest degree bound that guarantees stability for every saturated tree. Higher degrees can occur in stable examples: there are stable trees with two branching vertices of degree four or five and arbitrarily long connecting paths (Proposition 6.6). Their exceptional first moments depend on the dimensions of neighbouring factors and the arrangement of the branches.

Toric bundles with maximal Picard number. Let S_6 denote the blowup of $\mathbb{P}^2$ at its three torus-fixed points. Consider locally trivial toric bundles with fibre S_6^j over $\prod_{c=1}^{k} \mathbb{P}^{b_c}$, where $j, k, b_c \geq 1$, in the fan presentation (47). The twists are arbitrary integral vectors in the fibre lattice.

Theorem 1.3. *For every $n \geq 4$, the largest Picard number among smooth toric Fano n-folds in this bundle class with anticanonically stable tangent bundle is*

$$\left\lfloor \frac{5n}{4} \right\rfloor + 1.$$

For each base factor, the Fano condition bounds the number of nonzero twists by the dimension of that factor. A slope inequality for the relative tangent bundle requires each surface factor to have nonzero twists from at least two base factors. Stability further requires the incidence graph of these twists to be connected. Counting its edges gives the bound in Theorem 7.3. Bundles indexed by paths attain it in dimension $4j$, and the end attachments of Corollary 9.3 give the remaining dimensions.

For a tree Γ, assign one degree-six del Pezzo surface factor to each edge and a base factor $\mathbb{P}^{\deg(v)}$ to each vertex. Give each surface factor opposite primitive-ray twists at the two endpoints of its edge, as in Definition 8.1, and denote the bundle by $X(\Gamma)$.

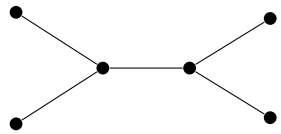
Theorem 1.4. *Every fixed finite tree of maximum degree three with at least one edge gives stable $T_{X(\Gamma)}$ after replacing each edge by an independently chosen sufficiently long path. A sufficient minimum length depends only on the number s of trivalent vertices and is $O(\log s)$ as $s \to \infty$. If Γ has j edges, then*

$$\dim X(\Gamma) = 4j, \qquad \rho(X(\Gamma)) = 5j + 1.$$

For each integer $q \geq 1$, every sufficiently large j admits at least q pairwise nonisomorphic toric fans of such stable extremizers.

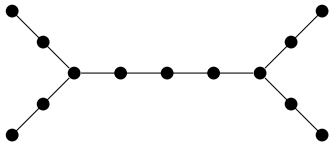
The subdivision assertion is Theorem 10.1. The graph can be recovered from the toric fan, so nonisomorphic graphs yield pairwise nonisomorphic fans (Lemma 8.4 and Corollary 10.2). This gives the asserted abundance, without classifying all bundles attaining the bound. The

Blowups: vertices carry projective factors



Stability without separating the branching vertices.

Bundles: edges carry del Pezzo factors



Stability and extremality after sufficiently long subdivisions.

FIGURE 1. Two uses of a subcubic tree. On the left an edge specifies a codimension-two blowup; on the right it specifies a fibre factor with opposite twists. The projective-space factors have dimensions equal to their vertex valences; for blowups, the leaf dimensions may be arbitrary positive integers. The right-hand subdivisions are schematic: the theorem requires the stated sufficient lengths.

proof uses exponential contraction of the functions that give volumes of polytope slices, together with a rank formula for subspaces meeting several branches. Figure 1 compares the bundle and blowup constructions. For the blowups, subdivision can destroy stability (Corollary 5.4).

Relation to earlier work. The toric subspace criterion and its formulation in terms of cone-volume concentration are due to [Kly90, HNS22]. Hering–Nill–Süß determine stability regions for smooth toric surfaces and varieties of Picard rank two. Dasgupta–Dey–Khan independently treat Picard rank two and classify the smooth toric Fano fourfolds of Picard rank three [DDK20]; the Picard-rank-at-most-two fourfold case is due to Biswas–Dey–Genc–Poddar [BDGP21]. Reynolds treats all 124 smooth toric Fano fourfolds and their unstable Harder–Narasimhan filtrations [Rey23]. Devey develops finite matroid tests for equivariant reflexive sheaves [Dev22]. We use saturated subsheaves, as required by the erratum [DDK21]. The single-edge blowup with dimensions $(1, 3)$ is already stable by [DDK20, Proposition 5.6.1(2)].

The families studied here have no bound on the dimension or the number of factors. For blowups, we prove the reduction to vertex inequalities, the exceptional mixed-intersection identity, and uniform estimates that establish stability or instability under the stated graph and dimension hypotheses. These estimates use Efron's monotonicity theorem through [SW18, Theorems 3.2–3.3]. For bundles, the relative tangent slope inequality follows from a divergence identity related to [HL14, Lemma 2.2 and Eq. (4.4)]. Although the unweighted fibre polytope is centred, weighting it by the volumes of the base slices can give a nonzero first moment. This first moment enters the relative tangent slope calculation. We use the fixed polarization $-K_X$; results for polarizations near a pulled-back or contracted class [NT24, CNT25] do not give the needed inequalities. The preceding paper [Wue26] constructs stable root-twisted bundles with $\rho = n + 2$; allowing the number of surface factors to grow produces the larger extremum and branching families here.

The bundle existence proofs use finite exact rational inequalities as premises for arguments covering all lengths. These arguments use the Birkhoff–Hopf contraction theorem [EN95], uniform error estimates, and induction. The appendices give the matrices, inequalities, and error estimates; the ancillary code reproduces the finite calculations. The blowup proofs use

integral identities, comparisons, and induction, with explicit polynomial arithmetic. Appendix F gives classification calculations, including the unique stable eightfold of Picard number eleven. Proposition F.3 completes the Picard-number-ten eightfold stratum: 64 stable, 692 strictly semistable and 5525 unstable tangent bundles. All strictly semistable members of this stratum are nontrivial toric products.

After the common stability criterion, Sections 3–6 develop the blowup results. Sections 7–10 treat the bundle bound and the construction of families attaining it. The appendices contain the path, end-attachment, and three-arm calculations, as well as further families and classification data. The Picard maxima proved here concern the two specified construction classes. Section 11 discusses the open problem of the unrestricted maximum.

Code and data. The Python programs, exact data and certificates used in this paper are available in the public companion repository. They are intended to accompany a future arXiv submission as ancillary files.

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2. Slope stability and symmetries of the rays

Let $X = X_\Sigma$ be a smooth toric Fano variety of dimension n, with lattice N, dual lattice M, and primitive ray generators $u_\rho \in N$ for $\rho \in \mathcal{R} := \Sigma(1)$. Write D_ρ for the corresponding invariant prime divisor, and put

$$d_\rho = D_\rho \cdot (-K_X)^{n-1}, \qquad D = (-K_X)^n = \sum_{\rho\in\mathcal{R}} d_\rho, \qquad \delta_\rho = \frac{n d_\rho}{D}.$$

Thus $d_\rho > 0$, $\delta_\rho > 0$, and $\sum_\rho \delta_\rho = n$. For a torsion-free sheaf $\mathcal{E}$, its slope is $\mu(\mathcal{E}) = c_1(\mathcal{E}) \cdot (-K_X)^{n-1}/\operatorname{rk}\mathcal{E}$. Stability, respectively semistability, means that every nonzero subsheaf of smaller rank has strictly smaller slope, respectively no larger slope. In particular, $\mu(T_X) = D/n$.

Lemma 2.1 (Toric slope criterion)**.** *The tangent bundle T_X is stable, respectively semistable, if and only if every nonzero proper complex subspace $V \subset N_\mathbb{C}$ satisfies*

$$w(V) := \sum_{\rho:\, u_\rho\in V} \delta_\rho < \dim V, \qquad \textit{respectively } w(V) \le \dim V. \tag{1}$$

It suffices to test subspaces spanned by ray generators.

Proof. The criterion is due to Hering–Nill–Süß [HNS22, Proposition 1.2 and Remark 2.6], obtained from Klyachko's filtrations [Kly90]. For completeness, the decreasing filtrations for T_X on $N_\mathbb{C}$ are

$$E^\rho(i) = \begin{cases} N_\mathbb{C}, & i \le 0, \\ \mathbb{C}u_\rho, & i = 1, \\ 0, & i \ge 2. \end{cases}$$

The saturated equivariant subsheaf $\mathcal{E}_V$ associated to V has induced filtrations $V \cap E^\rho(i)$ [DDK21, Corollary 0.0.2], rank $\dim V$, and degree $\sum_{u_\rho\in V} d_\rho$. Both slope stability and semistability can be tested on these subsheaves [HNS22, Proposition 2.3 and Remark 2.4]. If V contains rays, their span V' contains exactly the same rays and has dimension at most $\dim V$, so it has at least the same slope. If V contains no ray, then $w(V) = 0 < \dim V$. □

For a rational ray-generated subspace V, the saturated subsheaf $\mathcal{E}_V$ is also the toric foliation whose general leaves are orbits of the corresponding subtorus. Its first Chern class is

$$c_1(\mathcal{E}_V) = \sum_{u_\rho \in V} D_\rho.$$

This follows from the same induced filtrations. In particular, a line containing a single ray has first Chern class D_ρ; a line containing two opposite rays has the sum of the two divisors.

The same criterion can be expressed in terms of the anticanonical polytope. Let

$$P = \{m \in M_{\mathbb{R}} : \langle m, u_\rho \rangle \geq -1 \text{ for all } \rho\}$$

be the anticanonical polytope, with facets F_ρ. With lattice volume, $H^n = n!\,\mathrm{Vol}(P)$ and $D_\rho H^{n-1} = (n-1)!\,\mathrm{Vol}(F_\rho)$, where $H = -K_X$. The probability measure assigning to u_ρ the mass

$$\nu_P(\rho) = \frac{\mathrm{Vol}(\mathrm{conv}(0, F_\rho))}{\mathrm{Vol}(P)} = \frac{\delta_\rho}{n}$$

is the cone-volume measure on the ray directions. The slope criterion becomes $\nu_P(V) < \dim(V)/n$, the subspace concentration inequality of [HNS22]. We will calculate the divisor degrees explicitly for each of the two constructions.

Lemma 2.2. *The divisor degrees satisfy*

$$\sum_{\rho \in \mathcal{R}} d_\rho u_\rho = 0.$$

Proof. For $m \in M$, the principal divisor of the character χ^m is $\sum_\rho \langle m, u_\rho \rangle D_\rho$. Its intersection with $(-K_X)^{n-1}$ vanishes. Pairing with all $m \in M$ gives the assertion. □

2.1. **The ray matroid and invariant subspaces.** The *ray matroid* is the linear matroid on $\mathcal{R}$ whose rank function is

$$r(S) = \dim_{\mathbb{C}} \mathrm{span}\{u_\rho : \rho \in S\}, \qquad S \subseteq \mathcal{R}.$$

A circuit is a minimally linearly dependent set of ray generators. The components of the matroid are the equivalence classes generated by requiring all members of a circuit to belong to the same class; a ray in no circuit forms a singleton class. The matroid is *connected* if it has a single component.

Lemma 2.3. *If $C_1, \ldots, C_s$ are the components of the ray matroid and $V_a = \mathrm{span}\{u_\rho : \rho \in C_a\}$, then*

$$N_{\mathbb{C}} = V_1 \oplus \cdots \oplus V_s.$$

Every direct-sum decomposition of $N_{\mathbb{C}}$ in which each ray generator lies in a summand is obtained by grouping these component subspaces. In particular, stability of T_X implies connectedness of its ray matroid.

Proof. Choose a basis $B \subset \mathcal{R}$. For $\rho \notin B$, the ray u_ρ and the basis vectors occurring with nonzero coefficient in its expansion form a circuit. All those basis vectors therefore belong to the same component as ρ. Consequently V_a is spanned by the subset $B \cap C_a$ of this basis, proving the direct sum. Conversely, a circuit cannot meet two summands of any decomposition containing every ray in a summand: projecting its dependence to either summand would give a dependence on a proper subset of the circuit. Thus each component is contained in one summand. Since the rays span $N_{\mathbb{C}}$, each summand is the sum of the component subspaces it contains.

If $s > 1$, the equalities $\sum_a w(V_a) = n = \sum_a \dim V_a$ imply $w(V_a) \geq \dim V_a$ for at least one component, contradicting Lemma 2.1. □

Proposition 2.4 (Symmetry reduction)**.** *Let A be any subgroup of the lattice automorphisms preserving Σ. Then T_X is stable if and only if its ray matroid is connected and*

$$w(V) < \dim V$$

for every nonzero proper A-invariant subspace $V \subset N_{\mathbb{C}}$ spanned by ray generators.

Proof. The group A permutes the rays and preserves their divisor degrees, so $w(aV) = w(V)$ for $a \in A$. Necessity follows from Lemmas 2.1 and 2.3. For sufficiency, suppose that the ray matroid is connected and that the strict inequality holds for every subspace specified in the statement.

For any two subspaces $U, V \subseteq N_{\mathbb{C}}$, counting the rays in their union and intersection gives

$$w(U) + w(V) \leq w(U + V) + w(U \cap V). \tag{2}$$

Let

$$\lambda = \max_{0 \neq V \subseteq N_{\mathbb{C}}} \frac{w(V)}{\dim V}.$$

The maximum is attained among the finitely many subspaces spanned by rays, and $\lambda \geq 1$ because $w(N_{\mathbb{C}}) = n$. If U, V attain this maximum, then (2) and the bound $w(W) \leq \lambda \dim W$ imply that $U + V$ also attains it. They also imply $w(U \cap V) = \lambda \dim(U \cap V)$, including when the intersection is zero. Every maximizing subspace is spanned by rays, since $\lambda > 0$. If $\lambda > 1$, the sum of all maximizing subspaces is therefore a nonzero proper maximizing subspace. It is spanned by rays and is A-invariant, contrary to the hypothesis. Thus $\lambda = 1$, and T_X is semistable by Lemma 2.1.

Call a subspace V *tight* if $w(V) = \dim V$. Every nonzero tight subspace is spanned by rays: replacing it by the span of its rays would otherwise contradict $w(W) \leq \dim W$. The preceding argument shows that sums and intersections of tight subspaces are tight. Moreover, if U and V are tight, equality holds in (2). Since all ray weights are positive, every ray in $U + V$ then lies in U or in V.

There are finitely many nonzero tight subspaces. Let $U_1, \ldots, U_q$ be the minimal ones under inclusion. Distinct U_i have zero intersection, since an intersection is tight. Their sum is direct, as follows by induction. Put $W = U_1 + \cdots + U_{k-1}$. The intersection $U_k \cap W$ is tight, so minimality implies that it is either zero or all of U_k. In the latter case, choose a ray in U_k. Repeated use of the preceding observation about rays in a sum shows that this ray lies in some U_i with $i < k$, contradicting $U_i \cap U_k = 0$. Thus

$$S = U_1 \oplus \cdots \oplus U_q$$

is tight, is spanned by rays, and is A-invariant, since A permutes the minimal nonzero tight subspaces.

If $S \neq N_{\mathbb{C}}$, it contradicts the assumed strict inequality on invariant subspaces. If $S = N_{\mathbb{C}}$, every ray belongs to one of the U_i, again by the same observation. Connectedness of the ray matroid and Lemma 2.3 force $q = 1$. Hence $U_1 = N_{\mathbb{C}}$ is minimal among nonzero tight subspaces, so there is no proper nonzero tight subspace. All the inequalities in Lemma 2.1 are therefore strict. □

3. Blowups of products of projective spaces

Consider blowups of a product of projective spaces along invariant codimension-two centres, each defined by a ray from each of two factors. A simple graph records these pairs of factors. When this graph is a tree, Theorem 3.4 characterizes tangent stability by one divisor-degree inequality at each vertex. Its proof uses the ranks of subsets of rays and a divergence identity on the anticanonical polytope. Section 4 expresses the positive terms of this identity as exceptional intersection numbers. The estimates in Sections 5–6 give uniform stability results and local obstructions to stability.

3.1. **The fan and its primitive collections.** Let $G = (V, E)$ be a finite nonempty simple graph, and write $k = |V|$, $m = |E|$, and $d_v = \deg_G(v)$. Assign a positive integer b_v to each vertex. In

$$N = \bigoplus_{v \in V} \mathbb{Z}^{b_v},$$

let R_v be the $b_v + 1$ primitive ray generators of the standard fan of $\mathbb{P}^{b_v}$ in its summand. Thus $\sum_{r \in R_v} r = 0$. Suppose $d_v \leq b_v + 1$, and assign the edges incident to v injectively to rays $r_{v,e} \in R_v$. For every $e = \{v, w\}$, subdivide the original two-cone $\langle r_{v,e}, r_{w,e} \rangle$ by the ray

$$s_e = r_{v,e} + r_{w,e}. \tag{3}$$

Denote the resulting fan by $\Sigma(G, b)$ and its variety by $Y(G, b)$. The pairs of original rays are disjoint by injectivity. Permuting the rays within each projective-space factor carries any assignment to any other, so the resulting fan is independent of these choices up to isomorphism.

Proposition 3.1. *The fan $\Sigma(G, b)$ is independent of the order of subdivision and is smooth and projective. With $n = \sum_v b_v$, one has*

$$\dim Y(G, b) = n, \qquad \rho(Y(G, b)) = k + m. \tag{4}$$

Its primitive collections are precisely

$$\mathcal{C}_e = \{r_{v,e}, r_{w,e}\}, \qquad e = \{v, w\}, \tag{5}$$

$$\mathcal{P}_v(J) = \big(R_v \setminus \{r_{v,e} : e \in J\}\big) \cup \{s_e : e \in J\}, \qquad J \subseteq E(v), \tag{6}$$

where $E(v)$ is the set of edges incident to v. Their primitive relations are

$$r_{v,e} + r_{w,e} = s_e, \qquad \sum_{r \in \mathcal{P}_v(J)} r = \sum_{e = \{v,w\} \in J} r_{w,e}. \tag{7}$$

Every $Y(G, b)$ is weak Fano, and

$$Y(G, b) \text{ is Fano} \iff b_v \geq d_v \quad (v \in V). \tag{8}$$

If G is connected, $Y(G, b)$ admits no toric locally trivial bundle presentation with positive-dimensional complete fibre and positive-dimensional base.

Proof. Within each original maximal cone, the centres use disjoint pairs of basis vectors. Subdivision therefore amounts to subdividing separate two-dimensional orthants. These operations commute and preserve unimodularity, and the other original centres remain available for subdivision. Each step is a toric blowup along a smooth invariant centre, which also proves projectivity. The resulting fan has $n + k + m$ rays, proving (4).

The cones can be described as follows. For a subset A of the final rays, let B be its original rays and J the edges whose exceptional rays belong to A. Then A spans a cone if and only if

(1) B contains no endpoint pair $\mathcal{C}_e$;
(2) the union of B with both endpoints of every $e \in J$ contains no complete set R_v.

Indeed, the smallest original cone containing s_e contains both its endpoints, whereas the original endpoint pair is removed by subdivision. Both conditions are therefore necessary. Conversely, the second places the expanded set in an original product cone. The subdivisions within that cone act on disjoint coordinate pairs. The first condition then ensures that the chosen rays lie together in a cone of the subdivision.

The sets in (5) are minimal nonfaces. The expansion of $\mathcal{P}_v(J)$ contains R_v, and deleting any one member removes its unique representative of one ray of R_v. At any other vertex, that expansion contains at most one ray, because G is simple; it cannot contain the full set of at least two rays of that factor. There is also no original endpoint pair in $\mathcal{P}_v(J)$. The cone criterion therefore proves minimality, including the case in which every member is an exceptional ray. Conversely, a nonface containing no $\mathcal{C}_e$ expands to a set containing some R_v. Choosing one original or exceptional representative for each ray of R_v produces a subset $\mathcal{P}_v(J)$. Thus there are no further primitive collections.

The vector identities (7) follow from $\sum_{r\in R_v} r = 0$. On the right of the second identity there is at most one ray in each neighbouring factor. No two of these rays form another centre, since incidence rays are assigned injectively. They therefore span a cone of the final fan, and their positive sum lies in its relative interior. These are the primitive relations, of degrees 1 and $b_v+1-|J|$, respectively. Batyrev's primitive-collection criterion for nefness and ampleness [CLS11, Theorem 6.4.9] proves that $-K$ is nef and gives (8). The anticanonical polytope contains the origin in its interior and is bounded by completeness, so $-K$ is also big.

Finally, the cone complex of a torically locally trivial bundle is the join of its fibre and base cone complexes. Its rays partition into fibre rays and one lift of each base ray, and every primitive collection lies entirely in one part. Indeed, over each invariant affine chart the fan is the product of the fibre fan and the base cone, and these identifications agree on common faces. In the present fan, the primitive collection $R_v = \mathcal{P}_v(\varnothing)$ joins all original rays of one factor. The collections $\mathcal{C}_e$ join neighbouring factors, and $\mathcal{P}_v(\{e\})$ joins s_e to $b_v \geq 1$ original rays. For connected G, all rays consequently lie in one equivalence class generated by membership in a common primitive collection. They admit no partition into two nonempty parts respecting every primitive collection. Hence no choice of fibre and base gives a bundle presentation of the stated kind. □

The bundle assertion leaves open the existence of toric morphisms with special fibres or non-equivariant fibrations. Both the simplicity of G and the injectivity of the incidence assignments are required in the construction. For example, two parallel edges using the opposite pairs of rays of $\mathbb{P}^1 \times \mathbb{P}^1$ give the degree-six del Pezzo surface, which does not satisfy (8) with these vertex dimensions.

3.2. **Divisor degrees and a divergence identity.** For the remainder of this section, assume that $b_v \geq d_v$, and put $Y = Y(G, b)$. As in Section 2, normalize the divisor degrees by

$$\delta_r = \frac{nD_r \cdot (-K_Y)^{n-1}}{(-K_Y)^n}, \qquad \sum_r \delta_r = n. \tag{9}$$

They are all positive. Set $C_e = \delta_{s_e}$. There is a common value A_v for the degrees of the unassigned original rays at v, and

$$\delta_{r_{v,e}} = A_v - C_e, \qquad \sum_v (b_v + 1)A_v - \sum_e C_e = n. \tag{10}$$

Indeed, project the relation $\sum_r \delta_r r = 0$ of Lemma 2.2 onto the v-th lattice summand. The coefficient of an assigned ray is $\delta_{r_{v,e}} + C_e$, and that of an unassigned ray is its degree. Since the unique relation among R_v has equal coefficients, the first assertion follows. Summing the ray degrees gives the second. In particular,

$$0 < C_e < A_v \qquad (e \in E(v)). \tag{11}$$

There are $q_v = b_v + 1 - d_v \geq 1$ unassigned rays at v. Choose one as the negative sum ray and write the others as an integral basis $r_{v,1}, \ldots, r_{v,b_v}$. If $M = \mathrm{Hom}(N, \mathbb{Z})$, the affine coordinates $a_{v,j} = 1 + \langle x, r_{v,j} \rangle$ on $M_{\mathbb{R}}$ preserve lattice measure. We write $a_{v,e}$ for the coordinate assigned to the incidence (v, e). The anticanonical polytope P is

$$a_{v,j} \geq 0, \qquad \sum_{j=1}^{b_v} a_{v,j} \leq b_v + 1, \qquad a_{v,e} + a_{w,e} \geq 1 \quad (e = \{v, w\}). \tag{12}$$

We use lattice volume, without factorial normalization. If F_r is the facet corresponding to r, then

$$(-K_Y)^n = n! \,\mathrm{vol}(P), \qquad \delta_r = \frac{\mathrm{vol}(F_r)}{\mathrm{vol}(P)}. \tag{13}$$

The second formula follows from $D_r \cdot (-K_Y)^{n-1} = (n-1)! \,\mathrm{vol}(F_r)$.

Lemma 3.2. *For an edge $e = \{v, w\}$, define*

$$J_{v,e} = \frac{1}{\mathrm{vol}(P)} \int_{F_{s_e}} a_{v,e} \, d\,\mathrm{vol}\,.$$

Then

$$(b_v + 1)A_v = b_v + \sum_{e \in E(v)} J_{v,e}, \qquad J_{v,e} + J_{w,e} = C_e, \tag{14}$$

and $0 < J_{v,e} < C_e < A_v$. In particular, $A_v < 1$ at every vertex of degree one, for every positive value of b_v.

Proof. Consider the vector field equal to a_v in the v-th coordinate block and zero in every other block. Its divergence is b_v. Apply the divergence theorem. The coordinate facets have zero flux. The facet $\sum_j a_{v,j} = b_v + 1$ has flux $(b_v + 1)$ times its lattice volume, and the exceptional facet F_{s_e} has flux $-\int_{F_{s_e}} a_{v,e} \, d\,\mathrm{vol}$. All remaining facets have zero flux. The Euclidean length of a primitive integral normal occurs in the hypersurface measure and its reciprocal occurs in the unit-normal flux, so these factors cancel. Division by $\mathrm{vol}(P)$ proves the first identity.

On F_{s_e} one has $a_{v,e} + a_{w,e} = 1$. Its relative interior has both coordinates strictly between zero and one, since intersection with either coordinate facet has smaller dimension. This proves the second identity and the strict bounds on $J_{v,e}$; the remaining bound is (11). At a leaf, the first identity gives $(b_v + 1)A_v = b_v + J_{v,e} < b_v + A_v$, hence $A_v < 1$. □

3.3. **A cographic rank formula.** Construct a multigraph H with vertices o, the vertices v of G, and one vertex z_e for each edge of G. Its edges are indexed by the rays of $\Sigma(G,b)$ as follows:

- an assigned ray $r_{v,e}$ corresponds to an edge $z_e \to v$;
- each unassigned original ray at v corresponds to an edge $o \to v$;
- the exceptional ray s_e corresponds to an edge $o \to z_e$.

These orientations define the incidence matrix; connectivity is independent of their choice. Thus H has $k+m+1$ vertices and $n+k+m$ edges. It is connected, even if G is disconnected: each z_e joins the root, each nonisolated v joins some z_e, and an isolated v has unassigned rays joining it to the root.

Proposition 3.3. *For a subset S of the rays, let $H \setminus S$ denote the graph obtained by deleting its corresponding edges and retaining all vertices. Then*

$$r(S) = |S| + 1 - c(H \setminus S), \tag{15}$$

where c counts connected components, including isolated vertices. Consequently the ray matroid is the cographic matroid of H: its bases are exactly the complements of spanning trees of H.

Proof. Let B be the signed incidence matrix of H, with positive entries at the heads of oriented edges, and let $R : \mathbb{Q}^{E(H)} \to N_{\mathbb{Q}}$ send each edge basis vector to its ray. The row at v maps to $\sum_{r \in R_v} r = 0$, and the row at z_e maps to $s_e - r_{v,e} - r_{w,e} = 0$. Thus the row space of B lies in $\ker R$. Both spaces have dimension $k+m$: the incidence matrix has that rank because H is connected, and the original rays span $N_{\mathbb{Q}}$. Hence $\ker R = \operatorname{im} B^{\mathsf{T}}$.

A relation supported on S is therefore $B^{\mathsf{T}} f$, where f is constant on every component of $H \setminus S$. Modulo constant functions, this space has dimension $c(H \setminus S) - 1$. Subtracting from $|S|$ proves (15). Rational and complex ranks agree. A subset of n rays is a basis precisely when its complement is connected; that complement has $k+m$ edges on $k+m+1$ vertices, so it is a spanning tree. □

3.4. **The stability criterion for trees.**

Theorem 3.4. *Suppose G is a tree with at least one edge and $b_v \geq d_v$ for all v. Then $T_{Y(G,b)}$ is anticanonically stable if and only if $A_v < 1$ for every vertex, and is semistable if and only if $A_v \leq 1$ for every vertex. Thus it is strictly semistable when $\max_v A_v = 1$ and unstable when $\max_v A_v > 1$.*

Proof. Let K be the tree obtained by subdividing every edge of G at z_e, so H is obtained from K by adjoining the root and its edges. Give each edge of H its normalized ray degree. There are q_v root edges of weight A_v at v, one root edge of weight C_e at z_e, and the edge vz_e has weight $A_v - C_e$.

For a ray subset S, set $E' = E(H) \setminus S$, write $\delta(E')$ for its total weight, and put

$$\beta(E') = |E'| - |V(H)| + c(E').$$

This is the cycle rank of the subgraph with all vertices retained. By Proposition 3.3, $r(S) = n - \beta(E')$. Since the total weight is n, the slope inequality becomes

$$\delta(E') > \beta(E') \qquad \text{when } 0 < r(S) < n, \tag{16}$$

with a weak inequality for semistability.

For a nonempty connected vertex set Q of K, let $q(Q)$ count all root edges at its vertices, including multiplicity, and let $W(Q)$ be the sum of their weights and the weights of all K-edges internal to Q. Define

$$D(Q) = W(Q) - (q(Q) - 1).$$

For $Q = K$ this is zero, because the cycle rank and total weight of H both equal n. For $Q = \{z_e\}$ it is $C_e > 0$.

Otherwise the original vertices in Q form a nonempty connected subtree I of G. All exceptional vertices on internal edges of I belong to Q. Let ∂I be the edges with exactly one endpoint in I, write $v(e)$ for that endpoint, and set $t_e = 1$ if $z_e \in Q$, and $t_e = 0$ otherwise. Summing the contributions from original rays and applying (14) gives

$$\begin{aligned} W(Q) &= \sum_{v\in I}(b_v+1)A_v - \sum_{e\in\partial I}(1-t_e)A_{v(e)} - \sum_{e \text{ internal to } I} C_e \\ &= \sum_{v\in I} b_v + \sum_{e\in\partial I} J_{v(e),e} - \sum_{e\in\partial I}(1-t_e)A_{v(e)}. \end{aligned}$$

There are $|I|-1$ internal edges, and $\sum_{v\in I} d_v = 2(|I|-1) + |\partial I|$. Consequently

$$q(Q) - 1 = \sum_{v\in I} b_v - |\partial I| + \sum_{e\in\partial I} t_e,$$

and hence

$$D(Q) = \sum_{e\in\partial I}\big[J_{v(e),e} + (1-t_e)(1-A_{v(e)})\big]. \tag{17}$$

Assume now $A_v \le 1$ for every v. If I is proper, its boundary is nonempty, so this sum is strictly positive. If I contains every original vertex, connectivity forces $Q = K$. The exceptional singleton case was treated above. Hence $D(Q) > 0$ for every nonempty proper connected Q.

The selected non-root edges of E' form a forest in K. Each component has a connected vertex set Q and contains every K-edge internal to that set, since K is a tree. If it has h selected root edges, counted with multiplicity, it contributes $\max(h-1, 0)$ to $\beta(E')$. This count also applies when several components meet the root. For each component the first root edge makes the connection, and each remaining root edge adds a cycle. For $h = 0$, its contribution to weight minus cycle rank is nonnegative, and positive if it contains an edge. For $h \ge 1$, completing the missing root edges gives the exact contribution

$$D(Q) + \sum_{r \text{ a missing root edge at } Q} (1-\delta_r). \tag{18}$$

All root degrees are at most one; indeed $C_e < A_v \le 1$. Thus every component contribution is nonnegative, proving the weak inequalities in (16).

If every $A_v < 1$, all root degrees are strictly below one. Every proper component with a root edge has positive contribution. A full component $Q = K$ is also positive if a root edge is missing, by (18); if none is missing, $E' = E(H)$ and $S = \varnothing$. Components without root edges contribute positively whenever they contain an edge. The only other case with no positive contribution is the empty E', for which $r(S) = n$. Neither exceptional case lies in the range of ranks being tested, so all the required inequalities are strict.

Conversely, an unassigned ray of degree A_v belongs to a rank-one subspace of weight at least A_v. This is a proper subspace because $n \ge 2$. Stability therefore requires $A_v < 1$, and

semistability requires $A_v \le 1$. These necessity statements, the inequalities just proved, and Lemma 2.1 give all three alternatives. □

Proposition 3.3 gives the tangent subspace ranks for any simple graph. When the graph is a tree, (17) expresses the required inequalities as positive sums of vertex and exceptional contributions. This is where the tree hypothesis enters: it allows the vertex inequalities to control subsheaves of every rank.

Corollary 3.5. *For every $p, q \ge 1$, the blowup of $\mathbb{P}^p \times \mathbb{P}^q$ along the invariant codimension-two subvariety corresponding to a cone generated by one ray from each factor has anticanonically stable tangent bundle.*

Proof. Both vertices of its graph have degree one. Lemma 3.2 gives both strict inequalities required by Theorem 3.4. □

For $(p, q) = (1, 3)$, this recovers [DDK20, Proposition 5.6.1(2)].

3.5. **Stable paths and the largest Picard number.**

Theorem 3.6. *Let G be a path with $L \ge 1$ edges, endpoint dimensions $p, q \ge 1$, and dimension two at every internal vertex. Then $T_{Y(G,b)}$ is anticanonically stable, and*

$$\dim Y(G, b) = 2L + p + q - 2, \qquad \rho(Y(G, b)) = 2L + 1. \tag{19}$$

At each internal vertex its unassigned normalized degree satisfies $A_v \le 24/25$.

Proof. Fix the incidence coordinate a at one endpoint of an edge. Integrate over the branch on the opposite side and denote its volume by $\phi(a)$. We will prove that

$$\phi(a) > 0, \qquad \phi(a) = c := \phi(1) \ (a \ge 1), \qquad \phi(a) \ge ac \ (0 \le a \le 1). \tag{20}$$

A leaf of dimension b gives

$$\phi_b(a) = \frac{(b + \min(a, 1))^b}{b!}.$$

Indeed, subtracting the lower bound $\max(0, 1 - a)$ from its assigned coordinate leaves a simplex with the indicated volume. Bernoulli's inequality gives

$$\frac{\phi_b(a)}{\phi_b(1)} = \left(1 - \frac{1 - a}{b + 1}\right)^b \ge 1 - \frac{b}{b + 1}(1 - a) \ge a,$$

so (20) holds for every endpoint dimension.

At a degree-two vertex, integrate the coordinate assigned to the parent. An incoming function then contributes

$$g(a) = \int_0^{2+\min(a,1)} (2 + \min(a, 1) - t)\phi(t)\, dt. \tag{21}$$

For a positive function constant beyond one, set $I_0 = \int_0^1 \phi(t)\, dt$ and $I_1 = \int_0^1 t\phi(t)\, dt$. On $0 \le a \le 1$, this gives $g(a) = g_0 + g_1 a + g_2 a^2$, where

$$g_0 = 2I_0 - I_1 + c/2, \qquad g_1 = I_0 + c, \qquad g_2 = c/2.$$

All coefficients are positive, and $g_0 - g_2 = 2I_0 - I_1 > 0$. Hence

$$g(a) - ag(1) = (1 - a)(g_0 - ag_2) \ge 0.$$

The function is constant beyond one. Induction along a branch proves (20) for each incoming function; the initial leaf function need not be quadratic.

At an internal vertex let ϕ and ψ be the two incoming functions. The branches have disjoint coordinate blocks, so their product is the integrand over $a, b \geq 0$, $a + b \leq 3$. Distinguish the corresponding values of c, I_0 and I_1 by subscripts. Partitioning the simplex at $a = 1$ and $b = 1$ gives its volume V and the volume F of its unassigned facet:

$$V = I_{0,\phi} I_{0,\psi} + c_\psi(2I_{0,\phi} - I_{1,\phi}) + c_\phi(2I_{0,\psi} - I_{1,\psi}) + \tfrac{1}{2} c_\phi c_\psi, \tag{22}$$

$$F = c_\phi c_\psi + c_\psi I_{0,\phi} + c_\phi I_{0,\psi}. \tag{23}$$

The four terms of V are the contributions from the unit square, the two regions with exactly one coordinate at least one, and the remaining triangle, respectively. Eliminating one coordinate from $a + b = 3$ gives the lattice measure in the formula for F.

Define

$$i_\phi = I_{0,\phi}/c_\phi, \qquad h_\phi = (I_{0,\phi} - I_{1,\phi})/c_\phi,$$

and similarly for ψ. From (20), $i_\phi \geq 1/2$ and $h_\phi \geq 1/6$. Equations (22)–(23) become

$$\frac{F}{c_\phi c_\psi} = 1 + i_\phi + i_\psi, \qquad \frac{V - F}{c_\phi c_\psi} = i_\phi i_\psi + h_\phi + h_\psi - \frac{1}{2}.$$

Write $i_\phi = 1/2 + x$, $i_\psi = 1/2 + y$, $h_\phi = 1/6 + u$ and $h_\psi = 1/6 + v$, with $x, y, u, v \geq 0$. Then

$$\frac{24V - 25F}{c_\phi c_\psi} = 24xy + 11(x + y) + 24(u + v) \geq 0.$$

Thus (13) gives $A_v = F/V \leq 24/25 < 1$ at every internal vertex. Lemma 3.2 gives $A_v < 1$ at both endpoints. Stability follows from Theorem 3.4. This includes $L = 1$, for which there is no internal vertex. The dimension and Picard number follow from Proposition 3.1. □

Corollary 3.7. *For every $n \geq 1$, the largest Picard number among the smooth Fano n-folds $Y(G, b)$ with stable tangent bundle is*

$$2\left\lfloor \frac{n}{2} \right\rfloor + 1. \tag{24}$$

Here G ranges over the finite nonempty simple graphs with injective incidence-ray assignments in the construction above.

Proof. If G is disconnected, its fan is the product of the fans for its connected components. The tangent bundle is a direct sum of at least two positive-rank bundles; one summand has slope at least the slope of the whole bundle. Stability therefore forces connectedness. Consequently $k \leq m + 1$, whereas the Fano condition gives

$$n = \sum_v b_v \geq \sum_v d_v = 2m.$$

It follows that $\rho = k + m \leq 2m + 1 \leq 2\lfloor n/2 \rfloor + 1$.

For $n = 2L \geq 2$, take the length-L path with endpoint dimensions $(1, 1)$. For $n = 2L + 1 \geq 3$, take endpoint dimensions $(1, 2)$ instead. Theorem 3.6 gives stability, and (19) gives equality in (24). For $n = 1$, the single-vertex graph with dimension one gives $\mathbb{P}^1$, whose rank-one tangent bundle is stable and whose Picard number is one. □

The maximum is taken within the specified class of blowups of products of projective spaces. The class of bundles in Section 7 has a different maximum. The blowup maximum gives no upper bound for arbitrary smooth toric Fano varieties.

4. Exceptional intersections and tangent foliations

The first moments $J_{v,e}$ of the exceptional facets in Lemma 3.2 can be expressed as intersection numbers on the blowup centres. This expression also applies without a tree indexing the remaining factors. We identify the tangent subsheaf whose degree these moments determine and express its slope in terms of the exceptional contractions.

4.1. **The radial foliation.** Let B be a smooth projective toric variety and let $b \geq 2$. Choose a torus-fixed point $p \in \mathbb{P}^b$, distinct invariant hyperplanes $\mathcal{H}_1, \dots, \mathcal{H}_m$ through p, where $0 \leq m \leq b$, and distinct invariant prime divisors $D_1, \dots, D_m$ on B. Let

$$\pi : X \longrightarrow \mathbb{P}^b \times B$$

be the successive blowup of the strict transforms of $\mathcal{H}_i \times D_i$. Assume that X is Fano, and put

$$n = \dim X, \qquad H = -K_X, \qquad f = \mathrm{pr}_1 \circ \pi, \qquad L = f^*\mathcal{O}_{\mathbb{P}^b}(1). \tag{25}$$

The centres are defined by disjoint pairs of rays. Within each original smooth cone, subdivision acts on disjoint pairs of basis vectors. The subdivisions therefore commute and preserve the smoothness of the remaining centres. In particular, each exceptional divisor E_i can be contracted with all other blowups retained. The centres themselves may intersect.

Projection from p induces a rational map

$$X \dashrightarrow \mathbb{P}^{b-1} \times B.$$

Let $\mathcal{F}$ be the saturation in T_X of the tangent directions to its general fibres. Equivalently, $\mathcal{F}$ is the birational transform of the foliation by lines through p at fixed B-coordinate.

Proposition 4.1. *The sheaf $\mathcal{F}$ is a rank-one toric foliation and $\mathcal{F} \simeq \mathcal{O}_X(L)$. Its normalized slope is*

$$A := \frac{\mu_H(\mathcal{F})}{\mu_H(T_X)} = \frac{nLH^{n-1}}{H^n} = \frac{d}{dt} \log \mathrm{vol}(H + tL)\Big|_{t=0}. \tag{26}$$

In particular, $A \geq 1$ excludes stability of T_X, and $A > 1$ gives a strict rank-one destabilizer.

Proof. Choose basis rays $r_1, \dots, r_b$ for the hyperplanes through p and put $r_0 = -\sum_j r_j$. On the dense torus, the radial foliation has tangent line $\mathbb{C}r_0$. Since $b \geq 2$, no other ray of the projective-space factor lies on this line. Rays of B lie in the other lattice summand, and the new exceptional rays have nonzero components in both summands. Thus r_0 is the only ray on the line. The induced tangent filtrations from Lemma 2.1 give $c_1(\mathcal{F}) = D_{r_0}$. None of the blowup centres is contained in the invariant hyperplane opposite p. Its strict transform therefore has class $D_{r_0} = L$.

A saturated rank-one subsheaf of a locally free sheaf on a smooth variety is reflexive. Indeed, its quotient is torsion-free, so the depth lemma gives depth at least two in codimension at least two. It is consequently a line bundle, which proves the asserted identification. The foliation may still have singularities. The first equality in (26) now follows from $\mu_H(T_X) = H^n/n$. In a neighbourhood of the ample class H, the algebraic volume equals the top intersection number. Differentiation of $(H + tL)^n$ gives the last equality, and the slope conclusions follow. □

The hypothesis $b \geq 2$ is required because both original rays of a $\mathbb{P}^1$ factor lie on the same line and contribute to the corresponding tangent subsheaf. The leaf estimates of Lemma 3.2 are therefore separate from this foliation calculation.

4.2. **Mixed intersections on the exceptional centres.** Contract only E_i, obtaining $\pi_i : X \to Y_i$ with smooth centre Z_i of dimension $d = n-2$. The centre is the transverse intersection of the transforms of $\mathcal{H}_i \times B$ and $\mathbb{P}^b \times D_i$. Their normal line bundles on Z_i, in this order, will be denoted $N_{i,A}$ and $N_{i,B}$. Thus

$$N_{Z_i/Y_i} = N_{i,A} \oplus N_{i,B}, \qquad q_i : E_i = \mathbb{P}_{\text{lines}}(N_{i,A} \oplus N_{i,B}) \longrightarrow Z_i.$$

Let $S_{i,0}$ be the section corresponding to the normal direction $N_{i,B}$, and let $S_{i,1}$ correspond to $N_{i,A}$. The first section is the intersection with the strict transform of $\mathcal{H}_i \times B$. Identifying both sections with Z_i, put

$$M_{i,0} = H|_{S_{i,0}} = -K_{Z_i} + N_{i,A}, \qquad M_{i,1} = H|_{S_{i,1}} = -K_{Z_i} + N_{i,B}. \tag{27}$$

Both classes are ample. We use additive notation for line bundles. To verify the equalities, write $\xi = c_1(\mathcal{O}_{\mathbb{P}_{\text{lines}}(N)}(1))$. Then

$$E_i|_{E_i} = -\xi, \qquad H|_{E_i} = q_i^*(-K_{Y_i}|_{Z_i}) + \xi,$$

while ξ restricts to $-N_{i,B}$ on $S_{i,0}$ and to $-N_{i,A}$ on $S_{i,1}$. Adjunction gives (27).

In what follows, products of the $M_{i,j}$ denote top intersection numbers on Z_i. Define

$$Q_i = \sum_{k=0}^{n-2} (k+1) M_{i,0}^{n-2-k} M_{i,1}^k, \qquad C_i = \frac{n E_i H^{n-1}}{H^n}. \tag{28}$$

These definitions use the specified order of the two normal directions.

Proposition 4.2. *Let P be the anticanonical polytope of X and F_i its facet corresponding to E_i. On F_i, let t be the affine coordinate which is zero and one on the faces corresponding to $S_{i,0}$ and $S_{i,1}$, respectively. Then*

$$J_i := \frac{1}{\operatorname{vol}(P)} \int_{F_i} t \, d\operatorname{vol} = \frac{Q_i}{H^n}, \tag{29}$$

$$C_i = \frac{n}{H^n} \sum_{k=0}^{n-2} M_{i,0}^{n-2-k} M_{i,1}^k. \tag{30}$$

Interchanging the sections gives a positive contribution J_i^{opp} with $J_i + J_i^{\text{opp}} = C_i$. If $M_{i,1} - M_{i,0}$ is nef, then $J_i \geq C_i/2$; if $M_{i,0} - M_{i,1}$ is nef, the inequality reverses. Numerically equal endpoint classes give $J_i = C_i/2$.

Proof. Let a_i and c_i be the affine support coordinates of the two original rays defining the i-th blowup. The exceptional equation is $a_i + c_i = 1$, with $a_i, c_i \geq 0$, so $t = a_i$ has the stated endpoints. The original rays extend to an integral basis, so the exceptional facet has an integral tangent vector that increases a_i by one and decreases c_i by one. Its lattice measure consequently disintegrates as dt times lattice measure on the centre, with no index factor.

The slice at t is, up to translation, the moment polytope of

$$M_{i,t} = (1-t) M_{i,0} + t M_{i,1}.$$

Indeed, restrict the supporting inequalities of $H|_{E_i}$ to the two sections. Affine interpolation of their constants gives the supporting inequalities of $M_{i,t}$. Since both endpoint classes are ample, the normal fan along the whole segment is the fan of Z_i. The slice volume is consequently $M_{i,t}^{n-2}/(n-2)!$. Since $\operatorname{vol}(P) = H^n/n!$,

$$J_i = \frac{n(n-1)}{H^n} \int_0^1 t M_{i,t}^{n-2} \, dt. \tag{31}$$

For $d = n-2$, the coefficient of $M_{i,0}^{d-k}M_{i,1}^{k}$ in the integral is

$$\binom{d}{k}\int_0^1 t^{k+1}(1-t)^{d-k}\,dt = \frac{k+1}{(d+1)(d+2)}.$$

This gives (29). Omitting the factor t gives (30), which is also the projective-bundle formula for $(H|_{E_i})^{n-1}$. Replacing t by $1-t$ replaces the coefficient $k+1$ by $n-1-k$, so the two contributions sum to C_i. Their positivity follows from ampleness.

Finally, write $I_k = M_{i,0}^{d-k}M_{i,1}^{k}$. If $M_{i,1} - M_{i,0}$ is nef, then

$$I_{k+1} - I_k = M_{i,0}^{d-k-1}M_{i,1}^{k}(M_{i,1} - M_{i,0}) \geq 0.$$

Pairing k with $d-k$ in $\sum_k (k - d/2)I_k$ proves $J_i - C_i/2 \geq 0$. The reversed nef condition and numerical equality follow by the same argument. □

Theorem 4.3. *For the blowup and foliation in (25),*

$$(b+1)A = b + \sum_i J_i, \qquad A - 1 = \frac{\sum_i Q_i - H^n}{(b+1)H^n}. \tag{32}$$

Consequently $\sum_i Q_i \geq H^n$ excludes stability of T_X, and strict inequality makes $\mathcal{F}$ a strict destabilizer. Equivalently, the total contribution is

$$\begin{aligned}\sum_i J_i &= n\frac{f^*(-K_{\mathbb{P}^b})H^{n-1}}{H^n} - b \\ &= (n-b) - n\frac{(-K_{X/\mathbb{P}^b})H^{n-1}}{H^n}.\end{aligned} \tag{33}$$

Here $-K_{X/\mathbb{P}^b} = H - f^*(-K_{\mathbb{P}^b})$ denotes the relative anticanonical divisor class.

Proof. In central affine coordinates the anticanonical polytope has inequalities

$$a_j \geq 0, \qquad \sum_{j=1}^{b} a_j \leq b+1, \qquad a_i + c_i \geq 1 \quad (1 \leq i \leq m),$$

together with the inequalities from the rays of B. For the divergence theorem, use the vector field $\sum_j a_j \partial/\partial a_j$, whose divergence is b. The coordinate facets and the facets from B have zero flux. The opposite central facet contributes $(b+1)$ times its lattice volume, and F_i contributes $-\int_{F_i} a_i$. As in Lemma 3.2, the conversion between Euclidean and lattice facet measures cancels the normal-length factor. Division by $\mathrm{vol}(P)$ gives the first identity in (32). Proposition 4.2 gives the second, and Proposition 4.1 gives the slope conclusions. Substituting $f^*(-K_{\mathbb{P}^b}) = (b+1)L$ proves (33). □

The contribution of an exceptional divisor is its normalized degree C_i times the mean of t with respect to the slice-volume distribution. The polarizations on the two sections determine this mean. For example, if $M_{i,1} - M_{i,0}$ is nef for every i and $\sum_i C_i \geq 2$, stability fails; a strict weighted inequality gives instability. These nef assumptions are not required by the identity or by the tree theorems below. The dependence on other blowups is contained in Z_i and $M_{i,0}, M_{i,1}$. The inequality $\sum_i Q_i < H^n$ tests only the foliation $\mathcal{F}$, and does not control all tangent subsheaves.

4.3. **Slice volumes as intersection polynomials.** Suppose now that $m = b$ and $B = \prod_{i=1}^{b} B_i$, with each D_i pulled back from a smooth invariant prime divisor on B_i. Put $d_i = \dim B_i$ and $h_i = -K_{B_i}$. Assume that h_i and $h_i - D_i$ are ample, as well as $H = -K_X$. Define the positive continuous functions

$$p_i(a) = \begin{cases} (h_i - (1-a)D_i)^{d_i}/d_i!, & 0 \le a \le 1, \\ h_i^{d_i}/d_i!, & a \ge 1. \end{cases} \tag{34}$$

These are the volume functions along the indicated divisor segments on the base.

Proposition 4.4. *In this product-base setting, let*

$$Z(t) = \int_{\substack{a_i \ge 0 \\ \sum_i a_i \le t}} \prod_i p_i(a_i)\, da_1 \cdots da_b.$$

Then

$$A = \frac{Z'(b+1)}{Z(b+1)}, \qquad \frac{p_i'(a)}{p_i(a)} = \frac{d_i D_i (h_i - (1-a)D_i)^{d_i - 1}}{(h_i - (1-a)D_i)^{d_i}} \quad (0 < a < 1). \tag{35}$$

If q_i are positive continuous log-concave functions, constant beyond one, and every p_i/q_i is nonincreasing, then A is at most the corresponding ratio $Z_q'(b+1)/Z_q(b+1)$. Nondecreasing ratios give the reverse inequality.

Proof. At fixed $a_i \in [0,1]$, the exceptional constraint moves the facet of D_i inward by $1 - a_i$. The remaining polytope is therefore the moment polytope of $h_i - (1-a_i)D_i$. Ample endpoints place the whole segment in the ample cone, giving (34). For $a_i \ge 1$ the new constraint is redundant. Integration over the separate base factors gives the product integrand. Hence $Z(b+1)$ is the full polytope volume, and $Z'(b+1)$ is the lattice volume of the opposite central facet. Their ratio is A. Differentiating the intersection polynomial gives the second formula in (35). The comparison follows from Lemma 6.2, proved below using Efron's conditional-sum theorem. Log-concavity is required only for the comparison functions. □

For a vertex v of a blowup tree, deleting v separates the remaining construction into factors B_i. The divisor assigned to the deleted parent edge becomes an unassigned divisor D_i on its component. Its class is the pullback of a hyperplane at the root of that component. The classes $h_i - D_i$ are ample: after deleting the parent edge, every primitive relation of root type has degree at least two. Subtracting D_i lowers each such degree by one, while edge and non-root primitive relations are unchanged. Hence the branch functions in the tree calculations are given by the intersection polynomials (34).

When every nonleaf dimension equals its valence and the maximum degree is three, the proof of Theorem 6.1 bounds their logarithmic derivatives above by $3/(2+a)$. In the degree-four obstruction, each of four branch functions has nondecreasing ratio to (46). By Proposition 4.4, these comparisons can also be applied to other polarized base pairs without a graph construction. Unless the tree criterion applies, an upper bound controls only the radial foliation. A lower bound exceeding one proves instability of T_X.

It remains a separate structural question whether a large Picard number forces the contraction data or intersection inequality above. This description and (33) do not establish such an implication. In the bundle constructions of Section 8, edges represent twists. Their subdivision theorems therefore make no assertion about monotonicity of the exceptional contributions considered here.

5. Branching and neighbouring factor dimensions

Let $Y(G,b)$ be the blowup of Section 3, with distinct original rays assigned to the incidences at each vertex. Write $d_v = \deg_G(v)$ and assume throughout that $b_v \geq d_v$ and $b_v \geq 1$ for every vertex. Thus $Y(G,b)$ is Fano. We call a vertex v *saturated* when $b_v = d_v$, or equivalently when exactly one of its original rays is unassigned. The degree of this unassigned divisor can depend on blowups beyond the incident centres. We determine it first for saturated stars, which are stable in degrees four and five as well as in lower degrees. We then compare a saturated vertex with a star whose leaves have the same dimensions as its neighbours. Replacing the remaining geometry by these independent leaves can only decrease the unassigned divisor degree. This comparison applies to graphs with cycles and does not require the other vertices to be saturated.

5.1. **Stability of stars.** Let Y_m be the blowup associated with a star with $m \geq 1$ edges, central dimension m, and dimension one at each leaf. In the product lattice, choose bases $e_1, \dots, e_m$ for the central factor and u_i for leaf i. Put

$$e_0 = -\sum_{i=1}^{m} e_i, \qquad s_i = e_i + u_i.$$

Thus Y_m is obtained from $\mathbb{P}^m \times (\mathbb{P}^1)^m$ by subdividing the m original cones $\langle e_i, u_i \rangle$. Let A_m denote the normalized divisor degree of e_0, with the normalization used in Section 3.

Theorem 5.1. *For every $m \geq 1$, the variety Y_m is smooth Fano, has dimension $2m$ and Picard number $2m+1$, and admits no torus-equivariantly locally trivial bundle presentation with positive-dimensional complete fibre and positive-dimensional base. Its tangent bundle is anticanonically stable exactly when $1 \leq m \leq 5$. For every $m \geq 6$, the line $\mathbb{C}e_0$ defines a strictly destabilizing saturated subsheaf.*

Appendix A contains the proof, consisting of a finite sign calculation and a uniform estimate for the remaining values.

For the stable stars of valence four and five, the central degrees and the sums of the exceptional contributions are

$$A_4 = \frac{44080}{47119}, \qquad \sum_i J_{0,i} = \frac{31924}{47119} < 1; \qquad A_5 = \frac{100780}{103289}, \qquad \sum_i J_{0,i} = \frac{88235}{103289} < 1.$$

These values follow from the star-transition table in the appendix and (14). Thus four or five incident blowups need not make the radial foliation of Proposition 4.1 destabilizing. Whether the sum reaches one also depends on the neighbouring slice volumes.

For $m \geq 6$, this line gives the first term of the Harder–Narasimhan filtration (Corollary A.1).

5.2. **Comparison at a saturated vertex.** The following comparison extends the necessary condition from stars to arbitrary simple graphs in the construction, including disconnected graphs and graphs with cycles.

Theorem 5.2. *Let G be a finite simple graph with positive dimensions $b_w \geq d_w$ at every vertex, and use injective incidence assignments in the construction of $Y(G,b)$. Suppose v is saturated and $b_v = d_v = m \geq 1$. Let A_v be the normalized divisor degree of its unique*

unassigned original ray. Enumerate the neighbors as $w_1, \dots, w_m$ and put $c_i = b_{w_i}$. Define

$$\phi_c(a) = \frac{(c + \min(a, 1))^c}{c!}, \qquad Z_{\mathbf{c}}(t) = \int_{\substack{a_i \geq 0 \\ \sum_i a_i \leq t}} \prod_{i=1}^{m} \phi_{c_i}(a_i)\, da.$$

Then

$$A_v \geq \frac{Z'_{\mathbf{c}}(m+1)}{Z_{\mathbf{c}}(m+1)} \geq A_m.$$

The first ratio is nondecreasing in each neighboring dimension c_i. The dimensions at vertices other than v may exceed their degrees.

Proof. At every vertex w, choose an unassigned original ray as the negative sum ray, and use the other b_w rays as a lattice basis. This is possible because $b_w \geq d_w$. In affine coordinates $a_{w,j} = 1 + \langle x, r_{w,j} \rangle$, the anticanonical polytope is

$$a_{w,j} \geq 0, \qquad \sum_{j=1}^{b_w} a_{w,j} \leq b_w + 1, \qquad a_{w,e} + a_{z,e} \geq 1 \quad (e = \{w, z\}). \tag{36}$$

Here $a_{w,e}$ is the coordinate assigned to the incidence (w, e). The coordinate change preserves lattice volume.

Write $a = (a_1, \dots, a_m)$ for the coordinates at v. Omit the simplex inequality at v and integrate over all other vertices. Denote the resulting volume by $W(a)$ for $a_i \geq 0$. It is bounded above by the product of the other simplex volumes. Increasing a_i relaxes just one inequality, so the volume is coordinatewise nondecreasing. It becomes constant in each coordinate once that coordinate reaches one.

Moreover, $W(0) > 0$. At every other vertex choose its assigned coordinates equal to $1 + \varepsilon$ and its remaining coordinates equal to ε. For sufficiently small $\varepsilon > 0$, the sum at w is $d_w + b_w\varepsilon < b_w + 1$, and all mixed inequalities are strict. They remain strict in an open neighbourhood of these coordinates, of positive volume. By monotonicity, $W(a) > 0$ throughout the nonnegative orthant. To prove continuity of W, integrate over the fixed compact product of simplexes. As a converges, the indicator functions converge away from finitely many affine hyperplanes of measure zero, and dominated convergence applies.

We first prove that

$$R(a) = \frac{W(a)}{\prod_{i=1}^{m} \phi_{c_i}(a_i)} \tag{37}$$

is coordinatewise nondecreasing. Fix every central coordinate except a_i, and let w be the other endpoint of its edge. Put $b = b_w$. At w, let x be the coordinate assigned to this edge and let $z = (z_1, \dots, z_{b-1})$ be its remaining coordinates. Injectivity of the incidence assignments ensures that no other mixed inequality uses x. Simplicity ensures that w has no second edge to v.

Omit the simplex inequality at w and integrate over every vertex other than v, w. This gives a nonnegative function $\Psi(z)$ on the nonnegative orthant. Increasing any component of z enlarges the region of integration, so this function is coordinatewise nondecreasing. It may depend on the other fixed central coordinates but is independent of x and a_i. The integral includes all edges among the other vertices; no independence assumption on neighbouring components is needed.

For $0 \le a_i \le 1$, the interval for x is $[1 - a_i, b + 1 - \sum_j z_j]$. With $s = b + a_i$, integration over this interval and then the substitution $z = sy$ give

$$\begin{aligned} W(a) &= \int_{\substack{z_j \ge 0 \\ \sum_j z_j \le s}} (s - \textstyle\sum_j z_j)\Psi(z)\, dz \\ &= s^b \int_{\substack{y_j \ge 0 \\ \sum_j y_j \le 1}} (1 - \textstyle\sum_j y_j)\Psi(sy)\, dy. \end{aligned} \tag{38}$$

When $b = 1$, the last integral is zero-dimensional and the same identity holds. The integral factor is nondecreasing in s. Thus, with the other coordinates fixed and $0 \le a \le a' \le 1$,

$$\frac{W(a')}{W(a)} \ge \left(\frac{b + a'}{b + a}\right)^b = \frac{\phi_b(a')}{\phi_b(a)}.$$

Thus R is nondecreasing below one in each coordinate. Both W and ϕ_{c_i} are constant above one, and continuity gives monotonicity on the entire orthant. Moreover, R is bounded because each ϕ_{c_i} is bounded below by a positive constant.

For the comparison of simplex integrals, let $X_1, \dots, X_m$ be independent random variables with densities

$$f_i(x) = C_i e^{-x} \phi_{c_i}(x) \mathbf{1}_{\{x \ge 0\}},$$

where $C_i > 0$ is the normalizing constant. Each density is log-concave. On $(0, 1)$ its logarithmic derivative is $-1 + c_i/(c_i + x)$, which decreases; at one this derivative jumps downward to -1 and stays constant thereafter. Its support is the interval $[0, \infty)$.

Efron's conditional-sum theorem states that, for independent log-concave random variables, the conditional expectation of a bounded coordinatewise nondecreasing function is nondecreasing in their sum; the version for general log-concave densities and any finite number of variables is given in [SW18, Theorems 3.2–3.3]. Extend R to $\mathbb{R}^m$ by replacing each argument by its nonnegative part. The extension is bounded and coordinatewise nondecreasing. The theorem therefore implies that

$$h(s) = \mathbb{E}[R(X_1, \dots, X_m) \,|\, X_1 + \cdots + X_m = s]$$

has a nondecreasing version. When $m = 1$, the same assertion follows directly from $h(s) = R(s)$.

On the slice $\sum_i a_i = s$, the product exponential factor equals e^{-s} and cancels from the conditional expectation. Thus $h(s)$ is the ratio of the W-weighted slice volume to the $\prod_i \phi_{c_i}(a_i)$-weighted slice volume. These integrals are continuous for $s > 0$, by rescaling the compact sum simplex to a fixed simplex and applying dominated convergence. The denominator is positive. Their ratio is therefore continuous and agrees on all of $(0, \infty)$ with the nondecreasing version supplied by the theorem.

Set

$$F(t) = \int_{\substack{a_i \ge 0 \\ \sum_i a_i \le t}} W(a)\, da, \qquad q(t) = Z'_{\mathbf{c}}(t) > 0.$$

Using the lattice slice measure obtained by eliminating one coordinate, we have

$$F'(t) = q(t)h(t), \qquad F(t) = \int_0^t q(s)h(s)\, ds \le h(t)Z_{\mathbf{c}}(t).$$

All quantities are positive, so

$$\frac{F'(t)}{F(t)} \ge \frac{Z'_{\mathbf{c}}(t)}{Z_{\mathbf{c}}(t)}.$$

At $t = m + 1$, the denominator on the left is the volume of the anticanonical polytope. The numerator is the lattice volume of the unassigned facet at v. The ratio is A_v, which proves the first inequality.

For integers $b \geq c \geq 1$ and $0 < a < 1$,

$$\frac{d}{da} \log \frac{\phi_b(a)}{\phi_c(a)} = \frac{a(b-c)}{(b+a)(c+a)} \geq 0.$$

The ratio is constant above one. The same conditional-sum argument, applied to products of these ratios, proves monotonicity in the neighbouring dimensions. Taking every $c_i = 1$ gives $\phi_1(a) = 1 + \min(a, 1)$ and recovers the star ratio, proving the second comparison. □

Corollary 5.3. *In the Fano simple-graph mixed-blowup construction, a saturated vertex of degree at least six forces anticanonical tangent instability. In particular, if every vertex is saturated and the tangent bundle is stable, then the maximum graph degree is at most five. The star Y_5 attains degree five.*

Proof. Let v be saturated of degree $m \geq 6$. Theorems 5.1 and 5.2 give $A_v \geq A_m > 1$. Its unassigned ray is the negative sum of m basis vectors and is not collinear with another original ray in that factor. Original rays at other vertices lie in different direct summands, and exceptional rays have nonzero components in two summands. Thus its line has normalized ray weight exactly $A_v > 1$ and defines a strict rank-one destabilizer. The last assertion follows from Theorem 5.1. □

Corollary 5.4. *If a saturated degree-five vertex has at least four neighbors of factor dimension at least two, its unassigned divisor degree satisfies*

$$A_v \geq \frac{6244655501}{6233518181} > 1,$$

and its unassigned ray defines a strict rank-one destabilizer. Consequently, in a saturated simple graph with stable tangent bundle, each degree-five vertex has at least two degree-one neighbors.

Proof. By Theorem 5.2, it suffices to take neighbouring dimensions $(2, 2, 2, 2, 1)$. With total coordinate sum at most six, the simplex and facet volumes are

$$Z(6) = \frac{6233518181}{362880}, \qquad Z'(6) = \frac{892093643}{51840}.$$

These values can be calculated as follows. Let $c_{j,k}$ be the coefficient of $q^j u^k$ in

$$(2u^2 + 2u + 1 - q(3u + 1))^4 (u + 1 - q).$$

The Laplace transforms of ϕ_1, ϕ_2 are respectively $(s+1-e^{-s})/s^2$ and $(2s^2+2s+1-e^{-s}(3s+1))/s^3$. Thus

$$Z(6) = \sum_{j,k} c_{j,k} \frac{(6-j)^{14-k}}{(14-k)!}, \qquad Z'(6) = \sum_{j,k} c_{j,k} \frac{(6-j)^{13-k}}{(13-k)!},$$

where $0 \leq j \leq 5$ and $0 \leq k \leq 9$. These finite sums give the displayed values. The unassigned ray is the only ray on its line, by the same direct-summand argument as in Corollary 5.3. Its weight exceeds one, so the line strictly destabilizes. In the saturated case neighboring dimensions equal neighboring degrees; at most three of the five neighbors can therefore have degree at least two. □

Corollary 5.5. *A saturated five-arm tree is unstable whenever at least four of its arms have length at least two. In particular, subdivision of four arms of the stable five-arm star produces an unstable mixed blowup, and further subdivision of those arms preserves this obstruction.*

Proof. The central vertex has degree five and at least four degree-two neighbors, so Corollary 5.4 applies. □

Subdividing the indexing graph changes both the factor dimensions and the dimension of the variety; it is therefore distinct from subdividing a fan in fixed dimension. The corollary shows that sufficiently long arms do not ensure stability for arbitrary trees in this construction. The obstruction requires a saturated vertex and gives no corresponding degree bound when the factor dimension exceeds the degree.

6. Stability for trees of maximum degree three

Every tree of maximum degree three gives a stable blowup when each nonleaf dimension equals its valence, with arbitrary positive leaf dimensions. We prove this by bounding the branch volume functions under the integrations at vertices with one or two children. A vertex with four neighbours satisfying the lower bounds proved below has a destabilizing radial foliation. Thus three is the largest degree for which all saturated trees are stable. The stable degree-four and degree-five stars of Section 5 show that higher degrees can also occur. We give further stable examples with two branching vertices.

Theorem 6.1. *Let G be a finite tree with at least one edge and maximum degree at most three. Set $b_v = d_v$ at every nonleaf vertex, and choose any positive integer b_v at each leaf. Then the mixed blowup $Y(G,b)$ has anticanonically stable tangent bundle. If $M = |E(G)|$, then*

$$n = 2M + \sum_{v \text{ leaf}} (b_v - 1), \qquad \rho = 2M + 1.$$

It is smooth Fano and admits no torus-equivariantly locally trivial bundle presentation with positive-dimensional complete fibre and base. The normalized degrees of unassigned original divisors satisfy

$$A_v \le \frac{25950132}{26414113} < 1 \quad (d_v = 3), \qquad A_v \le \frac{24}{25} < 1 \quad (d_v = 2). \tag{39}$$

In particular, every saturated tree of maximum degree three gives such a stable variety with $n = 2M$ and $\rho = n + 1$.

The theorem imposes no condition on the distances between branching vertices or on the relative sizes of the components of $G - v$. The bounds in (39) apply to the stated individual divisor degrees. They provide no uniform slope gap for all proper subspaces, or for leaves of arbitrary dimension.

In the notation of Section 4, the two internal-vertex bounds imply

$$\sum_{e \ni v} J_{v,e} \le \frac{24558189}{26414113} < 1 \quad (d_v = 3), \qquad \sum_{e \ni v} J_{v,e} \le \frac{22}{25} < 1 \quad (d_v = 2).$$

These estimates bound the sum of the exceptional intersections with the specified order of normal directions, accounting for all the more distant branches. By Theorem 3.4, the resulting vertex inequalities imply stability of the tangent bundle.

6.1. **Comparison of simplex integrals.** The comparison will use Efron's conditional-sum theorem. Throughout this subsection, functions are positive and continuous on $[0,\infty)$ and constant on $[1,\infty)$. For $k \geq 1$, define

$$V_{p_1,\dots,p_k}(t) = \int_{\substack{x_i \geq 0 \\ \sum_i x_i \leq t}} \prod_i p_i(x_i)\,dx, \qquad F_{p_1,\dots,p_k}(t) = V'_{p_1,\dots,p_k}(t),$$

and

$$B_{p_1,\dots,p_k}(t) = \int_{\substack{x_i \geq 0 \\ \sum_i x_i \leq t}} (t - \textstyle\sum_i x_i) \prod_i p_i(x_i)\,dx.$$

The derivative F is the integral over the slice where the coordinates have fixed sum. Its lattice measure is obtained by eliminating one coordinate, whose coefficient is one.

Lemma 6.2. *Let p_i, q_i be functions as above, with q_i log-concave. If every p_i/q_i is nonincreasing, then*

$$\frac{B_{p_1,\dots,p_k}(t)}{B_{q_1,\dots,q_k}(t)} \quad \text{and} \quad \frac{F_{p_1,\dots,p_k}(t)}{F_{q_1,\dots,q_k}(t)}$$

are nonincreasing for $t > 0$, and

$$\frac{F_{p_1,\dots,p_k}(t)}{V_{p_1,\dots,p_k}(t)} \leq \frac{F_{q_1,\dots,q_k}(t)}{V_{q_1,\dots,q_k}(t)}. \tag{40}$$

If every p_i/q_i is nondecreasing, both monotonicity assertions and inequality (40) reverse.

Proof. Take independent variables X_i with densities proportional to $e^{-x}q_i(x)$ on $[0,\infty)$, and two further independent exponential variables of rate one. These densities are log-concave. The positive function $R(x) = \prod_i p_i(x_i)/q_i(x_i)$ is bounded, has the required coordinatewise monotonicity, and is independent of the two additional coordinates. Extend it to real arguments by taking their nonnegative parts. By Efron's conditional-sum theorem [SW18, Theorems 3.2–3.3], its conditional expectation is monotone in the total sum. The nonincreasing version follows by applying the theorem to $-R$.

On the slice with total sum t, the common exponential factor cancels. Integration over the two additional coordinates gives $t - \sum_i x_i$. The conditional expectation is therefore $B_p(t)/B_q(t)$. Without these two variables, the same argument gives $h(t) = F_p(t)/F_q(t)$; when $k = 1$, the assertion is precisely the assumed monotonicity of the ratio. Rescaling to a fixed compact simplex proves continuity of all slice ratios for $t > 0$. Their positive denominators ensure that the monotone versions supplied by the theorem agree with these ratios everywhere. In the nonincreasing case,

$$V_p(t) = \int_0^t F_q(s)h(s)\,ds \geq h(t)V_q(t), \qquad F_p(t) = F_q(t)h(t).$$

Division proves (40). The same calculation with the reversed integral inequality proves the other case. □

The finite rational calculations below follow from a coefficient identity. For a function p as above, define the formal power series

$$Q_p(z) = p(1) + z\sum_{j\geq 0} \frac{z^j}{j!} \int_0^1 (1-a)^j p(a)\,da. \tag{41}$$

Only finitely many coefficients are needed in each application.

Lemma 6.3. *For $k \geq 1$, $r \geq 0$, and an integer $s \geq 0$,*

$$\int_{\substack{a_i \geq 0 \\ \sum_i a_i \leq k+r}} \frac{(k+r-\sum_i a_i)^s}{s!} \prod_i p_i(a_i)\, da = [z^{k+s}] e^{rz} \prod_i Q_{p_i}(z). \tag{42}$$

For $r > 0$, the facet volume is

$$F_{p_1,\dots,p_k}(k+r) = [z^{k-1}] e^{rz} \prod_i Q_{p_i}(z).$$

Proof. Partition the domain according to the set S of coordinates in $[0,1]$. Subtract one from each of the remaining $k - |S|$ coordinates. Their integration leaves

$$\frac{(r + \sum_{i \in S}(1-a_i))^{k-|S|+s}}{(k-|S|+s)!} \prod_{i \notin S} p_i(1)$$

times the product of the weights of the coordinates in the unit interval. Express the power as a coefficient of an exponential and integrate these coordinates over $[0,1]^S$. Summing over S proves (42). Differentiate the case $s = 0$ with respect to r to obtain the facet formula. □

6.2. **The proof for maximum degree three.** Orient an edge towards a parent vertex. Fix the parent's incidence coordinate at a and let $p(a)$ be the volume obtained by integrating over the component on the other side of the edge. The function is positive and continuous, and is constant for $a \geq 1$. The only inequality involving the fixed coordinate is $a + x \geq 1$, which is redundant above one. Positivity follows by choosing all assigned coordinates slightly greater than one and all remaining coordinates small and positive. The inequalities $b_v \geq d_v$ ensure that such choices lie in the interior of every vertex simplex. Dominated convergence in the fixed compact product of simplexes gives continuity. Removing a vertex separates the tree into disjoint components, so fixing its coordinates gives a product of the corresponding branch functions.

For a leaf of dimension b, the incoming function is

$$\phi_b(a) = \frac{(b + \min(a,1))^b}{b!}.$$

At a nonleaf of degree two or three, integration of the coordinate assigned to the parent gives, for $0 \leq a \leq 1$,

$$T(p)(a) = B_p(2+a), \tag{43}$$

$$\mathcal{G}(p,q)(a) = B_{p,q}(3+a), \tag{44}$$

Extend both functions constantly above one. There are one and two child coordinates, respectively, and the intervals of integration in the parent coordinate have lengths $2 + a - x$ and $3 + a - x - y$.

Proof of Theorem 6.1. Put $P(a) = (2 + \min(a,1))^3$, and write $p \preceq P$ if p/P is nonincreasing. The function P is chosen for comparison and does not represent the volume of an additional projective-space factor. It is log-concave, since its logarithmic derivative equals $3/(2+a)$ below one and drops to zero above one. Every leaf satisfies $\phi_b \preceq P$, since

$$\frac{d}{da} \log \frac{\phi_b(a)}{P(a)} = \frac{b}{b+a} - \frac{3}{2+a} \leq 1 - \frac{3}{2+a} \leq 0 \qquad (0 < a < 1).$$

The inequality holds for every positive leaf dimension.

We show that both operations in (43)–(44) preserve this bound. Direct integration, or Lemma 6.3, gives

$$T(P)(a) = \frac{363}{10} + \frac{173}{4}a + \frac{27}{2}a^2,$$
$$\mathcal{G}(P,P)(a) = \frac{119659}{80} + \frac{148781}{80}a + \frac{3213}{4}a^2 + \frac{243}{2}a^3.$$

For a polynomial g of degree at most three,

$$\left(\frac{g(a)}{(2+a)^3}\right)' = \frac{(2+a)g'(a) - 3g(a)}{(2+a)^4}.$$

The numerator has the following coefficients and zero cubic coefficient.

g	1	a	a^2
$T(P)$	$-112/5$	$-65/2$	$-27/2$
$\mathcal{G}(P,P)$	$-12283/16$	$-20261/40$	$-297/4$

All are negative. Hence $T(P) \preceq P$ and $\mathcal{G}(P,P) \preceq P$. By Lemma 6.2, $T(p)/T(P)$ and $\mathcal{G}(p,q)/\mathcal{G}(P,P)$ are nonincreasing whenever $p, q \preceq P$. Multiplying positive nonincreasing ratios proves $T(p), \mathcal{G}(p,q) \preceq P$. Induction on the number of vertices in each oriented component gives $p \preceq P$ for every incoming function. No equality between the two child functions is required.

At a degree-three vertex, the simplex is bounded by total coordinate sum four. For three copies of P, its volume and facet volume are

$$V_{P,P,P}(4) = \frac{26414113}{320}, \qquad F_{P,P,P}(4) = \frac{6487533}{80}.$$

These values and the two preceding polynomials follow from

$$Q_P(z) = 27 + \frac{65}{4}z + \frac{131}{20}z^2 + \frac{233}{120}z^3 + O(z^4)$$

and Lemma 6.3. The volume and facet volume are the coefficients of z^3 and z^2 in $e^z Q_P(z)^3$, respectively. The two branch polynomials are $[z^2]e^{(1+a)z}Q_P(z)$ and $[z^3]e^{(1+a)z}Q_P(z)^2$. Inequality (40) therefore gives $A_v \le 25950132/26414113$. This bound is strictly below one, with difference $463981/26414113 > 0$.

At a degree-two vertex, each incoming function satisfies

$$\frac{p(a)}{p(1)} \ge \frac{P(a)}{P(1)} \ge a \quad (0 \le a \le 1), \qquad (2+a)^3 - 27a = (1-a)^2(a+8) \ge 0.$$

Thus both incoming functions satisfy (20). The calculation in (22)–(23) uses only these inequalities and the constant continuation, and gives $24V - 25F \ge 0$. Hence $A_v \le 24/25$. At every leaf, Lemma 3.2 gives $A_v < 1$, also when its dimension exceeds one. The criterion of Theorem 3.4 is therefore satisfied at every vertex. It gives the strict slope inequalities for all proper subspaces and hence for all subsheaves. The geometric assertions and counts follow from Proposition 3.1 and $\sum_v d_v = 2M$. □

6.3. **A local obstruction at degree four.**

Theorem 6.4. *Let $Y(G,b)$ be a Fano mixed blowup indexed by a finite tree, with $b_u \geq d_u$ at every vertex. Suppose that a vertex v and its four neighbors all have degree four and factor dimension four. Then*

$$A_v \geq \frac{40090063176909393024}{39498346443818421703} > 1. \tag{45}$$

The unassigned ray at v defines a strict rank-one destabilizer. No additional condition is imposed on the branches beyond these neighbors.

Proof. Let $\ell(a) = 1+\min(a,1)$. For every incoming branch function p in a Fano tree, the ratio p/ℓ is nondecreasing. To see this, let b be the dimension of its first vertex, x the coordinate assigned to the parent, and z its other $b-1$ coordinates. Integration over the more distant vertices gives a volume function $\Psi(z)$. It is coordinatewise nondecreasing because increasing a coordinate only relaxes the mixed inequalities. It is independent of x. For $0 \leq a \leq 1$, integration of x from $1-a$ to $b+1-\sum z_j$ and rescaling give

$$p(a) = (b+a)^b \int_{\substack{y\geq 0 \\ \sum_j y_j \leq 1}} (1-\textstyle\sum_j y_j)\Psi((b+a)y)\, dy.$$

The integral factor is nondecreasing; the same formula includes $b=1$ as a zero-dimensional integral. Moreover,

$$\frac{d}{da}\log\frac{(b+a)^b}{1+a} = \frac{a(b-1)}{(b+a)(1+a)} \geq 0.$$

Both functions are constant above one, which proves the claim.

At a degree-four neighbor w of v, the incoming function is $g_w(a) = B_{p_1,p_2,p_3}(4+a)$ on $[0,1]$, where p_i are its three child inputs. Apply the increasing case of Lemma 6.2 with reference inputs ℓ. This function is log-concave, and each p_i/ℓ is nondecreasing. It follows that g_w/g_4 is nondecreasing, where

$$g_4(a) = B_{\ell,\ell,\ell}(4+a) = \frac{5217+5965a+2610a^2+520a^3+40a^4}{120} \quad (0 \leq a \leq 1), \tag{46}$$

with constant continuation.

For a further application of the lemma, we check log-concavity of g_4. Writing $H = 120g_4$ on $[0,1]$, direct differentiation gives

$$\begin{aligned}(H')^2 - HH'' = {} & 8348485 + 14860260a + 11120040a^2 \\ & + 4474400a^3 + 1020000a^4 + 124800a^5 + 6400a^6.\end{aligned}$$

Its coefficients are positive. Also $H' > 0$, so the logarithmic derivative drops to zero at the constant continuation. Thus g_4 is log-concave on the whole half-line.

The four components of the tree minus v give four functions whose ratios to g_4 are nondecreasing. Their product is the volume over these components with the central coordinates fixed. The increasing case of (40) bounds A_v below by the ratio for four copies of g_4 with total coordinate sum at most five. The corresponding volumes are

$$V_0(5) = \frac{39498346443818421703}{23224320000}, \qquad F_0(5) = \frac{1044012061898682 11}{60480000}.$$

For their exact evaluation, formula (41) gives

$$Q_{g_4}(z) = \frac{598}{5} + \frac{3683}{48}z + \frac{513}{16}z^2 + \frac{195899}{20160}z^3 + \frac{11549}{5040}z^4 + O(z^5).$$

Lemma 6.3 identifies $V_0(5), F_0(5)$ with $[z^4]e^z Q_{g_4}(z)^4$ and $[z^3]e^z Q_{g_4}(z)^4$. Their ratio proves (45); its excess over one is $59171673309097 1321/39498346443818421703$. The unassigned ray is the negative sum of four independent basis vectors and is the only ray on its line: other original rays lie on different lines or in different vertex summands, and exceptional rays have nonzero components in two summands. That line has normalized weight exactly $A_v > 1$, proving strict instability. □

Corollary 6.5. *Three is the largest integer D for which every saturated mixed-blowup tree of maximum degree at most D has anticanonically stable tangent bundle. A counterexample at degree four has dimension* 32 *and Picard number* 33*.*

Proof. Theorem 6.1 proves the assertion for $D = 3$. For $D = 4$, take one degree-four vertex joined to four degree-four vertices, each with three additional pendant leaves, and saturate every vertex. There are sixteen edges, five dimension-four factors, and twelve unit leaves, so $n = 32$ and $\rho = 33$. All comparisons in Theorem 6.4 are equalities in this case. Thus (45) gives an unstable example of maximum degree four, which also excludes every $D \geq 4$. □

6.4. **Two branching vertices with arbitrary separation.** Stable trees may contain degree-four or degree-five vertices. In the following family, two branching vertices may be separated by an arbitrary distance, and each pendant edge has length one.

Proposition 6.6. *Let $d, e \in \{3, 4, 5\}$. Join vertices of degrees d, e by a path of $L \geq 1$ edges, and attach respectively $d-1, e-1$ unit-length pendant edges. Saturate every vertex. The resulting mixed blowup is smooth Fano, has anticanonically stable tangent bundle, and admits no nontrivial torus-equivariantly locally trivial bundle presentation. Writing $M = d+e+L-2$, its dimension and Picard number are $n = 2M$ and $\rho = 2M + 1$.*

In Appendix A, we prove the proposition by bounding the divisor degrees at degree-two vertices uniformly. At each branching vertex, the required inequality is the positivity of a linear functional of its incoming volume polynomial.

The pendant lengths in Proposition 6.6 cannot all be arbitrary at a degree-five centre, by Corollary 5.4. The degree-four obstruction also has a specific scope: its proof uses the factorization for trees and does not establish the corresponding assertion for simple graphs with cycles.

7. A sharp Picard bound for toric bundles

For smooth toric Fano bundles with fibre a product of degree-six del Pezzo surfaces and base a product of projective spaces, the maximal Picard number among varieties with stable tangent bundle is $\lfloor 5n/4 \rfloor + 1$ for every $n \geq 4$. We prove the upper bound for arbitrary integral twists. The Fano condition bounds the twisting at each base factor. Stability requires each surface factor to have nonzero twists from at least two base factors and excludes product decompositions. These conditions give the bound; the families constructed in the following sections attain it.

7.1. **Bundle fans and a relative tangent obstruction.** Let F be a smooth projective toric variety with lattice N_F, and let $B = \prod_{c=1}^{k} \mathbb{P}^{b_c}$, where $k, b_c \geq 1$. Throughout this section a *toric bundle with fibre F over B* means a bundle with the following fan presentation. Its lattice is $N_F \oplus \bigoplus_c \mathbb{Z}^{b_c}$. The fibre rays are those of F in N_F, and the remaining primitive ray

generators are

$$f_a^c = (0, e_a^c) \quad (1 \le a \le b_c), \qquad f_0^c = \left(t_c, -\sum_{a=1}^{b_c} e_a^c \right), \qquad t_c \in N_F. \tag{47}$$

For every maximal fibre cone and every choice of one omitted ray from each set $\{f_0^c, \dots, f_{b_c}^c\}$, their remaining generators form a maximal cone of X, and these are all the maximal cones. This fan description is a hypothesis throughout the section. It describes a locally trivial toric bundle [CLS11, Theorem 3.3.19] and does not cover arbitrary Mori fibre spaces. The vectors t_c specify the twisting. The displayed normalization is obtained by an integral shear from the lifts of the standard base rays.

Write $P_F = \{x : \langle x, v\rangle \ge -1 \text{ for every fibre ray } v\}$ when F is Fano. All volume measures below are lattice normalized, without a factorial. The following obstruction applies whenever P_F has barycentre zero. For the special case $F = \mathbb{P}^1$ and base dimension at least two, the strict inequality below recovers [BDGP21, Theorem 8.1].

Proposition 7.1 (Obstruction over projective space)**.** *Let F be a smooth toric Fano variety of positive dimension f, and let $\pi : X \longrightarrow \mathbb{P}^b$ be a toric bundle with the fan presentation (47) and fibre F, where $b \ge 1$. Assume that X is Fano and that the barycentre of the anticanonical polytope P_F is zero. Then*

$$\mu_{-K_X}(T_{X/\mathbb{P}^b}) \ge \mu_{-K_X}(T_X).$$

In particular, T_X is not anticanonically stable. The inequality is strict if $b \ge 2$ and the twisting vector t_1 in (47) is nonzero. It is an equality if $b = 1$ or $t_1 = 0$.

Proof. Write $n = f + b$ and specialize (47) to a single base factor. The lifted base rays are

$$(0, e_1), \dots, (0, e_b), \qquad (t, -e_1 - \cdots - e_b), \qquad t \in N_F.$$

Put $s(x) = \langle x, t\rangle$ and $L(x) = b + 1 + s(x)$ for $x \in P_F$. The anticanonical polytope of X is

$$P_X = \{(x, y) : x \in P_F,\ y_i \ge -1,\ \sum_{i=1}^{b} (y_i + 1) \le L(x)\}.$$

Ampleness of $-K_X$ implies $L(x) > 0$ on P_F. Indeed, the vertex of P_X corresponding to any maximal fibre cone together with the base cone generated by $e_1, \dots, e_b$ must satisfy the remaining base inequality strictly. This gives $L(x) > 0$ at every vertex of P_F, hence throughout P_F by affinity.

Let $A_q = \int_{P_F} L(x)^q \, dx$, with lattice-normalized Lebesgue measure. The slices are b-simplices, so

$$D := (-K_X)^n = \frac{n!}{b!} A_b.$$

Each of the $b + 1$ divisors associated with base rays has degree

$$d = \frac{(n-1)!}{(b-1)!} A_{b-1}.$$

For the facets $y_i = -1$, integrate the corresponding $(b-1)$-simplex volumes. All $b+1$ degrees agree by the principal divisor relations associated with the characters dual to e_i. The relative tangent bundle has rank f and first Chern class equal to the sum of the fibre-ray divisors.

Hence

$$\begin{aligned}\mu(T_{X/\mathbb{P}^b}) - \mu(T_X) &= \frac{D-(b+1)d}{f} - \frac{D}{n}\\ &= \frac{(n-1)!}{f(b-1)!}\,(A_b - (b+1)A_{b-1})\\ &= \frac{(n-1)!}{f(b-1)!}\int_{P_F} s(x)(b+1+s(x))^{b-1}\,dx.\end{aligned}$$

The barycentre assumption implies $\int_{P_F} s(x)\,dx = 0$. Subtracting $(b+1)^{b-1}\int s$ from the last integral rewrites it as

$$\int_{P_F} s(x)\left((b+1+s(x))^{b-1} - (b+1)^{b-1}\right)\,dx.$$

This integrand is nonnegative, because the function $z \mapsto z^{b-1}$ is nondecreasing for $z > 0$. If $b \geq 2$ and $t \neq 0$, it is strictly positive whenever $s(x) \neq 0$. Since P_F is full-dimensional, that set has positive measure. If $b = 1$ or $t = 0$, the integrand vanishes. □

Corollary 7.2 (The Assarf–Nill extremal bundles)**.** *Let X_m be the toric variety associated with the smooth Fano polytope B_m of Assarf–Nill* [AN16, Definition 41 and Proposition 43]*. Thus $\dim X_m = 3m$, $\rho(X_m) = 4m+1$, and X_m is a toric S_6^m-bundle over $\mathbb{P}^m$, with nonzero twist having one ray component in each hexagon factor. Then T_{X_m} is not anticanonically stable for every $m \geq 1$, and is unstable for every $m \geq 2$.*

Proof. The anticanonical polytope of S_6^m is centrally symmetric and thus has barycentre zero. Apply the proposition with $b = m$ and $f = 2m$. □

7.2. **Incidences of hexagon fibre factors.** Let S_6 be the blow-up of $\mathbb{P}^2$ at its three torus-fixed points. Its fan rays, in a suitable lattice basis r, q, are $\pm r, \pm q, \pm(r+q)$. Its anticanonical polytope is centrally symmetric. For fibre S_6^j, decompose $N_F = \bigoplus_{i=1}^j N_i$ and write $t_c = (t_{1c}, \ldots, t_{jc})$. The *incidence graph of the twisting* is the bipartite graph on the fibre factors and base factors, with an edge ic precisely when $t_{ic} \neq 0$. Thus its edges record which base factors twist a given surface factor.

Theorem 7.3 (Picard bound for hexagon bundles)**.** *Let $j, k \geq 1$, let $b_1, \ldots, b_k \geq 1$, and let X be a locally trivial toric S_6^j-bundle over $\prod_{c=1}^k \mathbb{P}^{b_c}$, with the fan presentation (47). Assume that X is smooth Fano and that T_X is anticanonically stable. Then*

$$\rho(X) \leq n + \lfloor n/4 \rfloor + 1, \qquad n = \dim X.$$

Here the twists in each hexagon lattice are arbitrary integral vectors; they are not required to have the graph form of Section 8. More precisely, if $s = \sum_c b_c$ and ℓ is the number of edges of the incidence graph, then

$$n = 2j + s, \qquad \rho(X) = 4j + k, \qquad 2j \leq \ell \leq s, \qquad j + k - 1 \leq \ell. \tag{48}$$

Proof. Use the lattice decomposition and incidence graph defined above. Let ℓ be the number of incidences and put $s = \sum_c b_c$. Counting rays gives

$$n = 2j + s, \qquad \rho(X) = 4j + k.$$

For each hexagon let ψ_i be the piecewise linear function equal to one on its primitive rays and linear on its fan cones. For an integral nonzero vector t, the positive integer $\psi_i(t)$ is the sum of its coefficients in the minimal containing unimodular cone. The rays of the cth base

factor form a primitive collection: any proper subset lies in a lifted base cone, while the whole set does not. Their sum is t_c in the fibre lattice. Expressing it in the minimal containing product cone gives a primitive relation of degree

$$b_c + 1 - \sum_i \psi_i(t_{ic}).$$

The Fano condition makes this degree a positive integer [CLS11, Theorem 6.4.9(b)]. Therefore

$$\#\{i : t_{ic} \neq 0\} \leq \sum_i \psi_i(t_{ic}) \leq b_c, \qquad \ell \leq s.$$

We next show that each hexagon vertex of the incidence graph has degree at least two. Let Q_i be its centrally symmetric anticanonical hexagon, and define

$$L_c(x) = b_c + 1 + \sum_i \langle x_i, t_{ic}\rangle, \qquad G(x) = \prod_{c=1}^{k} \frac{L_c(x)^{b_c}}{b_c!}, \qquad V = \int_{\prod_i Q_i} G(x)\, dx.$$

The Fano assumption gives $L_c > 0$ on the product of hexagons, by the same strict vertex-inequality argument as in the preceding proposition. The anticanonical polytope is a product of simplex slices over $\prod_i Q_i$, so V is its lattice volume. Since the base has positive dimension, $N_{i,\mathbb{C}}$ is a proper subspace, and no base ray lies in it. Its normalized ray-weight sum is

$$w(N_{i,\mathbb{C}}) = \frac{1}{V} \sum_{H \text{ facet of } Q_i} \int_{H \times \prod_{h \neq i} Q_h} G(x)\, d\sigma\, dx_{-i} = 2 + \frac{1}{V} \int_{\prod_i Q_i} x_i \cdot \nabla_i G(x)\, dx.$$

The second equality follows from the divergence theorem in the x_i variables. This slice-volume identity occurs in subspace concentration [HL14, Eq. (4.4)]. We use the symmetry of the fibre factor; the total polytope need not be centred. For a facet with primitive outward normal u and equation $\langle u, x_i\rangle = 1$, the normal component of x_i is $1/\|u\|$, exactly converting Euclidean facet measure to lattice facet measure. This gives the stated normalization.

If all t_{ic} vanish, then $w(N_{i,\mathbb{C}}) = 2$ and stability fails. If only t_{ic} is nonzero, write $z = \langle x_i, t_{ic}\rangle$ and $A = b_c + 1 + \sum_{h \neq i} \langle x_h, t_{hc}\rangle > 0$. Apart from a positive factor independent of x_i, the inner integral contributing to $\int x_i \cdot \nabla_i G$ is

$$b_c \int_{Q_i} z(A+z)^{b_c - 1}\, dx_i = b_c \int_{Q_i} z\big((A+z)^{b_c-1} - A^{b_c-1}\big)\, dx_i \geq 0.$$

Here $\int_{Q_i} z = 0$ by central symmetry, and the integrand is nonnegative by the same monotonicity argument. Consequently $w(N_{i,\mathbb{C}}) \geq 2$, again contrary to stability. Thus $\ell \geq 2j$.

Finally, the incidence graph is connected. Otherwise, grouping the lattice summands according to its components gives a direct sum of proper subspaces containing all ray generators, so the ray matroid is disconnected, contrary to Lemma 2.3. A connected graph on $j + k$ vertices has at least $j + k - 1$ edges. Combining the inequalities gives

$$s \geq \ell \geq 2j, \qquad s \geq \ell \geq j + k - 1.$$

Hence $n \geq 4j$ and

$$\rho(X) = 4j + k \leq 4j + s - j + 1 = n + j + 1 \leq n + \lfloor n/4 \rfloor + 1,$$

as claimed. □

Corollary 7.4 (Necessary conditions for equality)**.** *Under the hypotheses of Theorem 7.3, suppose that $n = 4j + r$ with $0 \le r \le 3$ and that $\rho(X) = \lfloor 5n/4 \rfloor + 1 = 5j + r + 1$. Then the incidence graph is a tree,*

$$k = j + r + 1, \qquad \ell = \sum_c b_c = 2j + r, \qquad \sum_{i=1}^{j} \big(\deg(i) - 2\big) = r.$$

Every nonzero t_{ic} is a primitive ray of its hexagon fan, and the number of incidences at each base vertex c is b_c.

Proof. Equality in the proof of Theorem 7.3 forces $s = \ell = j + k - 1$ and $j = \lfloor n/4 \rfloor$. Connectedness therefore makes the incidence graph a tree. The sum of the dimensions of the base factors equals the number of incidences, so each base inequality is an equality, and $\psi_i(t_{ic}) = 1$ for each nonzero twist. In a unimodular hexagon cone, a nonzero integral vector with coefficient sum one is a primitive ray. Finally $\sum_i \deg(i) = \ell$ gives the last formula. □

The coefficient $5/4$ comes from (48). Each surface factor contributes two dimensions and four Picard classes and requires at least two nonzero twists. The positivity of the base primitive relations bounds the number of these incidences by the total base dimension. The j surface factors therefore require at least $2j$ base dimensions. Connectedness bounds the number of base factors.

The equality conditions alone do not imply stability, since they do not bound the degrees of subspaces containing rays from several factors. Section 8 constructs the fans, and Section 10 proves stability for independently subdivided subcubic trees. Paths attain the bound when $4 \mid n$; the end attachments of Corollary 9.3 give attainment in every dimension $n \ge 4$. Thus Theorem 7.3 is sharp in the stated bundle class. No bound for arbitrary smooth toric Fano varieties is asserted.

8. Toric bundles associated with graphs

A graph with positive integers assigned to its vertices determines a toric fibre bundle as follows. Each edge corresponds to a degree-six del Pezzo surface in the fibre, and each vertex to a projective-space factor in the base. The endpoints of an edge specify the two base factors that twist the corresponding surface.

Definition 8.1. Let Γ be a finite nonempty graph without loops, with vertex set C and edge set I. Multiple edges are allowed. Write $k = |C|$ and $j = |I|$, choose an orientation $c_i^- \to c_i^+$ for every edge i, and assign an integer $b_c \ge 1$ to each vertex c. In the lattice

$$N = \bigoplus_{i \in I} \mathbb{Z}_i^2 \ \oplus \ \bigoplus_{c \in C} \mathbb{Z}^{b_c}$$

write $r_i = (1,0)_i$, $q_i = (0,1)_i$, and let $f_1^c, \dots, f_{b_c}^c$ be the standard basis of the last summand. Set

$$f_0^c = -\sum_{t=1}^{b_c} f_t^c + \sum_{c_i^+ = c} r_i - \sum_{c_i^- = c} r_i. \tag{49}$$

For each edge take the six rays generated by $\pm r_i, \pm q_i, \pm(r_i + q_i)$. Together with the rays generated by all f_t^c, these are the rays of Σ_Γ. Its maximal cones contain two cyclically adjacent hexagon rays for each i, and all but one of $f_0^c, \dots, f_{b_c}^c$ for each c; all their faces are included. We write $X(\Gamma, b)$, or $X(\Gamma)$ when the integers are understood, for the associated toric variety.

The fan is smooth and projective, as proved below. The surface with the indicated hexagon fan is the blow-up of $\mathbb{P}^2$ at its three torus-fixed points; we denote it by S_6, with the subscript indicating its anticanonical degree. Projection onto the last summands identifies $X(\Gamma)$ as a toric fibre bundle with fibre S_6^j over $\prod_c \mathbb{P}^{b_c}$. Reversing the orientation of an edge gives an isomorphic fan: multiplication by -1 on its summand $\mathbb{Z}_i^2$, and the identity on every other summand, gives the required lattice isomorphism. The variety is therefore independent of the edge orientations.

We use primitive collections to test ampleness. A *primitive collection* of a simplicial fan is a set of rays which does not span a cone, whereas every proper subset does. For a complete simplicial fan, its *primitive relation* expresses the sum of these primitive ray generators as a positive linear combination of the ray generators of the unique cone whose relative interior contains that sum. When the sum is zero, the cone on the right is the empty cone.

Lemma 8.2. *Suppose a complete simplicial fan Σ is split by a base fan Σ_B and a fibre fan Σ_F: there is a lattice projection with kernel N_F and a subfan $\widehat{\Sigma}_B$ projecting cone by cone isomorphically onto Σ_B, and the cones of Σ are exactly $\widehat{\sigma}+\tau$, with $\widehat{\sigma}\in\widehat{\Sigma}_B$ and $\tau\in\Sigma_F$. The primitive collections of Σ are precisely the fibre primitive collections and the lifts of the base primitive collections. Fibre primitive relations are unchanged. In a lifted base primitive relation, the coefficients of the lifted base rays on the right are those of the base relation; the other terms are a nonnegative combination of fibre rays.*

Proof. The two summands of $\widehat{\sigma}+\tau$ have independent linear spans: projection is injective on the first and vanishes on the second. Consequently a set of rays spans a cone exactly when its fibre part spans a fibre cone and its base part projects to a base cone. The simplicial complex of Σ is therefore the join of the complexes of Σ_F and Σ_B. A minimal nonface of a join is a minimal nonface of one factor, proving the assertion about collections.

Let x be the sum of the generators of a lifted base collection, and write its minimal containing cone as $\widehat{\sigma}+\tau$. Projection maps the relative interior of this cone onto the relative interior of σ. Thus σ is the minimal containing cone of the projected sum. Since the generators of σ are linearly independent, comparison of coefficients determines all base coefficients. The remaining terms in x are nonnegative multiples of fibre rays. If $x\in N_{F,\mathbb{R}}$, its minimal containing cone has $\widehat{\sigma}=0$, so the fibre relation is unchanged. □

Proposition 8.3. *The variety $X(\Gamma)$ is smooth and projective, and is a Zariski locally trivial toric fibre bundle*

$$X(\Gamma)\longrightarrow\prod_{c\in C}\mathbb{P}^{b_c}\qquad\text{with fibre } S_6^j.$$

Its dimension and Picard number are

$$n=2j+\sum_c b_c,\qquad \rho=4j+k. \tag{50}$$

Its ray matroid is connected if and only if Γ is connected. Furthermore,

$$X(\Gamma)\text{ is Fano}\iff \deg_\Gamma(c)\le b_c\quad\text{for every } c\in C, \tag{51}$$

where the degree counts edges with multiplicity.

Proof. To prove completeness and projectivity, let

$$Q=\{(a,z)\in\mathbb{R}^2:|a|\le 1,\ |z|\le 1,\ |a+z|\le 1\}, \tag{52}$$

the anticanonical polygon of S_6. In coordinates (m_i, y_t^c) on $M_{\mathbb{R}}$, with $m_i = (a_i, z_i)$, put $s_c = \sum_{c_i^+=c} a_i - \sum_{c_i^-=c} a_i$. Choose integers $A_c > \deg_\Gamma(c)$ and consider the polytope

$$m_i \in Q, \qquad y_t^c \geq -1, \qquad \sum_{t=1}^{b_c} (y_t^c + 1) \leq A_c + s_c.$$

Every slice over Q^j is a product of full-dimensional simplices, since $A_c + s_c > 0$. At any point at most b_c inequalities of the cth simplex are active. A vertex therefore requires $2j$ independent active hexagon inequalities, so its projection is a vertex of Q^j. Conversely, each such vertex and each choice of a vertex in every simplex give a vertex of the full polytope. Exactly n inequalities are active, and their inward normals are precisely the generators of a maximal cone in Definition 8.1. The fan of inward normals is therefore Σ_Γ, which is complete and projective. Each maximal cone is unimodular: its hexagon generators give a basis of $\bigoplus_i \mathbb{Z}_i^2$, and modulo that sublattice its remaining generators form a standard maximal cone of $\prod_c \mathbb{P}^{b_c}$. The same bases show that the polytope has integral vertices.

Projection to the base splits the fan in the sense of Lemma 8.2, with fibre the product hexagon fan. The toric local triviality theorem gives the bundle structure [CLS11, Theorem 3.3.19]. There are $6j + \sum_c (b_c + 1)$ rays. Subtracting n gives (50).

The fibre primitive collections are the nonadjacent pairs in a hexagon. Their relations are either $v + (-v) = 0$, of degree 2, or $v + v'' = v'$, of degree 1. Here the degree of a primitive relation is the number of terms on the left minus the sum of coefficients on the right. The other primitive collections, by Lemma 8.2, are $\{f_0^c, \ldots, f_{b_c}^c\}$, with relations

$$f_0^c + \cdots + f_{b_c}^c = \sum_{c_i^+=c} r_i + \sum_{c_i^-=c} (-r_i). \tag{53}$$

The right-hand side consists of one ray from each incident hexagon, so these rays span a cone and all coefficients are 1. Its degree is $b_c + 1 - \deg_\Gamma(c)$. Batyrev's primitive-collection criterion for ampleness, in the form of [CLS11, Theorem 6.4.9(b)], says that $-K_X$ is ample exactly when all these degrees are positive. Since the degrees are integers, this is (51).

For the matroid assertion, the six rays of each hexagon belong to one matroid component: the antipodal pairs and the relation $r_i + q_i - (r_i + q_i) = 0$ connect all six. For each c, the set

$$\{f_0^c, \ldots, f_{b_c}^c\} \cup \{r_i : i \text{ is incident to } c\}$$

is a circuit. Indeed, projection to $\mathbb{Z}^{b_c}$ forces the coefficients of all the f_t^c in any relation to agree; the independent r_i then determine all the other coefficients, and a nonzero relation has no zero coefficient. These circuits connect precisely the vertex and edge factors incident in Γ. Thus a connected graph gives a connected matroid. Distinct connected components of Γ use disjoint lattice summands, giving a direct sum of nonempty ray matroids and proving the converse. □

Lemma 8.4 (Recovery of the graph from the fan)**.** *Let Γ be a connected loopless graph with at least one edge. The fan Σ_Γ of Definition 8.1 determines Γ and the integers b_c up to isomorphism. In particular, isomorphic toric fans give isomorphic graphs with the same vertex dimensions.*

Proof. The fibre rays can be recovered without the given presentation. They are precisely the primitive ray generators whose negatives also occur in the fan. Every fibre ray has this property. A base ray cannot be opposite a fibre ray, since its projection to the base lattice is nonzero. Rays belonging to different base factors cannot be opposite, since their projections

lie in independent summands. Within a base factor of dimension at least two, the standard projective-space rays contain no opposite pair. If $b_c = 1$, its two lifted rays have sum

$$f_0^c + f_1^c = \sum_{c_i^+ = c} r_i - \sum_{c_i^- = c} r_i.$$

This sum is nonzero: connectedness and the existence of an edge exclude an isolated vertex, and the vectors from different incident edges belong to independent hexagon summands. The loopless hypothesis ensures that no edge contributes both signs at c.

Restrict the ray matroid to the recovered fibre rays. Its connected components are exactly the six-ray sets of the individual hexagons. Indeed, the opposite pairs and the circuit on the three ray lines connect all rays of one hexagon, whereas independent hexagon planes admit no circuit meeting two factors. This determines each plane W_i and their direct sum $W = \bigoplus_i W_i$.

The primitive collections consisting entirely of the remaining rays are precisely the groups $C_c = \{f_0^c, \dots, f_{b_c}^c\}$, by Lemma 8.2. Thus these groups, and the integers $b_c = |C_c| - 1$, are determined by the fan. Their vector sums lie in W:

$$s_c := \sum_{u \in C_c} u = \sum_{c_i^+ = c} r_i - \sum_{c_i^- = c} r_i.$$

The component of s_c in the recovered direct summand W_i is nonzero exactly when c is an endpoint of i. These components determine the incidence graph, including parallel edges, and hence Γ. Primitive ray generators, opposite pairs, matroid components, primitive collections, and these vector sums are all preserved by a lattice isomorphism of fans, proving the assertion. □

Corollary 8.5. *If Γ is connected and $X(\Gamma)$ is Fano, then*

$$\rho \le 5\left\lfloor \frac{n}{4} \right\rfloor + 1, \qquad \rho - n \le \frac{n}{4} + 1. \tag{54}$$

Equality in the first bound holds exactly when Γ is a tree with $j = \lfloor n/4 \rfloor$ edges. In that case

$$\sum_c \big(b_c - \deg_\Gamma(c)\big) = n - 4\lfloor n/4 \rfloor.$$

In particular, for $j \ge 1$ and $n = 4j$, equality is equivalent to a tree with $b_c = \deg_\Gamma(c)$ at every vertex.

Proof. Connectedness gives $k \le j + 1$, with equality exactly for a tree. The Fano condition gives $\sum_c b_c \ge \sum_c \deg_\Gamma(c) = 2j$. Hence $n \ge 4j$ and

$$\rho = 4j + k \le 5j + 1 \le 5\lfloor n/4 \rfloor + 1.$$

Equality in this chain holds precisely when $k = j + 1$ and $j = \lfloor n/4 \rfloor$, giving the stated characterization. The formula for the sum follows from $n - 4j = \sum_c (b_c - \deg_\Gamma(c))$. □

The corollary includes the graph with one vertex and no edges: $X(\Gamma) = \mathbb{P}^{b_c}$, $\rho = 1$, and equality in its first bound holds exactly for $1 \le b_c \le 3$. The empty graph is excluded throughout. If Γ is disconnected, the lattice and fan decompose over its connected components, so $X(\Gamma)$ is a product of positive-dimensional varieties. Its tangent bundle is not stable for any polarization: it splits into nonzero direct summands, and at least one summand has slope at least the slope of their direct sum. The bound (54) applies to this graph construction. No bound for arbitrary smooth toric Fano varieties is asserted.

Example 8.6. For $j \geq 1$, let Γ_j be the path with vertices $0, \dots, j$, edges $i : i \to i+1$ for $0 \leq i < j$, and weights $b_0 = b_j = 1$, $b_c = 2$ for $0 < c < j$. We denote $X(\Gamma_j)$ by X_j. It is Fano, has connected ray matroid, and satisfies

$$\dim X_j = 4j, \qquad \rho(X_j) = 5j + 1.$$

Thus these paths attain the bound in Corollary 8.5. To prove stability, we must also check the slope inequalities using the divisor degrees computed below; connectedness alone does not suffice.

8.1. **Anticanonical degrees.** Assume henceforth that $b_c \geq \deg_\Gamma(c)$. For $a = (a_i) \in [-1,1]^j$ define

$$s_c(a) = \sum_{c_i^+ = c} a_i - \sum_{c_i^- = c} a_i, \qquad L_c(a) = b_c + 1 + s_c(a), \qquad \Phi(a) = \prod_{c \in C} \frac{L_c(a)^{b_c}}{b_c!}. \tag{55}$$

In particular $L_c \geq 1$. All lattice volume measures below assign volume 1 to a fundamental parallelepiped, without a factorial. Put

$$d\nu(a) = \prod_{i \in I} (2 - |a_i|)\, da_i.$$

For a hexagon ray v in the ith factor, let $d\nu^{(i,v)}$ be obtained by replacing the ith factor of $d\nu$ with the measure dE_v on $[-1,1]$ specified by

$$\begin{aligned} dE_{r_i} &= \delta_{-1}, & dE_{-r_i} &= \delta_1, \\ dE_{r_i+q_i} = dE_{-q_i} &= \mathbf{1}_{[-1,0]}(a_i)\, da_i, & dE_{q_i} = dE_{-r_i-q_i} &= \mathbf{1}_{[0,1]}(a_i)\, da_i. \end{aligned} \tag{56}$$

Here $\delta_{\pm 1}$ denote unit point masses; this notation is separate from that for normalized divisor degrees. Each measure in (56) has total mass 1.

Proposition 8.7. *Write $D = (-K_X)^n$ and $d_v = D_v \cdot (-K_X)^{n-1}$ for $X = X(\Gamma)$. Then*

$$D = n! \int_{[-1,1]^j} \Phi \, d\nu, \tag{57}$$

$$d_{f_t^c} = (n-1)! \int_{[-1,1]^j} \frac{b_c}{L_c} \Phi \, d\nu \qquad (0 \leq t \leq b_c), \tag{58}$$

$$d_v = (n-1)! \int_{[-1,1]^j} \Phi \, d\nu^{(i,v)} \qquad (v \text{ in hexagon } i). \tag{59}$$

These formulas also hold for $j = 0$, with the integral over a point.

Proof. The moment polytope of $-K_X$ is given by

$$m_i = (a_i, z_i) \in Q, \qquad y_t^c \geq -1, \qquad -\sum_{t=1}^{b_c} y_t^c + s_c(a) \geq -1.$$

After translating y_t^c by 1, its slice over $(m_i) \in Q^j$ is

$$\prod_c \{(u_t^c) : u_t^c \geq 0,\ \sum_t u_t^c \leq L_c(a)\},$$

whose volume is $\Phi(a)$. For fixed $a \in [-1,1]$, the allowable z in (52) form an interval of length $2 - |a|$. Fubini's theorem and $D = n!\,\mathrm{vol}(P_{-K_X})$ therefore give (57).

For $t \geq 1$, the facet corresponding to f_t^c sets $u_t^c = 0$. It replaces one simplex volume by $L_c^{b_c - 1}/(b_c - 1)!$, multiplying the integrand by b_c/L_c. The facet corresponding to f_0^c sets

$\sum_t u_t^c = L_c(a)$; eliminating one coordinate gives the same volume. This elimination is an isomorphism of the affine facet lattice with the coordinate lattice, since the coefficient of the eliminated coordinate is 1 and s_c is integral affine. Its lattice measure therefore requires no additional factor. The standard facet-volume identity $d_v = (n-1)!\,\mathrm{vol}(F_v)$ proves (58), including $b_c = 1$.

A facet for a hexagon ray lies over the corresponding edge of Q. Each such edge has lattice length 1. The edges $a = -1$ and $a = 1$ project to the unit point masses in (56). On each other edge, projection to a is a lattice isomorphism onto an interval of length 1. Specifically, $a + z = -1$ and $z = 1$ give $[-1, 0]$, whereas $z = -1$ and $a + z = 1$ give $[0, 1]$. These are exactly the four interval measures in (56). Fubini's theorem on the facet gives (59). □

Remark 8.8. If $Z = \int \Phi\, d\nu$, the normalized degrees $\delta_v = nd_v/D$ are the respective integrals in (58) and (59), divided by Z. The factorials thus cancel in the slope criterion. The $b_c + 1$ rays associated to a vertex have equal degrees.

9. Families attaining the Picard bound

Paths and suitable modifications at their ends attain the Picard bound in every dimension at least four. Trees with one trivalent vertex give further examples and provide the initial case of the subdivision theorem in the next section. We state these results here; the degree calculations and uniform estimates are in Appendices B–D.

9.1. **Paths.** Take the graph of Section 8 to be a path with vertices $0, \dots, j$ and edges $i\colon i \to i+1$, $0 \le i < j$. Put $b_c = 2$ at the interior vertices and $b_0, b_j \in \{1, 2, 3\}$ at the ends. Denote the resulting variety by $X_j^{(b_0,b_j)}$. Its dimension and Picard number are

$$n = 4j - 2 + b_0 + b_j, \qquad \rho = 5j + 1. \tag{60}$$

The Fano condition holds at every vertex. We write $X_j = X_j^{(1,1)}$.

Theorem 9.1. *For every $j \ge 1$ and every $b_0, b_j \in \{1, 2, 3\}$, the tangent bundle of $X_j^{(b_0,b_j)}$ is slope stable with respect to its anticanonical polarization.*

The proof is in Appendix B.

9.2. **Attainment in every dimension.** Put $r = (1, 0)$ and $q = (0, 1)$ in the hexagon lattice and define

$$A_0 = (-r), \quad A_1 = (q, -r - q), \quad A_2 = (-r, q, -q), \quad A_3 = (r, -r, q, -r - q).$$

For $j \ge 1$, take j hexagon factors. Between hexagons $i - 1$ and i place a $\mathbb{P}^2$ base factor with twisting vector $r_{i-1} - r_i$. At the first hexagon attach one $\mathbb{P}^1$ base factor for each vector in A_a; at the last attach one for each vector in $-A_b$. Use the locally trivial toric bundle fan over the product of these projective spaces, with fibre S_6^j, and denote the resulting variety by $X_j(a, b)$.

Theorem 9.2. *For every $j \ge 1$ and*

$$(a, b) \in \{(1, 0), (2, 0), (1, 1), (3, 0), (1, 2)\},$$

$X_j(a, b)$ is smooth Fano and its tangent bundle is anticanonically stable. Moreover,

$$\dim X_j(a, b) = 4j + a + b, \qquad \rho(X_j(a, b)) = 5j + a + b + 1.$$

The proof is in Appendix C.

Corollary 9.3 (Sharpness in the bundle class)**.** *For every $n \geq 4$, the largest Picard number of a smooth toric Fano n-fold with stable tangent bundle among the locally trivial toric S_6^j-bundles over products of positive-dimensional projective spaces, with the fan presentation (47), is*

$$\left\lfloor \frac{5n}{4} \right\rfloor + 1.$$

Proof. Theorem 7.3 gives the upper bound. Write $n = 4j + r$, $0 \leq r \leq 3$. For $r = 0$ use the path X_j from Theorem 9.1. For $r = 1, 2, 3$ use respectively the pairs $(1, 0), (2, 0), (3, 0)$ in Theorem 9.2. Their Picard numbers are $5j + r + 1 = \lfloor 5n/4 \rfloor + 1$. □

9.3. **Trees with one trivalent vertex.** Trees with a trivalent vertex also give stable examples at the Picard bound, including when the three arms have different lengths. Let $\Gamma_{a,b,c}$ be the tree with one vertex of degree three and arms containing a, b, c edges. Put $b_v = \deg(v)$ and let $Y_{a,b,c} = X(\Gamma_{a,b,c})$ be the graph bundle of Section 8. Its base has one factor $\mathbb{P}^3$, three factors $\mathbb{P}^1$, and $a + b + c - 3$ factors $\mathbb{P}^2$.

Theorem 9.4. *For all integers $a, b, c \geq 2$, the tangent bundle of $Y_{a,b,c}$ is stable with respect to $-K_{Y_{a,b,c}}$. Moreover,*

$$\dim Y_{a,b,c} = 4(a + b + c), \qquad \rho(Y_{a,b,c}) = 5(a + b + c) + 1.$$

These examples attain the bundle bound in every dimension divisible by four and at least twenty-four. Their fans are not lattice-isomorphic to the path fans in the same dimension: the central $\mathbb{P}^3$ rays form a primitive collection of size four, whereas the path primitive collections have sizes two or three by Lemma 8.2.

The Fano property and connected ray matroid follow from Proposition 8.3. The proof in Appendix D allows the subspace selections on the three arms to vary independently. A cubic integral expresses their joint contribution at the central vertex. A contraction estimate bounds the effect of changing one arm on the other two, uniformly in all three lengths.

At a degree-two vertex, integration over the base simplex acts on a positive function by the operator

$$(\mathcal{T} f)(a') = \int_{-1}^{1} f(a) \frac{(3 + a - a')^2}{2} (2 - |a|)\, da.$$

Integration along a path iterates this positive integral operator. For positive continuous functions define

$$d_H(f, g) = \log \left(\sup_a \frac{f(a)}{g(a)} \sup_a \frac{g(a)}{f(a)} \right).$$

Lemma B.2 proves that for arbitrary positive continuous initial functions and every $i \geq 1$,

$$d_H(\mathcal{T}^i f, \mathcal{T}^i g) \leq \log(81/25)(2/7)^{i-1}. \tag{61}$$

The corresponding ray degrees are ratios of positive integrals. For each hexagon or interior-base ray type, with left and right chain indices $i, k \geq 1$, its normalized degree $\delta(i, k)$ consequently satisfies

$$|\log \delta(i, k) - \log \delta_\infty| \leq \log(81/25)\big((2/7)^{i-1} + (2/7)^{k-1}\big).$$

The limiting degree is independent of the data at the distant ends. A ray at an end has only a one-sided limiting degree, which can depend on the factor at that end. These estimates give convergence, without asserting monotonicity in the path lengths.

The deficit of a subspace is its dimension minus the sum of the normalized weights of the rays it contains. To deduce stability, we need a uniform bound on the deficits of all relevant

subspaces. The rank formula and gluing argument in Section 10 give such bounds when subtrees are joined.

10. Stability under edge subdivision

Every fixed finite tree of maximum degree three with at least one edge gives extremizers for the bundle bound after sufficiently long edge subdivisions. The divisor degrees become nearly independent of distant branches, and an inequality for joining subtrees bounds the deficits of subspaces meeting both sides of a path. The subdivision lengths may vary independently.

For a finite tree T of maximum degree three, suppress its degree-two vertices; the resulting tree will be called its skeleton. Let s be the number of trivalent vertices. If $s \geq 1$, the skeleton has $s+2$ leaves and $2s+1$ edges. Replacing its edges by paths of positive integral lengths gives a subdivision Γ. Throughout this section put $b_v = \deg_\Gamma(v)$ in Definition 8.1.

Theorem 10.1. *There are explicit integers L_s, depending only on the number $s \geq 1$ of trivalent vertices, such that $T_{X(\Gamma)}$ is anticanonically stable whenever every skeleton edge is replaced by a path of length at least L_s. The lengths may be chosen independently. One can take $L_1 = 2$ and $L_s = O(\log s)$ as $s \to \infty$. If j is the total number of edges of Γ, then*

$$\dim X(\Gamma) = 4j, \qquad \rho(X(\Gamma)) = 5j + 1.$$

In particular every fixed finite tree of maximum degree three and at least one edge gives stable tangent bundles after all sufficiently long independent edge subdivisions.

The case with no trivalent vertex is Theorem 9.1. These examples attain the Picard bound in Theorem 7.3. Since the fan determines the graph, different branching patterns give distinct extremizers in the same dimension.

Corollary 10.2 (Nonisomorphic extremal fans)**.** *For every integer $q \geq 1$, there is an integer J_q such that for every $j \geq J_q$ there are at least q pairwise nonisomorphic toric fans of smooth toric Fano varieties with stable tangent bundle, dimension $4j$, and Picard number $5j+1$. Each belongs to the bundle class of Theorem 7.3 and attains its maximum in that dimension.*

Proof. Choose q finite subcubic trees with an edge and with pairwise distinct numbers of trivalent vertices, all positive. Such trees exist: take a path of s vertices and attach leaves so that each path vertex has degree three, with the three-arm tree for $s = 1$. Subdivision preserves their different numbers of trivalent vertices.

For each chosen tree, first suppress degree-two vertices and give every skeleton edge a length satisfying Theorem 10.1. Let j_t be the resulting total length for the tth tree, and put $J_q = \max_t j_t$. For any $j \geq J_q$, increase the length of one skeleton edge in the tth tree by $j - j_t$. The stability theorem allows independent lengths, so every resulting graph bundle is stable. Each subdivided tree has j edges and $j+1$ vertices, and $\sum_v b_v = \sum_v \deg(v) = 2j$. Proposition 8.3 therefore gives dimension $4j$ and Picard number $5j+1$.

Subdivisions of the chosen trees are nonisomorphic because their numbers of trivalent vertices differ. Lemma 8.4 then excludes isomorphisms of their toric fans. Finally, Theorem 7.3 identifies $5j+1$ as the maximum within the bundle class. □

The corollary distinguishes isomorphism classes of toric fans. It gives neither a classification of all extremizers nor a growth rate for their number.

10.1. **Subdivision and dependence on distant branches.** By (61), the dependence on distant branches decreases exponentially along a subdivided edge. Stability also requires estimates for subspaces containing several ray types. We obtain them from the rank formula at a cut and induction on the number of trivalent vertices, starting with the quantitative form of Theorem 9.4. The finite rational comparisons for three-arm trees and paths are in Appendices D and B. Their error bounds are uniform over all allowed lengths. The proof gives the following sufficient lengths, without a claim of optimality:

s	1	2	3	4	5	10	25	50	100	1000
L_s	2	27	31	31	33	35	41	43	47	57.

The argument does not require an enumeration of tree shapes. The theorem applies to the bundles of Definition 8.1. In the blowup construction, subdivision can instead produce instability, as in Corollary 5.5.

We use the following slice functions. Put $d\nu(t) = (2 - |t|)\,dt$ on $[-1, 1]$ and $\mathcal{K}(s, t) = (3 + s - t)^2/2$. The right operator is

$$(\mathcal{T}_R h)(s) = \int_{-1}^{1} \mathcal{K}(s, t)h(t)\,d\nu(t).$$

Write h_∞ for its positive projective limit, obtained by iterating from $h_0(t) = 2 + t$ and normalizing at zero. By contraction, the limit is independent of the positive initial function. At a trivalent vertex, integrating the other two branches with incoming functions u, v gives the outgoing slice function

$$J_{u,v}(s) = \iint_{[-1,1]^2} \frac{(4 - s - t - z)^3}{6} u(t)v(z)\,d\nu(t)d\nu(z).$$

This is the cubic integral of (89). It depends on a boundary coordinate, unlike the scalar exceptional contributions $J_{v,e}$ in the blowup construction.

10.2. **Rank at a cut.** Consider the independent reflections of the hexagons and permutations of the untwisted base rays from Appendix D. These symmetries do not permute branches. An invariant ray-spanned subspace selects each hexagon as $H_e = 0, 1, 2$ (empty, distinguished line, or plane), and at each vertex selects the untwisted rays and twisted ray with indicators C_v, T_v. Delete the edges with $H_e > 0$. A component of the remaining forest is called complete for this selection if $C_v = T_v = 1$ at every vertex in it. If c_0 counts these components, the rank is

$$r = \sum_e H_e + \sum_v b_v C_v + \sum_v T_v - c_0. \tag{62}$$

Indeed, first quotient by the selected hexagon directions and untwisted base coordinates. A selected twisted ray with $C_v = 0$ has a base coordinate absent from the other selected rays, so it contributes one dimension. The remaining twisted rays are rows of the incidence matrix of the zero-edge forest. A dependence among selected rows extends by zero to a function constant on each component, and exists precisely on a complete component.

For a subtree rooted at a cut, postpone subtracting the relation of its root component. Write $a = 1$ if this component is complete, and $a = 0$ otherwise.

Let e indicate whether the selection is nonempty. Set $f = 1$ precisely when every hexagon is a plane, every vertex with $b_v \geq 2$ has $C_v = 1$, and every leaf has $C_v \vee T_v = 1$. These are precisely the local requirements for full rank when the tree is completed. They are necessary

because of the independent nontrivial representations at the hexagons and bases and the base coordinate belonging only to each leaf. They also suffice. The possible triples are

$$(a, e, f) = (0,0,0), (0,1,0), (1,1,0), (0,1,1), (1,1,1).$$

Let c_* count the complete components other than the root component. If $w_e(H)$ is the selected hexagon weight and δ_v one base-ray weight, define the open deficit

$$A = \sum_e (H_e - w_e(H_e)) + \sum_v (b_v C_v + T_v)(1 - \delta_v) - c_*.$$

Thus the open deficit is rank minus weight before subtracting the single boundary relation. Joining two rooted pieces with open deficits A, B and indicators a, b across a hexagon H gives final deficit

$$\begin{cases} A + B - ab, & H = 0, \\ A + B + H - w(H) - a - b, & H > 0. \end{cases} \tag{63}$$

For $H = 0$ the two root components merge and are complete exactly when $a = b = 1$; for $H > 0$ they close separately. The combined selection is nonempty exactly when $e \vee e' \vee (H > 0)$, and has full rank exactly when $f \wedge f' \wedge (H = 2)$.

10.3. **An inequality inherited from a completed tree.** Consider a finite subtree ending at a degree-two vertex adjacent to the cut. Give it the limiting degrees obtained by continuing the cut to an ordinary leaf whose distance tends to infinity. Equivalently, its incoming function at the cut is h_∞. Let α be its all-selected open deficit.

Lemma 10.3. *Suppose every sufficiently long completion to that leaf has deficit at least $\mu > 0$ for every invariant selection of rank strictly between zero and the ambient dimension. For the retained subtree,*

boundary condition	*inequality*
$a = 0,\ e = 1$	$A \geq \mu$
$a = 1,\ f = 0$	$A - \alpha \geq \mu$
$a = 1,\ f = 1$	$A - \alpha \geq 0.$

The empty selection has $A = 0$.

Proof. For $a = 0, e = 1$, select no ray in the continuation or its joining hexagon. The completed selection is proper and nonempty, including when the retained subtree has $f = 1$. Formula (63) makes its deficit exactly A. Pass to the limit in the finitely many retained degrees.

For $a = 1, f = 0$, select every ray in the continuation and take $H = 2$ at the cut. The retained subtree still prevents the selection from having full rank. Subtract the all-selected deficit, which is zero. The continuation and joining terms cancel exactly, leaving $A - \alpha$. Now pass to the limit; the cancellation avoids interchanging an unbounded sum with a limit. For $a = f = 1$, adding omitted rays preserves the open rank and boundary indicator and decreases the deficit, since their weights are positive. Thus selecting all rays minimizes the deficit under this boundary condition. □

10.4. **Deficit increments for the limiting degrees.** Let w_1, w_2 be the limiting weights of a distinguished hexagon line and full plane, and let δ be one limiting degree-two-base weight. Appendix B gives $w_2 + 3\delta = 4$. The rational moment matrices at index 40, with Lemma B.2, give

$$\frac{187}{100} < w_2 < \frac{15}{8}, \qquad 0 < w_1 < \frac{14}{25}. \tag{64}$$

The rational approximations are

$$w_1 \simeq 0.5579289459, \qquad w_2 \simeq 1.8740976113.$$

For the exact index-40 rational values, with $D = 11757/10000$, $B = 81$ and $y = 2D(2/7)^{39}$, the errors are at most $2B(y+y^2)$ and $6B(y+y^2)$, respectively; these bounds are smaller than the distances to the endpoints in (64).

Put $\beta = w_2/2$ and replace the open deficit by $A - \beta a$. Extending across one hexagon H and one new degree-two vertex C, T changes the boundary indicator to $a' = CTa$ for $H = 0$, and $a' = CT$ for $H > 0$. The adjusted increment is

$$(2C + T)(1 - \delta) + H - w(H) - a\mathbf{1}_{\{H>0\}} + \beta(a - a'). \tag{65}$$

For $C = T = 0$ its three values, indexed by $H = 0, 1, 2$, are

$$\begin{array}{c|ccc} a = 0 & 0 & 1 - w_1 & 2 - w_2 \\ a = 1 & \beta & \beta - w_1 & 1 - \beta. \end{array}$$

When $CT = 0$ and $(C, T) \neq (0, 0)$, add $(2C + T)(1 - \delta)$. For $C = T = 1$ the values are

$$\begin{array}{c|ccc} a = 0 & w_2 - 1 & \beta - w_1 & 1 - \beta \\ a = 1 & w_2 - 1 & w_2 - w_1 - 1 & 0. \end{array}$$

All nonzero entries exceed $1/16$ by (64); also $1 - \delta = (w_2 - 1)/3 > 0.29$. The only zero transitions are $(a, C, T, H) = (0, 0, 0, 0)$ and $(1, 1, 1, 2)$, corresponding to selecting no rays or all rays in one step. For the adjusted deficits, the joining correction is

$$\begin{cases} \beta(a + b) - ab, & H = 0, \\ H - w(H) + (\beta - 1)(a + b), & H > 0. \end{cases} \tag{66}$$

Its only zeros are $(a, b, H) = (0, 0, 0)$ and $(1, 1, 2)$; all other values exceed $1/16$. Consequently, inserting any number of steps with the limiting weights cannot decrease the adjusted deficit.

10.5. **Uniform comparison of divisor degrees.** Split an internal skeleton edge of length $2q + 1 + m$, $m \geq 0$, leaving q edges and q degree-two vertices at each end. Retain these two end paths. The middle consists of $m + 1$ hexagons and m degree-two bases. The two components, capped by a leaf on the chosen edge, have $s_1, s_2 \geq 1$ trivalent vertices, with $s_1 + s_2 = s$. Retain all their other edge lengths.

Define comparison weights by putting the incoming function h_∞ at each cut and using the limiting weights on the middle. Every individual ray weight is at most $B = 81$, provided all skeleton lengths are at least two. Indeed a junction slice function $J_{u,v}$ in (89) has oscillation ratio at most 27. Each hexagon has an endpoint of degree two or one, whose incoming function has oscillation at most nine. The facet measure has mass one and ν has mass three, giving $27 \cdot 9/3 = 81$. The base-ray bounds are at most $1, 2, 3$ at leaves, degree-two vertices and junctions.

Changing one incoming function by Hilbert distance η changes $J_{u,v}$ by at most η: normalize its pointwise ratio to lie in $[1, e^\eta]$ and integrate the positive cubic. At each degree-two vertex the distance contracts by $\tau = 2/7$. The path from the cut to each vertex is unique, so at each subsequent junction only one incoming function changes. The resulting change propagates along both outgoing branches; the two changes never meet again at a junction.

Set

$$x_q = D\tau^{q-1}, \qquad c_s = \frac{156s + 272}{5}, \qquad E_s(q) = Bc_s x_q(1 + 2x_q), \qquad D = \frac{11757}{10000}. \tag{67}$$

Lemma 10.4. *If all skeleton lengths are at least two and $2x_q \le 1$, the actual and comparison deficits of every selection differ by at most $E_s(q)$, independently of all subdivision lengths.*

Proof. The incoming function at either cut has traversed at least q degree-two vertices from the opposite junction. Its distance from the limit is at most $x = x_q$. Along the retained end path it contracts geometrically toward the junction. Six hexagon rays and three base rays per step give a sum of logarithmic degree errors at most $9x/(1-\tau)$. The base closest to the cut has error at most x; the adjacent retained hexagon has an additional contraction, which is not needed for this estimate.

A capped component with s_i junctions has $2s_i + 1$ skeleton edges. One contains the end path just counted; each other edge contributes at most $9x/(1-\tau)$, by contraction along its interior. This bound does not use the contraction before that edge is reached. Its s_i junction bases add $4s_i x$, and its $s_i + 1$ genuine leaf bases add $2(s_i + 1)x$. The total for the capped component is at most

$$\left\{ \frac{9(2s_i + 1)}{1-\tau} + 6s_i + 2 \right\} x.$$

Number the edges of the chosen path $i = 0, \dots, L-1$, where $L = 2q + 1 + m$, and its internal bases $c = 1, \dots, L-1$. On the middle edges $q \le i \le L - q - 1$, the omitted left and right distances are at most $D\tau^{i-1}$ and $D\tau^{L-2-i}$. On middle bases $q + 1 \le c \le L - q - 1$, they are at most $D\tau^{c-2}$ and $D\tau^{L-2-c}$. Each pair is at most $2x$; summing either set over its positions gives at most $2x/(1-\tau)$. The six hexagon and three base rays therefore add at most $18x/(1-\tau)$.

The sum of all logarithmic ray errors is consequently at most

$$\left\{ \frac{9(2s + 4)}{1-\tau} + 6s + 4 \right\} x = c_s x.$$

Each individual error η is at most $2x \le 1$, and $e^\eta - 1 \le \eta(1 + 2x)$. Multiplying by B bounds the sum of absolute ray-weight changes by $E_s(q)$. The selected rank is unchanged, so the same bound holds for every deficit. □

10.6. **Joining subtrees and induction.** Let α_i be the open deficit for the selection of all rays in component i. Join two identical copies of this component by a path of length $2q + 1$. Its actual all-selected deficit is zero. The comparison deficit is $2\alpha_i - w_2$. Lemma 10.4, applied with $2s_i$ junctions and the same q, gives

$$|\alpha_i - \beta| \le \tfrac{1}{2} E_{2s_i}(q).$$

This uses the degree identity for the doubled tree, without a stability assumption. Since c_s is affine,

$$\tfrac{1}{2}\{E_{2s_1}(q) + E_{2s_2}(q)\} = E_s(q). \tag{68}$$

Suppose the two families of completed components have uniform lower bounds for the deficits of invariant selections of rank strictly between zero and the ambient dimension, $\mu_1, \mu_2 \le \mu_0 := 2698/100000 < 1/16$. Lemma 10.3 and (68) show that a selection in a component whose boundary state has a positive inherited bound contributes at least that bound minus the error in its adjustment. A selection with $a = f = 1$ contributes at least minus that error, and an empty selection contributes zero. The intervening increments and the final joining correction are all nonnegative.

If neither selected boundary state has a positive inherited bound, a proper nonempty selection on the whole tree must contain a positive increment or joining correction. Otherwise

each zero transition preserves either the empty or the full selection, both excluded on the whole tree. Thus the comparison deficit is at least $\min(\mu_1, \mu_2) - E_s(q)$. Subtracting the degree-comparison error gives the joining inequality

$$\text{actual deficit} \ \geq \min(\mu_1, \mu_2) - 2E_s(q). \tag{69}$$

Proof of Theorem 10.1. For $s \geq 2$, let q_s be the least positive integer such that

$$2x_{q_s} \leq 1, \qquad E_s(q_s) \leq \frac{\mu_0}{4s(s-1)}, \qquad L_s = 2q_s + 1. \tag{70}$$

Put $L_1 = 2$ and $\mu_s = \frac{\mu_0}{2}(1+1/s)$. The minimum exists by geometric decay. These thresholds are nondecreasing: c_s increases and the allowed error decreases. We prove the stronger assertion that every invariant selection of rank strictly between zero and $4j$ has deficit at least μ_s. The case $s = 1$ is the quantitative conclusion of Theorem 9.4.

For $s \geq 2$, remove any internal skeleton edge and cap each exposed junction by a leaf. The two smaller trees have $s_1, s_2 \geq 1$ junctions, with $s_1 + s_2 = s$. Their unchanged lengths satisfy the inductive thresholds since $L_s \geq L_{s_i}$. Their new leaf edges may be continued to arbitrary sufficiently large lengths, giving the families required by Lemma 10.3. Retain q_s edges at either end of the chosen original edge; its length is at least $2q_s + 1$. Formula (69) now gives

$$\text{deficit} \geq \mu_{s-1} - 2E_s(q_s) \geq \mu_{s-1} - \frac{\mu_0}{2s(s-1)} = \mu_s > 0.$$

The independent local symmetries and connected ray matroid therefore give stability by Proposition 2.4.

For the growth bound, it suffices to choose

$$q \geq 1 + \left\lceil \log_{7/2}\left(\frac{8BDc_s s(s-1)}{\mu_0}\right)\right\rceil.$$

Then $2x_q \leq 1$ and $E_s(q) \leq 2Bc_s x_q \leq \mu_0/[4s(s-1)]$. Thus $L_s = O(\log s)$. The dimension and Picard number follow from Proposition 8.3, since $\sum_v b_v = 2j$ and a tree has $j + 1$ vertices. For a tree with original degree-two vertices, the skeleton lengths are sums of the original subdivision lengths. The case $s = 0$ is Theorem 9.1. □

The quantitative bound $\mu_s > 0.01349$ applies only to the locally invariant selections. The symmetry criterion gives stability for all subspaces, but no uniform quantitative bound for their deficits is asserted.

Corollary 10.5. *Suppose a tree has two trivalent vertices joined by a path of length at least 27, with two outer arms at each vertex. If all four outer arms have length at least two, the graph bundle with $b_v = \deg(v)$ has anticanonically stable tangent bundle.*

Proof. The two families of completed components are three-arm trees, so both inherited deficit bounds are μ_0 even when the outer lengths are less than L_2. Lemma 10.4 needs only outer lengths at least two. With $q = 13$, formula (69) gives

$$\mu_0 - 2E_2(13) = \frac{12211377739987820545958674 1}{59869134806427004500312500 00} > 0.02039.$$

□

11. Further questions

The Picard maxima for the two construction classes leave open the classification of their stable members. For blowups indexed by trees, the vertex criterion reduces this problem to divisor-degree inequalities. For bundles, the subdivision theorem constructs families attaining the maximum, while the classification of stable twists remains open.

Question 11.1. *Which saturated blowup trees of maximum degree four or five have stable tangent bundle? What additional subspace inequalities are needed for the same construction on graphs with cycles?*

The degree-four obstruction gives a sufficient condition for instability. Proposition 6.6 also provides stable examples with vertices of higher degree. The case of larger factor dimensions at nonleaf vertices is not covered by the subcubic stability theorem. The exceptional mixed-intersection formula of Section 4 applies without the tree hypothesis. It determines the slope of the specified foliation; higher-rank subsheaves require further inequalities.

Question 11.2. *Is there an absolute integer L_* such that every finite subcubic tree with an edge gives a stable graph bundle whenever all skeleton edges are subdivided to length at least L_*?*

The sufficient length in Theorem 10.1 grows logarithmically with the number of trivalent vertices. The theorem does not establish that this dependence is necessary. A further question is which primitive twisting vectors give stable bundles attaining the Picard bound. The necessary incidence conditions in Corollary 7.4 do not resolve this question.

Finally, let $R(n)$ be the largest Picard number of a smooth toric Fano n-fold with anticanonically stable tangent bundle. The bundle theorem gives the lower bound below. For comparison, a published splitting theorem supplies a general upper bound.

Corollary 11.3 (Consequence of Assarf–Nill)**.** *For $n \geq 4$,*

$$\left\lfloor \frac{5n}{4} \right\rfloor + 1 \leq R(n) \leq 2n - \left\lceil \frac{\sqrt{60n + 1309} - 37}{30} \right\rceil .$$

Proof. The lower bound is Theorem 1.3. For a stable X, put $k = 2n - \rho(X) \geq 0$, using Casagrande's bound [Cas06]. Assarf–Nill [AN16, Theorem 8] show that $n > 15k^2 + 37k + 1$ forces a factor S_6 in a product decomposition. For $n \geq 3$ this is a nontrivial product, whose tangent bundle is not stable. Thus $n \leq 15k^2 + 37k + 1$; solving for the nonnegative integer k gives the upper bound. □

Assarf–Nill [AN16, Conjecture 13] conjecture the stronger bound $\rho \leq (4n+3)/3$ for smooth toric Fano varieties of dimension at least three that are not nontrivial products. Tangent stability implies this nonproduct condition.

Question 11.4. *Does every smooth toric Fano n-fold with anticanonically stable tangent bundle satisfy $\rho \leq (4n+3)/3$ for $n \geq 3$? What is $\limsup_{n\to\infty} R(n)/n$?*

The results here give an asymptotic lower ratio of $5/4$; the proposed upper ratio $4/3$ remains conjectural. To deduce the upper bound from the toric slope criterion, one would need to show that a Picard number exceeding the bound forces a proper subspace to violate the required degree inequality. The exceptional-intersection identity and the incidence argument for the specified bundles do not establish this implication.

Appendix A. Calculations for stars and two branching vertices

We prove Theorem 5.1 and Proposition 6.6 using the notation of Sections 5 and 6.

A.1. **Stability of stars.**

Proof. The geometric assertions follow from Proposition 3.1. Lemma 3.2 gives unassigned divisor degree strictly less than one at every leaf. By Theorem 3.4, stability is therefore equivalent to $A_m < 1$. This also applies to $m = 1$, when both vertices are leaves.

Use affine coordinates $a_i = 1 + \langle x, e_i\rangle$ and coordinates $y_i = \langle x, u_i\rangle$. The anticanonical polytope is

$$a_i \geq 0, \qquad \sum_{i=1}^{m} a_i \leq m+1, \qquad -\min(a_i, 1) \leq y_i \leq 1.$$

Define, for $a \geq 0$ and $t > 0$,

$$\ell(a) = 1 + \min(a, 1), \qquad Z_m(t) = \int_{\substack{a_i \geq 0 \\ \sum_i a_i \leq t}} \prod_{i=1}^{m} \ell(a_i)\, da_1 \cdots da_m. \tag{71}$$

The lattice volume of the polytope is $Z_m(m+1)$. Eliminating one coordinate from $\sum_i a_i = t$ gives the lattice measure on its central facet, whose volume is $Z'_m(t)$. Consequently

$$A_m = \frac{Z'_m(m+1)}{Z_m(m+1)}. \tag{72}$$

The use of lattice measure eliminates any Euclidean normal-length factor.

For $t > m$, the integral has the finite expression

$$Z_m(t) = \sum_{j=0}^{m} (-1)^j \binom{m}{j} \sum_{h=0}^{m-j} \binom{m-j}{h} \frac{(t-j)^{2m-h}}{(2m-h)!}. \tag{73}$$

Indeed, the Laplace transform of ℓ is $(\lambda + 1 - e^{-\lambda})/\lambda^2$. The simplex integral in (71) has transform $(\lambda + 1 - e^{-\lambda})^m/\lambda^{2m+1}$; expanding the numerator and inverting each term proves (73). Differentiation gives the exact values of $1 - A_m$ in Table 1. Thus stability holds for $1 \leq m \leq 5$, and $A_m > 1$ for $6 \leq m \leq 14$. Each entry compares the individual divisor degree A_m with one. At $m = 1$, the line $\mathbb{C}e_0$ also contains e_1, so the table entry is not the weight deficit of that line.

It remains to prove $A_m > 1$ for all $m \geq 15$. Set $t = m + 1$ and

$$J_m(t) = \int_{\substack{a_i \geq 0,\ \sum_i a_i \leq t \\ 0 \leq a_1 \leq 1}} a_1 \prod_{i=2}^{m} \ell(a_i)\, da_1 \cdots da_m.$$

Rescale the simplex in (71) to a fixed simplex and differentiate. Since $\ell'(a) = 1$ on $(0, 1)$ and is zero on $(1, \infty)$, symmetry among the coordinates gives

$$tZ'_m(t) = mZ_m(t) + mJ_m(t). \tag{74}$$

The derivative of ℓ fails to exist only on a set of measure zero, which contributes no additional term.

For every integer $p \geq 1$, $s > 0$, and $0 < \theta \leq 1$, the inequalities $\theta\ell(a) \leq \ell(\theta a) \leq \ell(a)$ imply

$$\theta^{2p} Z_p(s) \leq Z_p(\theta s) \leq \theta^p Z_p(s).$$

TABLE 1. Exact values of $1 - A_m$, obtained from (73) and its derivative at $t = m + 1$.

m	$1 - A_m$
1	3/7
2	7/31
3	149/1193
4	3039/47119
5	2509/103289
6	−13135/2987177
7	−41300337/1594796251
8	−8432407681/197900582879
9	−92165038835/1646595334057
10	−1447101945437/21629833922043
11	−289731964699063/3811832962784653
12	−244482437035005131/2920544373257729893
13	−1113498721218169506 9/12329450011419556514 3
14	−9019201501156191732 47/9391892098065855984793

Using the upper bound for Z_m and the lower bound for J_m yields

$$\begin{aligned} Z_m(t) &= \int_0^t \ell(a) Z_{m-1}(t-a)\, da \\ &\le 2Z_{m-1}(t) \int_0^t (1 - a/t)^{m-1}\, da = \frac{2t}{m} Z_{m-1}(t), \\ J_m(t) &= \int_0^1 a Z_{m-1}(t-a)\, da \\ &\ge Z_{m-1}(t) \int_0^1 a(1 - a/t)^{2(m-1)}\, da. \end{aligned}$$

For $0 \le a \le 1$, the inequality $\log(1 - x) \ge -x/(1 - x)$ gives

$$2(m-1)\log(1 - a/t) \ge -\frac{2(m-1)a}{t-a} \ge -2a.$$

Hence

$$\int_0^1 a(1 - a/t)^{2(m-1)}\, da \ge \int_0^1 a e^{-2a}\, da = \frac{1 - 3e^{-2}}{4} > \frac{1}{7}.$$

For the strict inequality, use $e^2 > 7$. The terms through degree four in its power series sum to seven, and every remaining term is positive. It follows that $J_m(t)/Z_m(t) > m/(14t)$. Substituting in (74), with $t = m + 1$, proves

$$A_m - 1 = \frac{mJ_m(t)/Z_m(t) - 1}{m+1} > \frac{m^2 - 14m - 14}{14(m+1)^2} > 0 \qquad (m \ge 15). \tag{75}$$

The numerator on the right is one at $m = 15$ and increases thereafter.

For $m \ge 6$, the ray e_0 is the only ray on its line. The other central rays are not collinear with the negative sum of at least two basis vectors, and every other ray has a nonzero leaf component. Thus $\mathbb{C}e_0$ has normalized ray weight $A_m > 1$ and gives the stated strict destabilizer. The finite calculation and the tree criterion complete the proof. □

Corollary A.1. *For $m \geq 6$, the first term of the Harder–Narasimhan filtration of T_{Y_m} is the line bundle $\mathcal{O}_{Y_m}(D_{e_0})$.*

Proof. Put $A = A_m$, and let B, C be the normalized degrees of $-u_i, s_i$, respectively. The degrees of u_i, e_i are $B - C, A - C$, so $(m+1)A + 2mB - mC = 2m$. On the simplex $\Delta = \{a_i \geq 0 : \sum_i a_i \leq m+1\}$, put $F(a) = \prod_{j \neq i} \ell(a_j)$ and $Z = Z_m(m+1)$. The leaf facets give

$$B = \frac{\int_\Delta F\, da}{Z}, \qquad C = \frac{\int_{\Delta \cap \{a_i \leq 1\}} F\, da}{Z}.$$

Since $\ell \leq 2$, strictly on an open set, $B > 1/2$. Also $Z(1 - 2B + C) = \int_{\Delta \cap \{a_i \leq 1\}} a_i F\, da > 0$. Thus $2B - C < 1$, and positivity of $B - C$ gives $C < 2B - C < 1$.

Uniqueness of the first Harder–Narasimhan term implies its invariance under the torus and the permutations of the arms. Its generic subspace is spanned by rays. Otherwise, replacing it by the span of the rays it contains would preserve its positive degree and decrease its rank, contrary to maximality of the slope. Define

$$L = \mathbb{C}e_0, \quad \mathcal{E} = \operatorname{span}(e_i), \quad \mathcal{U} = \operatorname{span}(u_i), \quad \mathcal{S} = \operatorname{span}(s_i).$$

An invariant subspace spanned by rays is generated by complete ray orbits. The positive and negative leaf rays have the same span. Any two of $\mathcal{E}, \mathcal{U}, \mathcal{S}$ span the ambient space, and $L \subset \mathcal{E}$. Thus the nonzero proper possibilities are $L, \mathcal{E}, \mathcal{U}, \mathcal{S}, L + \mathcal{U}, L + \mathcal{S}$. Their normalized slopes at the three rank-m spaces are

$$\widehat{\mu}(\mathcal{E}) = 2 - 2B < 1, \qquad \widehat{\mu}(\mathcal{U}) = 2B - C < 1, \qquad \widehat{\mu}(\mathcal{S}) = C < 1.$$

The last two spaces have slopes $(A + m(2B - C))/(m+1)$ and $(A + mC)/(m+1)$, both strictly below $A > 1$. Thus the generic subspace of the first term is L. Since the term is saturated and has rank one, it is reflexive and hence a line bundle on the smooth variety. Its induced filtration has a single jump, at e_0 with multiplicity one, giving the line bundle $\mathcal{O}_{Y_m}(D_{e_0})$. □

Rank-one maximal destabilizing subsheaves also occur in the four-dimensional examples of Reynolds [Rey23, Appendix A, p. 68].

A.2. **Two branching vertices.**

Proof. Let ℓ be the volume function for a leaf of dimension one. A branching vertex of degree d with $d - 1$ pendant leaves contributes the function

$$g_d(a) = [z^d] e^{(1+a)z} Q_\ell(z)^{d-1} \qquad (0 \leq a \leq 1),$$

by Lemma 6.3. It is constant above one. The three polynomials are

$$g_3(a) = \frac{137 + 155a + 60a^2 + 8a^3}{12},$$
$$g_4(a) = \frac{5217 + 5965a + 2610a^2 + 520a^3 + 40a^4}{120},$$
$$g_5(a) = \frac{122889 + 141357a + 66120a^2 + 15760a^3 + 1920a^4 + 96a^5}{720}.$$

At the degree-d centre, the volume function from the connecting path is $T^{L-1} g_e$. For any positive function p, write $c = p(1)$, $i = \int_0^1 p(a)\, da$, and $j = \int_0^1 ap(a)\, da$. Then

$$T(p)(a) = (2i - j + c/2) + (i + c)a + (c/2)a^2. \tag{76}$$

All coefficients are positive, since $2i - j > 0$.

First consider the degree-two vertices of the connecting path. The polynomials g_d are convex on $[0,1]$, and their endpoint differences are

$$g_3(1) - g_3'(1) = \frac{61}{12}, \qquad g_4(1) - g_4'(1) = \frac{1447}{120}, \qquad g_5(1) - g_5'(1) = \frac{3821}{144}.$$

The tangent-line inequality at one gives $g_d(a) \geq a g_d(1)$. Equation (76) also gives

$$T(p)(a) - aT(p)(1) = (1-a)(2i - j + c/2 - ac/2) \geq 0.$$

Every incoming function at a degree-two vertex of the connecting path therefore satisfies (20). The calculation in the proof of Theorem 3.6 gives $A_v \leq 24/25 < 1$ at all these vertices, for every L.

It remains to treat the two branching vertices. For $d \in \{3,4,5\}$ define the linear functional

$$D_d(p) = [z^d]e^z Q_\ell(z)^{d-1} Q_p(z) - [z^{d-1}]e^z Q_\ell(z)^{d-1} Q_p(z). \tag{77}$$

This is $V - F$ at the degree-d centre. It is positive exactly when $A_v < 1$. In the following calculation, each monomial a^k is extended by the constant one above one. Then

$$T(a^k) = \left(\frac{2}{k+1} - \frac{1}{k+2} + \frac{1}{2}\right) + \left(\frac{1}{k+1} + 1\right) a + \frac{1}{2}a^2.$$

Table 2 lists the values of $D_d(T(a^k))$ obtained by exact evaluation of (41) and (77).

TABLE 2. Values of $D_d(T(a^k))$, with each monomial extended constantly above one.

d	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
3	4477/360	611/80	1483/240	7909/1440	367/72	24403/5040
4	362/15	19847/1440	15361/1440	44209/4800	15077/1800	79019/10080
5	102241/3360	725819/60480	398891/60480	55289/13440	410239/151200	15739/8640

For $L = 2$, the final application of T acts on g_e, which has degree at most five and positive coefficients. For $L \geq 3$, it acts on $T^{L-2}g_e$, a quadratic with positive coefficients by (76). In both cases the function is a nonnegative linear combination of the constantly extended monomials $1, a, \ldots, a^5$, with at least one positive coefficient. Linearity and Table 2 prove $D_d(T^{L-1}g_e) > 0$ for every $L \geq 2$. Interchanging d, e proves the inequality at the other centre.

For $L = 1$, the two centres have incoming functions g_e, g_d. Table 3 lists the six cases up to interchange. Their positive values are obtained from the displayed polynomials and (77) by division by the corresponding positive volumes.

Every pendant leaf satisfies $A_v < 1$ by Lemma 3.2. Theorem 3.4 now proves stability. The remaining assertions follow from Proposition 3.1. □

APPENDIX B. PATH DEGREES, RANKS AND UNIFORM ESTIMATES

This appendix proves Theorem 9.1 for the varieties $X_j^{(b_0,b_j)}$ of Section 9. The proof is computer assisted: fixed rational matrices give the divisor degrees, and a recurrence with three states gives the relevant subspace ranks. A contraction estimate reduces the assertion for arbitrary path lengths to a finite calculation. We give the matrices, recurrence, rational bounds and error estimate below.

TABLE 3. The minimum of one minus the two normalized divisor degrees at adjacent branching vertices.

(d,e)	$\min(1-A_{\text{left}}, 1-A_{\text{right}})$
$(3,3)$	104355/994027
$(3,4)$	2643419/54754807
$(3,5)$	14018811/1317077851
$(4,4)$	97814081/2152964421
$(4,5)$	1281377927/155298531575
$(5,5)$	8349076421/1244343460105

B.1. **Divisor degrees along the path.** Write $a_i = \langle m_i, r_i\rangle \in [-1,1]$ for the coordinate on the ith dual hexagon and $d\nu(a) = (2-|a|)\,da$. The factors in the volume formula are

$$L_0 = b_0 + 1 - a_0, \qquad L_c = 3 + a_{c-1} - a_c \ (1 \le c < j), \qquad L_j = b_j + 1 + a_{j-1}.$$

Set

$$\mathcal{K}(a,a') = \frac{(3+a-a')^2}{2}, \qquad \ell_0(a) = \frac{(b_0+1-a)^{b_0}}{b_0!}, \qquad h_0(a) = \frac{(b_j+1+a)^{b_j}}{b_j!}. \tag{78}$$

Define positive functions recursively by

$$\ell_{i+1}(a') = \int \ell_i(a)\mathcal{K}(a,a')\,d\nu(a), \qquad h_{k+1}(a) = \int \mathcal{K}(a,a')h_k(a')\,d\nu(a'). \tag{79}$$

All integrals without limits in this section are over $[-1,1]$. If Z_j is the volume of the moment polytope, then

$$Z_j = \int \ell_i(a)h_{j-1-i}(a)\,d\nu(a), \qquad D = n!Z_j.$$

For a hexagon ray v at position i, and an interior base ray at position c, respectively, the normalized degrees are

$$\delta_{i,v} = \frac{\int \ell_i h_{j-1-i}\,dE_v}{\int \ell_i h_{j-1-i}\,d\nu}, \tag{80}$$

$$\delta_c = \frac{\iint \ell_{c-1}(a)(3+a-a')h_{j-1-c}(a')\,d\nu(a)d\nu(a')}{\iint \ell_{c-1}(a)\mathcal{K}(a,a')h_{j-1-c}(a')\,d\nu(a)d\nu(a')}. \tag{81}$$

Here E_v is the push-forward of lattice length on the corresponding hexagon edge. For $v = r, -r$ it is unit mass at $-1, 1$, respectively; for the other four rays it is Lebesgue measure on $[-1,0]$ or $[0,1]$, twice each. At the left end, the numerator for a single ray is obtained by replacing ℓ_0 with $(b_0+1-a)^{b_0-1}/(b_0-1)!$; the right end is analogous. These formulas follow by integrating the product of simplex volumes in Section 8.

In the basis $1, a, a^2$, the moment matrix and the coefficient matrix of the kernel are

$$W = \begin{pmatrix} 3 & 0 & 5/6 \\ 0 & 5/6 & 0 \\ 5/6 & 0 & 7/15 \end{pmatrix}, \qquad \kappa = \begin{pmatrix} 9/2 & -3 & 1/2 \\ 3 & -1 & 0 \\ 1/2 & 0 & 0 \end{pmatrix}. \tag{82}$$

For cubic end factors, append a fourth row and column, using

$$W_{pq} = I_{p+q}, \qquad I_t = \begin{cases} 0 & t \text{ odd}, \\ 2\big(2/(t+1) - 1/(t+2)\big) & t \text{ even}, \end{cases} \qquad 0 \le p, q \le 3,$$

and extend κ by zero. If λ, η are the coefficient vectors of ℓ_0, h_0, then

$$Z_j = \lambda^{\mathsf{T}}(W\kappa)^{j-1}W\eta. \tag{83}$$

Replace the relevant W by $W_v = (\int a^{p+q}\, dE_v)_{pq}$ to compute a hexagon degree. For an interior base degree replace the relevant κ by the coefficient matrix of $3+a-a'$. The four edge moment sequences are $(-1)^t$, 1, $(-1)^t/(t+1)$, and $1/(t+1)$. Thus every finite-path degree used below is rational and is determined by the displayed data.

B.2. **Invariant subspaces and ranks.** Let A be generated by the independent hexagon reflections $(x, y) \mapsto (x - y, -y)$, the permutations of the two untwisted rays at each interior base factor, and the permutations of the b untwisted rays at each end $\mathbb{P}^b$. These preserve the fan and the degrees. By Proposition 2.4, it suffices to test the A-invariant ray-spanned subspaces, since the path has connected ray matroid.

Adjoining all rays in the span of a ray subset preserves its rank and increases its weight. The two nontrivial reflection orbits in a hexagon each span its plane. Hence it suffices to consider the following *configurations*: choose, at each hexagon, nothing, the pair $\{r_i, -r_i\}$, or all six rays; at each base factor, choose all its untwisted rays or none, and choose its twisted ray independently. Configurations need not be closed under taking the rays in their span, but include every invariant closed set. Positivity for all nonempty configurations of proper rank is therefore equivalent to stability.

Write $H_i \in \{0, 1, 2\}$ for the hexagon choice, and $(C_c, T_c) \in \{0, 1\}^2$ for the choices of the untwisted rays and twisted ray at base vertex c. Let $g_c = \sum_{t=1}^{b_c} f_t^c$. The nontrivial representation contributed by a selected hexagon plane has dimension one; that contributed by selected untwisted base rays has dimension $b_c - 1$. For $b_c = 1$ this latter contribution is zero. The remaining vectors lie in the space with basis $r_0, \dots, r_{j-1}, g_0, \dots, g_j$, where

$$f_0^0 = -g_0 - r_0, \qquad f_0^c = -g_c + r_{c-1} - r_c \ (1 \le c < j), \qquad f_0^j = -g_j + r_{j-1}. \tag{84}$$

Lemma B.1. *The rank of a configuration is computed as follows. A selected hexagon line or plane contributes H_i; selected untwisted rays at vertex c contribute $b_c C_c$. Each selected twisted ray with $C_c = 0$ contributes one further dimension. For the remaining twisted rays, form a graph with vertices $r_0, \dots, r_{j-1}$ and one additional vertex $*$. Identify every selected r_i with $*$. Include $r_{c-1}r_c$ when $C_cT_c = 1$, and include the edges $*r_0$ and $r_{j-1}*$ when $C_0T_0 = 1$ and $C_jT_j = 1$. Add the rank of this graphic matroid to the preceding contributions.*

A configuration spans the whole lattice space if and only if every hexagon is a plane, every interior base has $C_c = 1$, and each end has $C_c = 1$ or has $b_c = 1$ and $T_c = 1$.

Proof. The nontrivial representations of the independent group factors are distinct and occur with multiplicity one. After accounting for them and the selected basis vectors r_i, g_c, a twisted ray whose g_c is not selected has a coordinate occurring in no other remaining vector. It adds one dimension and may be removed. The other vectors in (84), modulo the selected basis vectors, are precisely the incidence vectors of the stated graph. Their rank is its graphic-matroid rank. Full rank requires each nontrivial representation, hence every hexagon plane and every interior untwisted pair. All r_i are then selected, and the only remaining condition is to obtain each end's base space, as stated. □

The minimum can be computed by the following recurrence. After processing hexagon i, the state of r_i is G if it is selected, g if it is unselected but joined to $*$ through the processed edges, and u otherwise. A period consists of an interior base factor followed by its right-hand

hexagon. Let w_h be the weight of hexagon choice h, where $w_0 = 0$, and let b be the weight of one base ray. For a choice (C, T, h) put

$$L = CT, \qquad a(C,T) = 2C + T(1-C).$$

The rank increment, new state, and weight increment are given by

(85)

condition	rank increment	new state	weight increment
$h \geq 1$	$a(C,T) + h + L\,\mathbf{1}_{s=u}$	G	$(2C+T)b + w_h$
$h = 0,\ L = 1$	$a(C,T) + 1$	g if $s \neq u$, u otherwise	$(2C+T)b$
$h = 0,\ L = 0$	$a(C,T)$	u	$(2C+T)b$

The left end contributes $b_0 C_0 + T_0(1 - C_0)$ dimensions before the first hexagon; its possible remaining edge starts at $*$. The right end contributes $b_j C_j + T_j(1-C_j) + C_j T_j \mathbf{1}_{s=u}$ dimensions. Its weight is $(b_j C_j + T_j)\delta_j$, and similarly at the left. To process the initial hexagon, use (85) with incoming state G, $L = C_0 T_0$, and $a(C,T) = 0$, since the left base contribution has already been included; only the hexagon weight is subtracted at this step.

Use binary indicators e, f to record whether any ray has been selected and whether the full-rank conditions on the processed factors still hold. After choosing the left end, initialize

$$e = C_0 \vee T_0, \qquad f = C_0 \vee (b_0 = 1 \wedge T_0).$$

After the initial hexagon, replace these by $e \vee (H_0 > 0)$ and $f \wedge (H_0 = 2)$. Each interior period updates them by

$$e \longmapsto e \vee C \vee T \vee (h > 0), \qquad f \longmapsto f \wedge C \wedge (h = 2).$$

At the right end, the final values are $e \vee C_j \vee T_j$ and $f \wedge \big(C_j \vee (b_j = 1 \wedge T_j)\big)$. At each step retain the least cost, defined as rank minus weight, for each state and pair of indicators. At the end discard final $e = 0$ and final $f = 1$. This gives the exact *minimum invariant rank deficit*, denoted $m_A(j; b_0, b_j)$. Its positivity suffices for stability. The recurrence does not assert that this minimum equals the minimum over arbitrary subspaces.

B.3. **Uniform convergence.** For positive functions on $[-1, 1]$ define Hilbert's projective distance by

$$d_H(f,g) = \log\left(\sup_a \frac{f(a)}{g(a)} \sup_a \frac{g(a)}{f(a)}\right), \qquad \Delta = \log(81/25), \quad \tau = 2/7.$$

Lemma B.2. *For $i, i' \geq 1$ one has $d_H(\ell_i, \ell_{i'}) \leq \Delta \tau^{\min(i,i')-1}$, and the same holds for the right-hand functions. The bound also compares chains starting from different end factors. For a hexagon or interior-base degree ratio, viewed as a function $\delta(i,k)$ of its left and right chain indices,*

$$|\log \delta(i,k) - \log \delta(i',k')| \leq d_H(\ell_i, \ell_{i'}) + d_H(h_k, h_{k'}).$$

The limits $\delta(i,\infty)$, $\delta(\infty,k)$, and $\delta_\infty = \delta(\infty,\infty)$ exist. In particular,

$$|\log \delta(i,k) - \log \delta_\infty| \leq \Delta(\tau^{i-1} + \tau^{k-1}) \qquad (i,k \geq 1),$$
$$|\log \delta(i,k) - \log \delta(i,\infty)| \leq \Delta \tau^{k-1} \qquad (i \geq 0,\ k \geq 1).$$

The limits in which both chain indices tend to infinity are independent of the end factors. An end-ray degree has a one-sided limit, obtained by sending the opposite chain index to infinity; its logarithmic error satisfies the corresponding one-index bound above, with the end factor fixed. This limit may depend on that end factor. Every individual ray weight, including all these limiting weights, is at most 27.

Proof. The Birkhoff–Hopf theorem for a strictly positive continuous kernel gives projective diameter equal to the logarithm of its maximal cross-ratio and contraction factor $\tanh(\Delta/4)$ [EN95, Theorems 3.5 and 6.3]. The measure ν has full support. For $a > b$, the cross-ratio of $3 + a - a'$ increases with a' and decreases with b'; at $a' = 1, b' = -1$ it is

$$\frac{(2+a)(4+b)}{(4+a)(2+b)} \leq \frac{9}{5}.$$

This increases with a and decreases with b, and equality holds at $a = 1, b = -1$. Exchanging the pairs treats $a < b$. Squaring gives $81/25$ for $\mathcal{K}$, and $\tanh(\Delta/4) = 2/7$. One application puts any two positive initial functions at distance at most Δ; each subsequent application contracts this distance by τ.

Each degree ratio is the ratio of two integrals of the chain functions against positive measures. If $m \leq f/\widetilde{f} \leq M$, replacing f by $\widetilde{f}$ changes such a ratio by a factor between m/M and M/m. Applying this to the two chain functions proves the logarithmic estimate. The degree ratios are therefore Cauchy, which proves the stated limits without a spectral assumption.

For the last assertion, each initial end polynomial has maximum divided by minimum at most 9. For fixed a, the ratio of the largest to the smallest value of $\mathcal{K}(a, a')$ as a' varies is at most 9; the transposed kernel has the same property. All chain functions therefore have oscillation ratio at most 9. A hexagon edge measure has total mass 1 and ν has mass 3, so (80) is at most $9^2/3 = 27$. An interior base weight is an average of $2/(3 + a - a') \leq 2$, and an end weight is an average of $b/(b + 1 \pm a) \leq 1$. The bounds persist under limits. □

B.4. **The finite calculation and its uniform error bound.** With the limiting weights, an interior period has total weight exactly 4. Indeed, Lemma B.2 bounds the difference between the total weight of a path and $(j - 1)$ times its limiting period weight by a constant independent of j: the geometric tails are summable and the end factors have bounded weight. The total weight is $n = 4j - 2 + b_0 + b_j$, so division by j gives the assertion.

Consequently the all-selected period at state G has cost zero, as does the empty period at state u. For the limiting period recurrence, every other simple directed cycle has cost greater than 0.1259. To verify this by a finite rational calculation with a controlled error, substitute the chain indices $(80, 80)$ for (∞, ∞) in (80)–(81), enumerate the 36 transitions of (85), and then the simple cycles of length at most three. Table 4 lists the least transition costs after removing the two zero loops. Each positive entry lies within 10^{-6} of the displayed number; the limiting entries marked 0 are exactly zero. There is no transition from u to g. Thus none of the remaining zero transitions belongs to a zero-cost cycle.

TABLE 4. Least limiting transition costs, with the all-selected $G \to G$ and empty $u \to u$ loops removed.

	G	g	u
G	0.125902	0.874098	0
g	0	0.874098	0
u	0.125902	—	0.291366

For the finite comparison, consider a path of length $j = 2q + m$, where $q = 40$ and $m \geq 0$. A period has index p when it consists of the base factor p and hexagon p. Define comparison weights by using $\delta(i, \infty)$ at the left end and periods $1, \ldots, q-1$, δ_∞ at periods $q, \ldots, q+m-1$, and $\delta(\infty, k)$ at periods $q+m, \ldots, j-1$ and the right end. The initial hexagon belongs to the left end. Let $M(m)$ denote the costs of configurations with these comparison weights.

Each of the two end regions contains at most $9q+4$ rays. At a hexagon p the chain indices are $(p, j-1-p)$; at an interior base factor p they are $(p-1, j-1-p)$. Put

$$\overline{\Delta} = \frac{11757}{10000} > \Delta, \quad x = \overline{\Delta}\tau^{q-2}, \quad y = 2\overline{\Delta}\tau^{79}, \quad s = \frac{2x}{1-\tau}.$$

The strict upper bound on Δ follows, using rational arithmetic, from $\sum_{r=0}^{7} \overline{\Delta}^r/r! > 81/25$. Since $e^t - 1 \le t + t^2$ for $0 \le t \le 1$, Lemma B.2 bounds the sum of all weight changes between the original path and $M(m)$ by

$$E_{\mathrm{path}} = 27\{9s(1+2x) + 2(9q+4)(x+x^2)\}. \tag{86}$$

The first term sums the two geometric tails in the central periods. The second bounds the contributions from both end regions. It uses the weaker index $q-2$, which also applies to the first base factor of the right region when $m = 0$. This bound is uniform in m and in the configuration selected.

For the four comparison paths $M(0), \dots, M(3)$, replacing every infinite chain index by 80 changes the cost of any configuration by at most

$$E_{\mathrm{rat}} = 27\{9(2q+4)+4\}(y+y^2) < 6 \cdot 10^{-39}. \tag{87}$$

The same bound also applies to a cycle of length at most three. After these substitutions, all computations use rational numbers only. Their all-selected costs have absolute value below 10^{-36}. Thus the all-selected cost of $M(m)$ has absolute value below $2 \cdot 10^{-36}$ for every m, because adding a limiting full period changes that cost by zero.

Table 5 gives rational lower bounds from the recurrence, first for every original path $1 \le j \le 84$, and then for each of the four rational comparison paths. The bounds in the table are terminating decimals, not rounded approximations to the minima. Reversing the path exchanges its ends, so the six rows cover all nine ordered end types.

TABLE 5. Lower bounds for the minimum invariant rank deficit of nonempty configurations of proper rank. The comparison column uses chain index 80 in place of infinity.

(b_0, b_j)	every original path $1 \le j \le 84$	each comparison path $M(0), \dots, M(3)$
$(1,1)$	0.029505	0.029505
$(2,1)$	0.029505	0.029505
$(2,2)$	0.058900	0.058900
$(3,1)$	0.003851	0.029505
$(3,2)$	0.035003	0.039341
$(3,3)$	0.039341	0.039341

The calculation is specified by (82), the moment sequences following it, the end polynomials (78), and the recurrence (85), with the two binary indicators described there. The accompanying program reproduces these tables using exact rational arithmetic; Appendix F describes additional independent checks.

Proof of Theorem 9.1. The finite calculation proves positivity for $j \le 84$. For the remaining lengths use $M(m)$ as above. Every configuration of proper nonzero rank in $M(0), \dots, M(3)$ costs more than $0.0295 - E_{\mathrm{rat}}$. The empty configuration costs zero. A full-rank configuration costs at least the all-selected configuration, because all weights are positive, and hence at least $-2 \cdot 10^{-36}$.

For $m \geq 4$, two boundaries of the central periods have the same state. Delete the periods between them. The remaining configuration belongs to $M(m')$ with $m' < m$ and has the same rank increments outside the deleted walk. If the walk consists only of the all-selected zero loop or only of the empty zero loop, deletion preserves both its cost and whether the whole configuration is nonempty of proper rank. In the first case all deleted hexagons and base factors satisfy the full-rank conditions; the remaining state G prevents the new configuration from being empty. In the second case the remaining state u prevents it from having full rank. If the walk is not of either kind, its decomposition into simple cycles contains a positive-cost cycle and its cost exceeds 0.1259.

Induction therefore gives a lower bound $0.0295 - E_{\mathrm{rat}} - 2 \cdot 10^{-36}$ for every nonempty proper-rank configuration of $M(m)$. Subtracting the error between the original and comparison weights, every relevant rank deficit is greater than

$$0.0295 - \left(E_{\mathrm{path}} + 2E_{\mathrm{rat}} + 2 \cdot 10^{-36}\right) > 0.0295 - 6 \cdot 10^{-17} > 0.$$

Proposition 2.4 gives stability. □

Corollary B.3. *For every $n \geq 4$ there is a smooth toric Fano n-fold with anticanonically stable tangent bundle and Picard number $5\lfloor n/4 \rfloor + 1$. This is the largest Picard number of a stable example among the connected Fano hexagon bundles over graphs defined in Section 8.*

Proof. Write $n = 4j + t$, $0 \leq t \leq 3$. Choose the end types $(1,1), (2,1), (2,2), (3,2)$ for $t = 0, 1, 2, 3$, respectively. Equation (60) gives the claimed dimension and Picard number, and Theorem 9.1 gives stability. The upper bound within this class is the graph bound of Section 8. □

Appendix C. Proof for the end attachments

The proof of Theorem 9.2 uses the varieties and lists of twisting vectors of Section 9.

C.1. **Geometry and divisor degrees.** The lattice is the direct sum of the hexagon lattices and the base lattices. In a base factor of dimension d, the lifted rays are $e_1, \ldots, e_d, t - \sum e_i$, where t is its stated twisting vector. Maximal cones contain adjacent ray pairs in each hexagon and omit one ray in each base factor. The simplex lengths in the anticanonical polytope are at least one, so its normal fan is the displayed complete bundle fan. The block triangular matrices of its maximal cones are unimodular. The primitive collections are the hexagon primitive collections and the lifted primitive collections of the base. Each leaf relation has degree $2 - 1 = 1$, and each internal base relation has degree $3 - 2 = 1$. The fibre primitive relations have degrees one and two. Thus the variety is Fano. The incidence graph is connected; within each hexagon the rays form one matroid component, and each lifted base relation connects its base coordinates to all the incident hexagon components. Hence the ray matroid is connected.

The number of base factors is $j + a + b + 1$ and their total dimension is $2j + a + b$. This gives the dimension and Picard number. For stability, we compute the divisor degrees and subspace ranks as follows. The dual anticanonical hexagon is

$$Q = \{(x, z) : |x|, |z|, |x + z| \leq 1\}.$$

Its vertices, cyclically ordered, are $(1,0), (1,-1), (0,-1), (-1,0), (-1,1), (0,1)$. For $A_a = (t_1, \ldots, t_{a+1})$ set

$$P_a(x,z) = \prod_{\ell=1}^{a+1} (2 + \langle (x,z), t_\ell \rangle), \qquad m_a(p) = \int_Q x^p P_a(x,z)\, dx\, dz \quad (0 \leq p \leq 2).$$

The three-entry moment vectors are

a	$m_a(0)$	$m_a(1)$	$m_a(2)$
0	6	$-5/6$	$5/3$
1	$139/12$	$-5/3$	$10/3$
2	$67/3$	$-31/10$	$31/5$
3	43	$-86/15$	$1373/120$

For a hexagon ray h, define $m_{a,h}$ by the same formula integrated over its facet $\langle (x,z),h\rangle = -1$ with lattice measure. For either ray of leaf ℓ, define $m_{a,\ell}$ by omitting the factor $2+\langle (x,z),t_\ell\rangle$ from P_a. These integrals determine every modified moment vector exactly. Reflection changes a right-hand moment vector v_p to $(-1)^p v_p$. The reflected local hexagon ray h is the global ray $-h$.

Write

$$W=\begin{pmatrix} 3 & 0 & 5/6\\ 0 & 5/6 & 0\\ 5/6 & 0 & 7/15\end{pmatrix},\quad \kappa=\begin{pmatrix} 9/2 & -3 & 1/2\\ 3 & -1 & 0\\ 1/2 & 0 & 0\end{pmatrix},\quad \kappa_1=\begin{pmatrix} 3 & -1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}.$$

These are respectively the moments of $(2-|x|)dx$, the coefficients of $(3+x-x')^2/2$, and the coefficients of $3+x-x'$. For $j\ge 2$, let $L_0=m_a$ be a row vector and let R_0 be the reflected m_b as a column vector. Set

$$L_{i+1}=L_i\kappa W,\qquad R_{i+1}=W\kappa R_i.$$

The anticanonical polytope volume is $Z=L_0\kappa R_{j-2}$, and $(-K_X)^n=n!Z$. Normalized weights are facet volumes divided by Z. For the left end replace L_0 by its modified vector; for the right replace R_0 by its reflected modified vector. For the internal base factor c, $1\le c\le j-1$, each of its three rays has weight

$$\beta_c=\frac{L_{c-1}\kappa_1 R_{j-1-c}}{Z}.$$

For an internal hexagon i, $1\le i\le j-2$, replace W in $L_{i-1}\kappa W\kappa R_{j-2-i}$ by its facet moment matrix W_h. Its denominator is Z. These formulas follow by integrating the simplex slices and applying Fubini. For $j=1$, the volume is the integral of $P_a(x,z)P_b(-x,-z)$ over Q.

C.2. **End subspaces and the rank recurrence.** An end of type a has lattice $E_a=\mathbb{Z}^2\oplus\mathbb{Z}^{a+1}$ and rays

$$(h,0)\ (h\in\{\pm r,\pm q,\pm(r+q)\}),\quad (0,e_\ell),\ (t_\ell,-e_\ell)\quad (1\le\ell\le a+1).$$

Enumerate the ray-spanned subspaces by starting with zero, adjoining each ray in turn, and closing under inclusion of all rays in the span. This enumeration uses exact linear algebra and gives respectively $13,34,92,241$ subspaces for $a=0,1,2,3$, including zero and the whole space. The corresponding ray subsets and ranks are included in the ancillary data. For such an end subspace S, record its rank, selected ray weight, whether $r\in S$, and whether $S+\mathbb{C}r=E_{a,\mathbb{C}}$. This enumeration requires no symmetry of either end.

For the symmetry reduction, use only the independent reflections of the internal hexagons fixing r_i, and the transpositions of the two untwisted rays of each internal $\mathbb{P}^2$. These preserve the entire fan, including the end pieces. Every invariant ray-spanned subspace is represented by an end choice on each side, choices $H_i\in\{0,1,2\}$ of empty set, distinguished line, or whole plane in each internal hexagon, and binary choices C_c,T_c for the two untwisted base rays and the twisted ray. Passing to closure can only increase weight at fixed rank, so minimizing over all these configurations suffices.

The outgoing boundary state is G if its distinguished ray was selected, g if that line is spanned through the preceding relations, and u if it is not spanned. Initialize at the left end with G when $r \in S$ and u otherwise, with rank $\dim S$. Adjoining an internal base and the following hexagon adds the rank

$$d = 2C + T(1 - C) + \begin{cases} H + \mathbf{1}_{CT=1,\, s=u}, & H \geq 1, \\ CT, & H = 0, \end{cases}$$

and gives state G when $H \geq 1$, state g when $H = 0$, $CT = 1$, and $s \neq u$, and state u otherwise. Its selected weight is $(2C + T)\beta + \omega(H)$, where $\omega(H)$ is the empty, line, or plane weight. These formulas follow from the path relation $-g_c + r_{c-1} - r_c$ modulo the previously selected subspaces.

For the selected subspace V in the processed ambient space E, define binary indicators $\eta = \mathbf{1}_{V \neq 0}$ and $F = \mathbf{1}_{E/(V + \mathbb{C}r) = 0}$, where r is the outgoing distinguished ray. Initially $F = \mathbf{1}_{S + \mathbb{C}r = E_{a,\mathbb{C}}}$. At an internal step, $\eta' = \mathbf{1}_{\eta = 1 \text{ or } d > 0}$ and

$$F' = F \text{ and } \big(\mathbf{1}_{s=u} + 4 - d = \mathbf{1}_{s'=u}\big).$$

Indeed, if F holds, the old rank deficiency is exactly $\mathbf{1}_{s=u}$. Otherwise, a missing direction independent of the boundary can never be supplied by the remaining chain. This also proves that $F = 0$ implies $F' = 0$.

At the right end, choose a subspace S_R, put $e = \dim S_R$, and let g_R mean $r \in S_R$. The final base factor and right end add rank

$$d_R = 2C + T(1 - C) + e + \mathbf{1}_{CT=1}\mathbf{1}_{s=u \text{ or } \neg g_R}.$$

The whole configuration has full rank precisely when

$$F \text{ and } \big(\mathbf{1}_{s=u} + 2 + (b + 3) - d_R = 0\big).$$

The formula for d_R follows because the remaining relation $r_{j-2} - r_{j-1}$ is dependent exactly when both boundary lines are already spanned. Minimize rank minus selected weight, retaining only the least cost for each state and pair of binary indicators. Discard the empty and full-rank final configurations. The rank argument above proves this recurrence directly; its justification does not depend on comparisons with numerically computed ranks.

C.3. **Uniform comparison and stability.** The internal kernel is positive and has Hilbert projective contraction factor $\tau = 2/7$ and diameter $\Delta = \log(81/25)$, as in the path theorem. The end polynomials are positive on Q. Equivalently, their moment vectors are those of the positive density

$$g_a(x) = \frac{1}{2 - |x|} \int_{\{z:(x,z) \in Q\}} P_a(x, z)\, dz$$

against $(2 - |x|)dx$. After one application of the kernel the outgoing functions have oscillation ratio at most nine. Thus they converge to the same limiting functions as the original path ends. Every internal hexagon weight is at most 27, and every base weight is at most two. At an end, $1 \leq P_a \leq 3^4 = 81$ and the incoming function has oscillation ratio at most nine. Since a hexagon facet has lattice length one and $\mathrm{vol}(Q) = 3$, every end hexagon weight is at most $81 \cdot 9/3 = 243$. End leaf weights are at most one. The same bounds hold for the limiting weights and the finite comparison weights.

Take $q_0 = 20$, comparison index $K = 40$, and $j = 2q_0 + m$. The left end and periods $1, \ldots, q_0 - 1$ retain their left chain indices and use the limiting right function. Periods $q_0, \ldots, q_0 + m - 1$ use both limiting functions. The remaining periods and the right end

retain their right indices and use the limiting left function. Each end region contains at most $9q_0 + 10$ rays. Write

$$\bar{\Delta} = 11757/10000, \quad x = \bar{\Delta}\tau^{q_0-3}, \quad y = 2\bar{\Delta}\tau^{K-2}, \quad B = 243.$$

The weaker exponents allow for the first application of the kernel from an end piece, including the adjacent base factor. The contraction bound and $e^t - 1 \le t + t^2$ give the following uniform error between the original and comparison weights:

$$E_{\mathrm{path}} = B\left\{9\frac{2x}{1-\tau}(1+2x) + 2(9q_0+10)(x+x^2)\right\}.$$

For each of the four comparison paths $m = 0, 1, 2, 3$, replacing infinite indices by K changes every configuration cost by at most

$$E_{\mathrm{rat}} = B\{9(2q_0+4)+10\}(y+y^2).$$

This bound also applies to every simple internal transition cycle.

The limiting all-selected period has weight four. Indeed, the convergence errors are summable, so the difference between the total weight and the number of periods times the limiting weight is bounded independently of j. The total weight is $4j + a + b$. Hence the all-selected G loop and empty u loop have exactly zero cost. All other simple cycles are the same as in the original path theorem and have cost exceeding 0.1259. The rational comparison calculation gives the following strict lower bounds, written as terminating decimals rather than rounded minima: Their all-selected comparison costs have absolute value less than $6 \cdot 10^{-25}$. The

TABLE 6. Strict lower bounds for nonempty proper-rank configurations.

(a, b)	every original $2 \le j \le 44$	each comparison $m = 0, 1, 2, 3$
$(1, 0)$	0.029505	0.029505
$(2, 0)$	0.029486	0.029486
$(1, 1)$	0.030639	0.030639
$(3, 0)$	0.028404	0.028404
$(1, 2)$	0.029486	0.029486

exact rational bounds give

$$E_{\mathrm{path}} + 2E_{\mathrm{rat}} + 6 \cdot 10^{-25} < 0.00006523.$$

The finite inputs are specified by the displayed rays, moment integrals, matrices, recurrence, and constants. The accompanying programs evaluate them using rational arithmetic. The $j = 1$ cases are checked by positive rational decompositions into ray bases with strongly connected exchange graphs, as in Lemma F.4. Their normalized degrees also agree with an independent calculation by direct integration of the product of the stated linear factors over Q and its facets. The decompositions are supplied in the ancillary data.

Completion of the proof of Theorem 9.2. Apply the cycle-deletion argument of Appendix B to the central periods. A walk between repeated states either contains a positive-cost simple cycle, or consists solely of one of the two zero loops. Deleting all-selected G loops preserves nonemptiness and proper rank. Deleting empty u loops does likewise: the boundary at which deletion occurs is an internal hexagon, so state u implies that this hexagon's independent nondistinguished direction is missing; the shortened configuration cannot become full rank. Thus the same zero-loop argument applies in the presence of the end pieces. A full-rank

reduced configuration has cost at least that of the all-selected configuration, by positivity of weights. Deleting cycles therefore reduces every comparison configuration to $m \leq 3$, or supplies the positive cycle bound. Subtracting all errors, every nonempty proper-rank configuration at arbitrary length has deficit

$$> 0.028404 - 0.00006523 = 0.02833877 > 0.$$

Together with the finite cases and connectedness of the ray matroid, the symmetry criterion proves the theorem. □

Appendix D. Proof for three independent arms

This appendix proves Theorem 9.4 for the varieties $Y_{a,b,c}$ constructed in Section 9.

D.1. **Independent arm subspaces and the central relation.** Consider the independent local reflections $r \mapsto r$, $q \mapsto -r - q$, the untwisted-ray transpositions at internal $\mathbb{P}^2$ factors, and the permutations of the three central untwisted rays. They preserve the distinguished vectors and the sums of untwisted base coordinates, hence every twist. They also preserve adjacent hexagon pairs and omitted base-ray sets. They therefore generate a group A of fan automorphisms for arbitrary arm lengths.

Every A-invariant closed ray set selects each hexagon as empty, its distinguished line, or its whole plane, denoted $H = 0, 1, 2$. At an internal base the two untwisted rays are selected together, and the twisted ray independently, with indicators C, T. At a leaf its two rays are chosen independently. At the center C_*, T_* select all three untwisted rays and the twisted ray. These configurations include every invariant flat, with independent choices on the three arms.

Apply the path recurrence to each arm from the leaf to the center, using (85). Its state $s = G, g, u$ records whether the outgoing distinguished line is selected, is spanned through earlier relations, or is unspanned. Let e indicate a nonempty arm selection. Let $f = 1$ when every hexagon is a plane, every internal base has $C = 1$, and the leaf has at least one ray selected. The reachable terminal triples are

$$(G, 1, 0), \quad (G, 1, 1), \quad (g, 1, 0), \quad (u, 0, 0), \quad (u, 1, 0).$$

Lemma D.1. *Suppose the three arm selections have ranks r_i, deficits v_i, and terminal triples (s_i, e_i, f_i). Adding the central choice gives rank*

$$\sum_{i=1}^{3} r_i + 3C_* + T_*(1 - C_*) + C_*T_*\mathbf{1}_{\{\exists i: s_i = u\}}$$

and deficit

$$\sum_{i=1}^{3} v_i + 3C_* + T_*(1 - C_*) + C_*T_*\mathbf{1}_{\{\exists i: s_i = u\}} - (3C_* + T_*)\delta_*. \tag{88}$$

Here δ_* is the normalized degree of one central ray. The selection is empty exactly when all e_i, C_*, T_* vanish, and has full rank exactly when $C_* = f_1 = f_2 = f_3 = 1$.

Proof. The arm spans lie in disjoint coordinate summands. The central untwisted rays add $3C_*$ dimensions. If $C_* = 0$ and $T_* = 1$, the twisted ray has a new central coordinate and adds one dimension. If $C_* = T_* = 1$, its remaining image is minus the sum of the terminal distinguished vectors in the direct sum of the three arm quotients. This sum is nonzero

precisely when at least one summand is nonzero, giving the displayed rank. Subtracting selected weights proves (88).

Full rank requires each hexagon's nontrivial reflection representation, each internal base's nontrivial transposition representation, and the central base's two-dimensional standard representation. Thus every hexagon must be a plane, every internal untwisted pair must be selected, and $C_* = 1$. The remaining leaf coordinate requires at least one leaf ray. These conditions are also sufficient. The condition for emptiness follows from the definition of the indicators. □

Minimize each arm deficit for its five terminal triples, and combine the results for the four central choices. This gives $5^3 \cdot 4 = 500$ cases before excluding the empty and full-rank selections. Positivity of the remaining deficits implies stability by Proposition 2.4. All deficit bounds below refer to these A-invariant configurations.

D.2. **The mixed cubic integral.** Write $L_1 = a, L_2 = b, L_3 = c$. On each arm number the hexagons from the center by $p = 0, \dots, L_i - 1$. With $d\nu(t) = (2 - |t|)\, dt$ on $[-1, 1]$ and the kernel $\mathcal{K}$ of (78), put

$$h_0(t) = 2 + t, \qquad h_{k+1}(s) = \int \mathcal{K}(s, t) h_k(t)\, d\nu(t).$$

For positive functions u, v define

$$J_{u,v}(s) = \iint \frac{(4 - s - t - z)^3}{6} u(t)v(z)\, d\nu(t)d\nu(z), \tag{89}$$

$$J^-_{u,v}(s) = \iint \frac{(4 - s - t - z)^2}{2} u(t)v(z)\, d\nu(t)d\nu(z). \tag{90}$$

Set $h^{(i)} = h_{L_i - 1}$. The left initial function on arm i is $\ell_{0,i} = J_{h^{(j)}, h^{(k)}}$, where $\{i, j, k\} = \{1, 2, 3\}$. Propagate it by the left operator in (79). The central ray degree is

$$\delta_* = \frac{\int J^-_{h^{(j)}, h^{(k)}}(s) h^{(i)}(s)\, d\nu(s)}{\int J_{h^{(j)}, h^{(k)}}(s) h^{(i)}(s)\, d\nu(s)}. \tag{91}$$

The value is independent of i. The other degrees are the path ratios (80)–(81), using left index p and right index $L_i - 1 - p$ for hexagon p, and left index $c - 1$ and right index $L_i - 1 - c$ for internal base c. The leaf numerator replaces $2 + t$ by 1. These formulas follow by integrating the other two arms in Proposition 8.7.

To compute the coefficients of these polynomials, put $M_r(u) = \int t^r u(t)\, d\nu(t)$. The coefficient of s^d in $J_{u,v}$ is

$$\sum_{\substack{r,t \ge 0 \\ d + r + t \le 3}} \frac{(-1)^{d+r+t} 4^{3-d-r-t} M_r(u) M_t(v)}{d! r! t! (3 - d - r - t)!}. \tag{92}$$

Replacing 3 by 2 gives $J^-_{u,v}$. Thus the four-by-four moment matrices of Appendix B determine every finite degree exactly.

D.3. **Uniform estimates with different arm lengths.** Retain $\Delta = \log(81/25)$ and $\tau = 2/7$. Lemma B.2 applies to any positive initial function. The left and right chains have positive projective limits ℓ_∞, h_∞, independent of their initial functions, with index-k error at most $\Delta \tau^{k-1}$ for $k \ge 1$. Each degree ratio has logarithmic error at most the sum of the projective errors of its factors.

Every individual ray weight in the original configurations and the comparison configurations below is at most $B = 81$. Indeed, for fixed t, z the oscillation ratio of $(4 - s - t - z)^3$ is at most 27, and this persists under positive integration. Right functions have oscillation ratio at most 9. Hexagon facet measures have mass one and ν has mass three, giving $27 \cdot 9/3 = 81$. Internal, leaf and central base weights are positive averages bounded by 2, 1 and 3 respectively.

Fix $q = 16$. Call an arm short if $2 \le L_i < 2q$, and otherwise write $L_i = 2q + m_i$, $m_i \ge 0$. First replace the incoming function $h^{(i)}$ of every long arm by h_∞, keeping short incoming functions unchanged. Denote these functions by $\widehat{h}^{(i)}$. Use them in the central degree and in each initial left function; for the moment retain every original right chain on the arms.

Put $\bar{\Delta} = 11757/10000 > \Delta$, using the rational exponential estimate of Appendix B. For a long arm set $\epsilon_i = \bar{\Delta}\tau^{L_i-2}$, and for a short arm set $\epsilon_i = 0$. Positivity of the mixed integral gives

$$d_H\big(J_{h^{(j)},h^{(k)}}, J_{\widehat{h}^{(j)},\widehat{h}^{(k)}}\big) \le \epsilon_j + \epsilon_k.$$

Let $z = 3\bar{\Delta}\tau^{2q-2}$. Each sum on the right is at most z. After propagation along arm i, the logarithmic weight change is at most $z\tau^p$ for hexagon p, $z\tau^{c-1}$ for internal base c, and $z\tau^{L_i-1}$ for the leaf. The central logarithmic change is at most $\sum_i \epsilon_i \le z$.

Using $e^t - 1 \le t + t^2$ for $0 \le t \le 1$, sum the six hexagon rays, three internal-base rays and two leaf rays on each arm. Their geometric sum is bounded by $9/(1-\tau) + 2$. Including the four central rays gives the uniform error bound

$$E_J = B\left(10 + \frac{27}{1-\tau}\right)(z + z^2). \tag{93}$$

The bound is therefore independent of the number of rays on the other arms.

For the next comparison, divide each long arm into three regions. Take an internal period to be base c followed by hexagon $c-1$ and give both the index $p = c - 1$; hexagon p has the same region index. For $p < q$, keep the finite left function and use h_∞ on the right. For $q \le p < q + m_i$, use both limiting functions. For $p \ge q + m_i$, keep the right function and use ℓ_∞ on the left. The leaf uses the limiting left function. Short arms retain the weights obtained in the preceding comparison at the central vertex. Write $M(m_1, \dots, m_k)$ for this comparison, where k is the number of long arms. Its initial functions at the central vertex depend on the short lengths and the set of long arms, but not on any m_i.

Put $x = \bar{\Delta}\tau^{q-2}$ and $S = 2x/(1-\tau)$. Each boundary region has at most $9q + 4$ rays. The omitted chain contributes a logarithmic error of at most x. In a middle period p the error is at most

$$t_p = \bar{\Delta}(\tau^{p-1} + \tau^{L_i-3-p}), \qquad t_p \le 2x, \qquad \sum_p t_p \le S.$$

There are at most nine rays per period. Summing over at most three long arms gives

$$E_Z = 3B\{9S(1 + 2x) + 2(9q + 4)(x + x^2)\}. \tag{94}$$

Hence every selected-ray deficit changes by at most $E_J + E_Z$ from the original degrees to the comparison weights, uniformly in all arm lengths.

D.4. **Finite comparisons and independent arm shortening.** Consider the thirty short lengths $2, \dots, 31$ and the four long-arm middle lengths $m = 0, 1, 2, 3$. Their unordered triples give 7140 finite comparisons. This ordering identifies repeated geometries; the selections on the three arms remain independent in every case.

For rational evaluation put $P = 40$. Replace h_∞ by h_P and ℓ_∞ by the index-P left chain starting at $2 - s$. The common limit is independent of this initial choice. With $y = 3\bar{\Delta}\tau^{P-1}$, each ray degree has logarithmic error at most y: a finite mixed left function changes by at most $2\bar{\Delta}\tau^{P-1}$, and an omitted right or left limit by at most $\bar{\Delta}\tau^{P-1}$. A finite comparison has at most $27(2q+3)+1$ rays, so its deficit error is bounded by

$$E_R = B\{27(2q+3)+1\}(y+y^2). \tag{95}$$

Let μ be the smallest proper nonempty rational deficit in these comparisons, and let F_0 be the largest absolute all-selected cost. Table 7 records the finite minima by number of long arms; each displayed lower bound is rounded down. The arm recurrence, the five terminal states, (88), and the mixed moment formula specify the finite calculation.

TABLE 7. Strict rational lower bounds for proper nonempty configurations in the finite mixed-arm comparisons. All short lengths are $2, \dots, 31$; each long arm has middle length $0, 1, 2, 3$.

Long arms	Comparisons	Lower bound
0	4960	0.029198008
1	1860	0.029198008
2	300	0.029198008
3	20	0.029505518

The exact rational calculations give

$$\mu > 0.029198, \quad \gamma > 0.1259, \quad F_0 < 10^{-15},$$

and

$$E_J < 6.506 \cdot 10^{-13}, \qquad E_Z < 0.002216796, \qquad E_R < 1.634 \cdot 10^{-16}.$$

With the limiting weights, the empty u-loop and all-selected G-loop have cost zero. The latter assertion follows because a limiting period has total weight four, as proved in Appendix B. The same period calculation gives a positive rational simple-cycle gap γ, after the two zero loops are removed. A simple cycle uses at most three nine-ray periods, so its true gap is at least $\gamma - E_R$.

Set $F = F_0 + E_R$. Inserting a full middle period changes neither the all-selected cost nor the data at the central vertex of a comparison with limiting weights. Therefore the all-selected cost is at least $-F$ for every choice of middle lengths. Every full-rank selection has cost at least this value: adding its omitted positive-weight rays changes no rank. Empty selections have cost zero.

Proof of Theorem 9.4. We first show that every proper nonempty comparison with limiting weights has deficit at least

$$\min\{\mu - E_R,\ \gamma - E_R - F\}.$$

Induct on the sum of its middle lengths. If each is at most three, the finite comparison applies. Otherwise choose an arm with at least four middle periods. Two of its boundary states agree among G, g, u. Delete the closed walk between them on this arm. The retained indices from the center and leaf are unchanged, as are the functions at the central vertex, so the resulting weights are those of the shorter comparison.

If the deleted walk contains only the empty u-loop, it removes no selected ray, and the retained u boundary prevents full rank. If it contains only the full G-loop, the retained G

boundary prevents emptiness, and deleting selected planes and untwisted pairs removes no failure of a full-rank condition. Proper nonemptiness is therefore preserved in both zero-cost cases, and induction applies.

Every other closed walk contains a positive simple cycle, with cost at least $\gamma - E_R$. The remaining configuration is proper nonempty, empty, or full rank. Apply induction, the zero empty cost, or the bound $-F$, respectively. This proves the comparison inequality. The argument uses the gap for one arm at a time and requires neither equal lengths nor simultaneous deletion on different arms.

Subtracting the error between the original and comparison weights gives every proper nonempty A-invariant configuration deficit at least

$$\min\{\mu - E_R,\ \gamma - E_R - F\} - E_J - E_Z > 0.02698.$$

The symmetry criterion proves anticanonical stability. The dimension and Picard number follow from the graph construction. □

Appendix E. Stability for cycles

For cycles, symmetry reduces the number of subspaces needed to test stability. Four inequalities for the divisor degrees suffice; a recurrence proves them by induction.

Theorem E.1. *Let C_j be a cycle with $j \geq 3$ vertices and edges, or the graph with two vertices and two parallel edges when $j = 2$. Put $b_c = 2$ at every vertex. Then $X(C_j)$ is a smooth toric Fano variety with stable tangent bundle, dimension $4j$, and Picard number $5j$.*

Proof. The geometric assertions and connectedness of the ray matroid follow from Proposition 8.3. Number vertices and edges modulo j, orienting edge i from i to $i+1$, and write $g_c = f_1^c + f_2^c$. The twisted base ray at c is

$$f_0^c = -g_c + r_{c-1} - r_c.$$

Rotation sends vertex c to $c+1$ and hexagon i to $i+1$, preserving the coordinates in each transported summand. Reflection sends c to $-c$ and i to $-i-1$, and negates both coordinates in each transported hexagon. Both maps send the displayed twisted ray to the twisted ray at the image vertex. They preserve adjacent pairs of hexagon rays and the choices of an omitted base ray, hence preserve the fan. The same reflection works for $j = 2$, where it exchanges the parallel edges and fixes the vertices.

Together with the independent hexagon reflections $r_i \mapsto r_i$, $q_i \mapsto -r_i - q_i$ and the transpositions of f_1^c, f_2^c, these maps have four ray orbits: the $2j$ distinguished hexagon rays $\pm r_i$, the other $4j$ hexagon rays, the $2j$ untwisted base rays, and the j twisted base rays. A ray-spanned invariant subspace containing any nondistinguished hexagon ray must contain the whole plane of every hexagon. Thus it suffices to consider a common hexagon choice $H \in \{0, 1, 2\}$, for empty, line, or plane, and common binary choices C, T for untwisted and twisted base rays.

Let a, b, c be the normalized degrees of a distinguished hexagon ray, a nondistinguished hexagon ray, and a base ray, respectively. Symmetry makes the first two degrees independent of the hexagon factor. Principal divisor relations give equal degrees for all three base rays at a vertex, so there is only one base value c. Their sum satisfies

$$2a + 4b + 3c = 4. \tag{96}$$

The nine proper nonempty configurations are listed in Table 8. For their ranks, selected untwisted base rays contribute $2jC$, and hexagon choices contribute jH. If $C = 0$, each

selected twisted ray has an independent base coordinate. If $C = T = 1$, its residual vector is $r_{c-1} - r_c$: these vectors span a space of dimension $j - 1$ when $H = 0$, and vanish modulo the selected hexagon lines when $H > 0$. These contributions give the ranks in the table. The other three configurations are empty or have full rank.

TABLE 8. Proper cycle configurations and their weights.

H	C	T	Rank	Weight divided by j
0	0	1	j	c
0	1	0	$2j$	$2c$
0	1	1	$3j-1$	$3c$
1	0	0	j	$2a$
1	0	1	$2j$	$2a+c$
1	1	0	$3j$	$2a+2c$
1	1	1	$3j$	$2a+3c$
2	0	0	$2j$	$2a+4b$
2	0	1	$3j$	$2a+4b+c$

It remains to prove the bounds

$$a < \frac{1}{2}, \qquad b > \frac{1}{4}, \qquad \frac{2}{3} < c < \frac{5}{6}. \tag{97}$$

They imply every strict weight inequality in the table. For example, $3jc < (5/2)j \le 3j - 1$ for $j \ge 2$; the row $(1,1,1)$ follows from $2a + 3c = 4 - 4b < 3$; and the rows with $H = 2$ follow from $2a + 4b = 4 - 3c$. The remaining inequalities follow directly from (97). Stability then follows from Proposition 2.4.

Let W, κ be the three-dimensional matrices in (82), and put $M = W\kappa$. For $0 \le p, q \le 2$, set

$$(W_r)_{pq} = (-1)^{p+q}, \qquad (W_-)_{pq} = \frac{(-1)^{p+q}}{p+q+1}, \qquad \kappa_1 = \begin{pmatrix} 3 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Expanding the cyclic product of kernels and integrating each variable gives a trace of the corresponding matrix product. Thus the volume and facet integrals of Section 8 are

$$Z_j = \operatorname{tr}(M^j), \qquad A_j = \operatorname{tr}(W_r \kappa M^{j-1}),$$
$$B_j = \operatorname{tr}(W_- \kappa M^{j-1}), \qquad C'_j = \operatorname{tr}(W \kappa_1 M^{j-1}).$$

Here $a = A_j/Z_j$, $b = B_j/Z_j$, $c = C'_j/Z_j$. The denominator is the integral of the strictly positive cyclic kernel, so $Z_j > 0$. Define four rational sequences by

$$s_j^{(1)} = \tfrac{1}{2} Z_j - A_j, \quad s_j^{(2)} = B_j - \tfrac{1}{4} Z_j, \quad s_j^{(3)} = C'_j - \tfrac{2}{3} Z_j, \quad s_j^{(4)} = \tfrac{5}{6} Z_j - C'_j.$$

The characteristic polynomial of M is

$$x^3 - \frac{27}{2} x^2 + \frac{2491}{240} x - \frac{127}{864}.$$

By Cayley–Hamilton, each sequence satisfies

$$s_{j+3} = \frac{27}{2} s_{j+2} - \frac{2491}{240} s_{j+1} + \frac{127}{864} s_j. \tag{98}$$

TABLE 9. Exact starting values for the four cycle inequalities.

j	$s_j^{(1)}$	$s_j^{(2)}$	$s_j^{(3)}$	$s_j^{(4)}$
1	$\frac{9}{4}$	$\frac{9}{8}$	0	$\frac{9}{4}$
2	$\frac{8431}{240}$	$\frac{6149}{480}$	$\frac{1141}{180}$	$\frac{2963}{144}$
3	$\frac{324403}{720}$	$\frac{464599}{2880}$	$\frac{184207}{2160}$	$\frac{1100717}{4320}$

The starting values obtained from these matrices are listed in Table 9. Index one denotes a formal trace value, not a looped graph construction.

All first terms are nonnegative, all second and third terms are positive, and componentwise

$$s_3 - 12s_2 = \left(\frac{20887}{720}, \frac{21871}{2880}, \frac{19903}{2160}, \frac{34037}{4320}\right) > 0.$$

Suppose $s_k > 12s_{k-1} > 0$ and $s_{k-2} \geq 0$. Recurrence (98) gives

$$s_{k+1} > \left(\frac{27}{2} - \frac{2491}{2880}\right) s_k > 12s_k > 0,$$

since $27/2 - 2491/2880 - 12 = 1829/2880 > 0$. Induction proves positivity for each sequence at every $j \geq 2$. Their positivity is equivalent to (97), which proves stability. □

The matrices and initial values above specify the induction. Exact computations reproduce the initial values and compare the trace degrees with the independently integrated graph formulas for $j = 2, 3$.[1] These computations verify the finite matrix calculations. The conclusion for every length follows from the induction.

APPENDIX F. THE FINITE CLASSIFICATIONS

The path family shows that $\rho \leq n+2$ can fail for varieties with stable tangent bundle. The finite classifications of smooth toric Fano varieties determine the first dimension in which this occurs. Throughout this section stability refers to the polarization $-K_X$.

Theorem F.1. *Let X be a smooth toric Fano variety of dimension $n \leq 7$. If T_X is stable, then $\rho(X) \leq n+2$. For each $n = 4, 5, 6, 7$ this bound is attained, by respectively $3, 1, 10, 15$ isomorphism classes.*

Theorem F.2. *The largest Picard number of a smooth toric Fano eightfold with stable tangent bundle is 11. Exactly one isomorphism class attains it: the path variety X_2 of Example 8.6.*

Proposition F.3. *Among the 6281 smooth toric Fano eightfolds of Picard number ten, exactly 64 have stable tangent bundle, 692 have strictly semistable tangent bundle, and 5525 have unstable tangent bundle, with respect to $-K_X$. Every strictly semistable member is a nontrivial toric product. Consequently none of these 692 members has a stable tangent bundle for any ample polarization.*

[1] The checks are implemented in `code/compute06_cycle_tables.py` in the ancillary package.

Proof. The complete eighteen-ray stratum of the input classification is checked by the finite procedure below. Each verdict is supplied either by a strict rational slope witness or by a positive rational expression in matroid bases and its exact exchange graph. All fans and divisor degrees are recomputed independently. The resulting counts are

	stable	strictly semistable	unstable
$\rho = 10$	64	692	5525.

For every strictly semistable record, the ray configuration splits as a direct sum and its complete fan is checked to be the Cartesian product of the factor fans. A maximal-cone lattice basis makes this an integral lattice splitting. The numbers of records with respectively $2, 3, 4, 5, 6, 7$ finest factors are $464, 170, 44, 10, 3, 1$. The accompanying data give all factors and cones. Finally, the tangent bundle of a nontrivial product is a direct sum of nonzero subbundles; at least one summand has slope at least the rank-weighted average at every ample class, precluding stability. □

F.1. **A stability criterion using matroid bases.** For $S \subseteq \mathcal{R}$, put $\delta(S) = \sum_{\rho\in S} \delta_\rho$. Lemma 2.1 is equivalent to

$$\delta(S) < r(S) \quad \text{for every nonempty proper } S \subset \mathcal{R}, \tag{99}$$

with weak inequalities for semistability. Indeed, if $r(S) < n$, apply the subspace criterion to the span of S. If $r(S) = n$, positivity of the weights gives $\delta(S) < \delta(\mathcal{R}) = n$. Conversely, for a proper subspace V containing rays, use the set of all rays in V; its rank is at most $\dim V$.

A basis of the ray matroid is a subset $B \subset \mathcal{R}$ for which $\{u_\rho : \rho \in B\}$ is a basis of $N_{\mathbb{C}}$. Write $\mathbf{1}_B$ for its incidence vector. The convex hull of these vectors is the *base polytope* of the ray matroid.

Lemma F.4. *Suppose the normalized degree vector admits a decomposition*

$$\delta = \sum_{k=1}^{\ell} \lambda_k \mathbf{1}_{B_k}, \qquad \lambda_k > 0, \qquad \sum_{k=1}^{\ell} \lambda_k = 1,$$

where each B_k is a basis of the ray matroid. Then T_X is semistable. Form a directed graph on $\mathcal{R}$ with an edge $e \longrightarrow e'$ whenever, for some k, one has $e \notin B_k$, $e' \in B_k$, and $(B_k \setminus \{e'\}) \cup \{e\}$ is also a basis. A subset S is tight, meaning $\delta(S) = r(S)$, if and only if no directed edge leaves S. Thus T_X is stable if and only if this graph is strongly connected. Otherwise every proper sink strong component is a tight flat and proves strict semistability.

Proof. For every $S \subseteq \mathcal{R}$,

$$\delta(S) = \sum_k \lambda_k |B_k \cap S| \le r(S),$$

because $B_k \cap S$ is independent. Equality holds precisely when $|B_k \cap S| = r(S)$ for every k. If S is tight, then $B_k \cap S$ is a basis of $\operatorname{span} S$ for every k. The expansion in B_k of a ray in S therefore uses only members of S, so no exchange edge leaves S. Conversely, suppose no edge leaves S. If $B_k \cap S$ failed to span S, some $e \in S \setminus B_k$ would have a nonzero coefficient at an $e' \in B_k \setminus S$ in its basis expansion. Replacing e' by e would give an outgoing edge, a contradiction. Thus every $B_k \cap S$ has rank $r(S)$ and S is tight.

If the graph is strongly connected, there is no proper nonempty tight set, proving stability. Otherwise a sink strong component is such a set. It is a flat: adding any ray in its span would increase its positive weight without changing rank, contrary to the already proved semistability. Positivity also makes its rank less than n. The argument is independent of Proposition 2.4. □

In particular, all tight sets are precisely the outgoing-closed unions of strong components. Ordering the components by reversed reachability identifies this lattice with an order-ideal lattice. The block ranks are quotient dimensions obtained from nested tight sets; they need not equal the dimensions spanned by individual blocks of rays.

F.2. **Coverage and exact slope comparisons.** We use the complete classification of smooth Fano polytopes obtained by Øbro's algorithm [Øbr07], with the low-dimensional data from polyDB and the seven- and eight-dimensional lists distributed by Paffenholz [Paf]. A smooth Fano polytope is the convex hull of the primitive ray generators, and its facets determine the maximal cones. The facet normals of the dual moment polytope thus give the same ray configuration. We use this correspondence to compare the two data distributions, which adopt different polytope conventions.

The computations through dimension six are those of [Wue26]. In dimension four, the classification by tangent bundle stability was already obtained by Reynolds [Rey23, Appendix A]; our four-dimensional calculation independently verifies that classification. The additional varieties examined here are

n	Picard numbers examined	number of varieties
7	$\rho \geq 9$	1277
8	$\rho \geq 11$	1582
8	$\rho = 10$	6281.

Thus the calculation includes all sevenfolds at or above $\rho = n + 2$. The original eight-dimensional calculation covers $\rho \geq n + 3$; Proposition F.3 adds a complete stability classification at $\rho = n + 2$. The higher-rank screening records described below retain their stated distinction between strict and equality witnesses. No stability classification in dimension nine is used.

For the new Picard-number-ten stratum, the fan and degree calculation also has a complete check independent of facet triangulations. Every proposed facet consists of eight rays with an integral inverse matrix. Its supporting functional is exactly one on these rays and strictly less than one on all others. Every seven-ray ridge occurs twice. Since the adjacency graph of facets of a polytope is connected, this nonempty collection, closed across every ridge, is the full boundary. The unimodular ray bases therefore give the complete smooth Fano fan.

If $u_{\sigma,i}$ is the basis dual to the rays of a maximal cone, choose an integral ζ for which every $w_{\sigma,i} = \langle u_{\sigma,i}, \zeta \rangle$ is nonzero. Equivariant localization gives

$$\begin{aligned} (-K_X)^8 &= \sum_{\sigma} \frac{(\sum_i w_{\sigma,i})^8}{\prod_i w_{\sigma,i}}, \\ D_\rho(-K_X)^7 &= \sum_{\sigma \ni \rho} \frac{w_{\sigma,\rho}(\sum_i w_{\sigma,i})^7}{\prod_i w_{\sigma,i}}. \end{aligned} \tag{100}$$

The second computation uses another generic integral vector and its own exact inverses. It reproduces all 113058 divisor degrees and checks all 2443089 maximal cones of the stratum. The slope and product-fan checks then prove Proposition F.3.

For the original higher-Picard screening, the intersection numbers d_v are computed as normalized facet volumes of the moment polytope. Candidate facets and vertices obtained numerically are checked by rational linear algebra before evaluating the volumes. The identities

$$\sum_v d_v = D, \qquad \sum_v d_v v = 0$$

provide consistency checks, while each final slope comparison is a comparison of rational numbers. To disprove stability it suffices to exhibit a proper ray-spanned subspace W for which

$$\frac{\sum_{v\in W} d_v}{\dim W} \geq \frac{D}{n}. \tag{101}$$

We first examine subspaces spanned by at most four rays. If none satisfies the displayed inequality, we examine all ray-spanned subspaces. Thus stability is concluded only after the full subspace test when the initial search is inconclusive.

The non-stability records distinguish strict inequality from equality. A strict inequality in (101) proves instability. An equality excludes stability but does not exclude the existence of a different subspace with strictly larger slope. We use the term *equality witness* for the latter records, without asserting semistability in all such cases.

Proof of Theorem F.1. The bound in dimensions at most six follows from the complete low-dimensional calculation cited above and, in dimensions one and two, from the classifications of smooth toric Fano curves and surfaces. The numbers of varieties with $\rho \geq n+3$ in dimensions $4, 5, 6$ are $2, 7, 52$; none has stable tangent bundle. Of the 1277 sevenfolds examined, 1013 have $\rho = 9$ and 264 have $\rho \geq 10$. Among the latter, 100 admit a strict witness and 164 have a recorded equality witness. Hence none is stable. Among the former, exactly 15 are stable. Together with the stable counts $3, 1, 10$ at $\rho = n + 2$ in dimensions $4, 5, 6$, this proves both assertions. □

ρ	total	stable	unstable	equality witness
11	1286	1	665	620
12	243	0	85	158
13	40	0	12	28
14	11	0	0	11
15	1	0	0	1
16	1	0	0	1

TABLE 10. All smooth toric Fano eightfolds with $\rho \geq 11$. The final column excludes stability but does not assert semistability.

Proof of Theorem F.2. Table 10 accounts for all 1582 varieties with $\rho \geq 11$: one is stable, 762 have a strict destabilizing witness, and 819 have an equality witness. The stable variety has $\rho = 11$ and identifier `P8D.smooth_fano_8d.gz.389596` in Paffenholz's list. Its anticanonical degree is $D = 5753440$.

To identify this variety with X_2, order the coordinates of its lattice as

$$(r_0, q_0, r_1, q_1, f_1^0, f_1^1, f_2^1, f_1^2),$$

with the orientations in Example 8.6. In the database coordinates the lattice map is

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{102}$$

A direct calculation gives $\det A = 1$ and shows that A maps the 19 ray generators of X_2 bijectively onto those of the listed variety. Since both fans are the fans over the faces of their smooth Fano polytopes, this ray bijection identifies the fans as well.

Stability can also be verified directly from the database rays, independently of the path-family proof. Its rational base-polytope expression consists of 19 bases with positive rational coefficients summing to 1. The weighted sum of their incidence vectors equals $(\delta_v)_v$, and their directed exchange graph is strongly connected. Lemma F.4 therefore proves stability. These checks identify the stable variety. The absence of any other stable variety among the examined records proves uniqueness and maximality. □

F.3. **The varieties of Picard number $n+2$.** Define the even-dimensional del Pezzo variety V_{2m} as the smooth toric Fano variety whose fan is the fan over the faces of the polytope

$$\operatorname{conv}\{\pm e_1, \dots, \pm e_{2m}, \pm(e_1 + \cdots + e_{2m})\} \subset \mathbb{Z}^{2m}_{\mathbb{R}}. \tag{103}$$

This is the del Pezzo variety of Voskresenskiĭ and Klyachko [VK85]. It has $4m+2$ rays and Picard number $2m+2$; in particular $V_2 = S_6$. A toric fibre bundle with this fibre means a Zariski locally trivial toric morphism with fibre V_{2m}, equivalently a fan split by the base fan and the fan of V_{2m} as in Lemma 8.2.

Theorem F.5. *Suppose $4 \le n \le 7$, X is a smooth toric Fano n-fold, $\rho(X) = n+2$, and T_X is stable. Then either n is even and $X \cong V_n$, or X is a toric fibre bundle with fibre V_{2m}, $0 < 2m < n$, over a smooth projective toric variety Y satisfying $\rho(Y) = \dim Y$. The numbers of isomorphism classes are as follows:*

n	V_n	V_2-*bundles*	V_4-*bundles*	*total*
4	1	2	0	3
5	0	1	0	1
6	1	7	2	10
7	0	14	1	15.

Proof. For the 29 stable varieties counted in Theorem F.1, we examine rational subspaces spanned by rays in even dimension. The fibre configurations are identified with (103) in integral coordinates. In each of the 27 proper-fibre cases, the non-fibre rays project bijectively onto the rays of a smooth complete quotient fan. Moreover, the maximal cones of the original fan are exactly all sums of a fibre maximal cone and a chosen lift of a quotient maximal cone. Thus the full split-fan condition holds, and the toric local triviality theorem applies. The other two configurations are V_4 and V_6 themselves.

The base is projective: a base maximal cone lifts to a cone of the original fan, and the orbit closure associated to any fixed fibre maximal cone is a section isomorphic to the base.

Since X is projective, so is this closed subvariety. Finally, counting fibre and base rays in a split fan gives

$$\rho(X) = \rho(V_{2m}) + \rho(Y), \qquad \dim X = 2m + \dim Y.$$

Substituting $\rho(X) = n + 2$ and $\rho(V_{2m}) = 2m + 2$ yields $\rho(Y) = \dim Y$. □

This theorem describes the stable varieties in the specified part of the finite classification. It does not give a sufficient condition for a del Pezzo bundle of these dimensions to be stable. The base need not be a product of projective lines: in dimension six, stable hexagon bundles already occur over other smooth toric bases of dimension and Picard number four.

F.4. **Reproducibility.** The programs and proof data are available at https://github.com/bwuebben/toric-tangent-stability/tree/main/paper2/paper/anc. For a future arXiv submission, the package is intended to accompany the manuscript as ancillary files. The package contains the Python programs in `code/`, the exact inputs and outputs in `data/`, and the rational basis decompositions in `certificates/`. The file `README.txt` gives the requirements, commands and scope of each calculation. Run the commands below from the package directory (`paper2/paper/anc/` in the public repository).

The ancillary package includes standalone rational-arithmetic programs for the uniform path, end-attachment, branching and cycle calculations. Their inputs are the rays, moment matrices, rank recurrences and comparison bounds stated above. In particular, the end-attachment and branching programs reproduce the finite minima, all-selected costs and error bounds. The unequal-arm calculation evaluates all 500 central choices for each of 7140 mixed comparisons, including every short-length case required by the contraction argument. The general-tree theorem extends these calculations by induction on trees, without enumerating their shapes. The additional arithmetic computations are the limiting transition inequalities and the explicit subdivision thresholds. From the ancillary directory, the commands

```
python3 -S code/end_attachment_proof.py --uniform
python3 -S code/unequal_spider_proof.py --uniform
python3 -S code/two_junction_gluing.py
python3 -S code/subcubic_thresholds.py
python3 -S code/research_graph_targets.py
```

require only the Python standard library. The package includes the exact rational outputs and 93 positive basis decompositions for independently checking selected finite examples. Running `python3 -S code/verify_certificate.py certificates/*.json` checks all these decompositions. Both the rational verification and the uniform stability proofs are independent of the numerical searches used to find these decompositions.

For the mixed-blowup construction, the tree criterion, the all-length path theorem and the saturated-vertex comparison are proved by identities and inequalities without a finite graph enumeration. The fourteen exact fractions used in the star-transition proof are reproduced directly from (73) and its derivative by

```
python3 -S code/mixed_star_table.py --check-json
  data/mixed_star_table/signs.json
```

The second line continues the same command. This calculation covers only $m \le 14$; the estimate (75) proves the assertion for every $m \ge 15$.

The finite rational inputs for the tree stability theorem and the local degree-four and degree-five obstructions are reproduced by

```
python3 -S code/mixed_tree_arithmetic.py --check-json
  data/mixed_tree_arithmetic/signs.json
```

This also checks the eighteen positive linear-functional values and six adjacent pairs in Proposition 6.6. The conclusions for arbitrary trees and connecting lengths use the monotonicity and induction arguments in Section 6.

The ancillary data distinguish the input polytopes from the varieties for which the calculation decides stability. The seven-dimensional records cover all 1277 varieties with $\rho \geq 9$; the original eight-dimensional screening records cover the 1582 with $\rho \geq 11$. The separate `census8_certificates.json.gz` contains the full 6281-record $\rho = 10$ classification, its rational basis decompositions or strict witnesses, and the product factors of all 692 semistable cases. The records include identifiers, ray coordinates, divisor degrees and the subspaces proving non-stability. The stable-boundary list is restricted to $\rho = n + 2$ in dimensions four through seven. In particular it does not include X_2, whose Picard number is $n + 3$.

For the stable eightfold, the rational expression in 19 bases can be checked using only the ray matrix, the normalized degrees and rational arithmetic. Verify that each determinant is nonzero, each coefficient is positive, and their sum is 1. Then check the weighted incidence-vector identity and the strong connectivity of the directed exchange graph. This verification requires no fan symmetry. The identification matrix (102) gives a separate comparison with the graph construction.[2]

The complete Picard-number-ten verification is run from the ancillary directory by `python3 -B code/verify_census8.py`. It checks the raw input, every smooth Fano fan, all divisor degrees using (100), each exact stability verdict, and all semistable product fans. The program's requirements and coverage are specified in `README-census8.txt`. The original classification's completeness and nonrepetition are inputs to this finite verification, not conclusions of the numerical hull routine.

Appendix G. Further examples and relative tangent slopes

The preceding methods also give stability results for del Pezzo varieties and formulas for relative tangent slopes of toric bundles. All slopes in this appendix are taken with respect to the anticanonical polarization of the total space.

G.1. **Del Pezzo and pseudo del Pezzo varieties.** Let $n = 2m \geq 2$, let $e_1, \dots, e_n$ be the standard basis of $\mathbb{Z}^n$, and put $s = e_1 + \cdots + e_n$. The del Pezzo variety V_n and the pseudo del Pezzo variety $\widetilde{V}_n$ have ray generators

$$V_n: \quad \pm e_1, \dots, \pm e_n, \pm s, \qquad \widetilde{V}_n: \quad \pm e_1, \dots, \pm e_n, s,$$

respectively; their fans consist of the cones over the faces of the convex hulls of these generators [VK85, Ewa88, Nil06]. Thus V_2 is the del Pezzo surface of degree six and $\widetilde{V}_2$ the one of degree seven.

Proposition G.1. *For every $m \geq 1$, both V_{2m} and $\widetilde{V}_{2m}$ are smooth toric Fano varieties with stable tangent bundles. Their Picard numbers are $2m + 2$ and $2m + 1$, respectively.*

Proof. Their anticanonical polytopes are

$$[-1, 1]^n \cap \left\{ \left| \sum_i x_i \right| \leq 1 \right\}, \qquad [-1, 1]^n \cap \left\{ \sum_i x_i \geq -1 \right\}.$$

[2]The computations are implemented by `screen_high_rank.py`, `matroid_certificate.py` and `verify_certificate.py`, all in the ancillary package's `code/` directory. The first two programs require NumPy and SciPy; the exact certificate verifier uses only the Python standard library. The stable-eightfold expression is retained as `P8D_389596.json`. The ancillary tables contain only completed calculations.

Here n is even, so no cube vertex lies on a cutting hyperplane. The remaining cube vertices have coordinate sum zero in the first case and nonnegative coordinate sum in the second. Every new vertex lies on a cube edge: one coordinate is zero and the remaining coordinates are ± 1 with sum 1 or -1 in the first case, and sum -1 in the second. The primitive normals of the active facets are a signed coordinate basis at an old vertex. At a new vertex they are $n-1$ signed coordinate vectors together with s or $-s$; their determinant is ± 1. Both polytopes are therefore integral and simple with unimodular normal cones. The origin is interior, and all displayed inequalities define facets. Thus both varieties are smooth and Fano.

For V_n, use the equivalent lattice presentation $\mathbb{Z}^{n+1}/\mathbb{Z}(e_0 + \cdots + e_n)$, with rays $\pm e_i$. The symmetric group S_{n+1} acts through its irreducible standard representation on $N_{\mathbb{C}}$. Its ray matroid is connected: opposite rays are parallel and $\{e_0, \dots, e_n\}$ is a circuit meeting every parallel class. Proposition 2.4 proves stability.

For $\widetilde{V}_n$, the group S_n permuting the coordinate vectors preserves the fan. Its only nonzero proper invariant subspaces are $\mathbb{C}s$ and $\{\sum_i x_i = 0\}$; this follows from the decomposition into the trivial and standard representations, which are non-isomorphic irreducibles, including when $n = 2$. The latter hyperplane contains no ray. The circuit $\{e_1, \dots, e_n, s\}$ and the opposite pairs show that the ray matroid is connected. Write $d_{e_i} = p$, $d_{-e_i} = q$, and $d_s = h$. Lemma 2.2 gives $q = p + h$, so

$$D = n(p+q) + h = n(2p+h) + h > nh.$$

Thus the only invariant subspace spanned by rays, $\mathbb{C}s$, satisfies the strict slope inequality. Proposition 2.4 applies. Finally, subtracting the dimension from the number of rays gives the two Picard numbers. □

G.2. **Toric bundles over $\mathbb{P}^1$.** Let $Z = X_{\Sigma_Z}$ be a positive-dimensional smooth projective toric variety with lattice N_Z, and let $t \in N_Z$. In $N_Z \oplus \mathbb{Z}$, take the cones generated by a cone of Σ_Z and either $(t, 1)$ or $(0, -1)$, together with all their faces. Denote the resulting variety by X_t; projection to the last coordinate gives a toric bundle $X_t \to \mathbb{P}^1$ with fibre Z.

Proposition G.2. *The fan of X_t is smooth and complete. The variety X_t is Fano if and only if Z is Fano and*

$$t \in Q \cap N_Z, \qquad Q = \operatorname{conv}\{u_\rho : \rho \in \Sigma_Z(1)\}.$$

When Z is Fano, $Q \cap N_Z$ consists exactly of the origin and the vertices of Q.

Proof. On the upper half-space, the shear $(x, y) \mapsto (x - yt, y)$ identifies the cones with the product of Σ_Z and the nonnegative ray. On the lower half-space they already have product form. The fan is therefore complete, and each maximal cone is unimodular. If X_t is Fano, restriction of $-K_{X_t}$ to a fibre gives $-K_Z$, so Z is Fano.

Now assume that Z is Fano. For a facet F of Q, let $u_F \in M_Z$ satisfy $\langle u_F, v\rangle = 1$ on F and be strictly less than one at every other vertex. The linear forms supporting the two proposed facets over F are

$$(u_F, -1), \qquad (u_F, 1 - \langle u_F, t\rangle).$$

Evaluating each form on the opposite apex gives the strict convexity condition $\langle u_F, t\rangle < 2$. Since this quantity is integral, the condition is equivalent to $\langle u_F, t\rangle \le 1$ for every facet, hence to $t \in Q$. These are exactly the strict support-function inequalities for ampleness of $-K_{X_t}$. Finally, the simplices $\operatorname{conv}(0, F)$ cover Q and are unimodular, so they contain no lattice points other than their vertices. □

G.3. **Bundles over products of projective lines.** Let F be a smooth toric Fano variety of dimension $f \geq 1$, and set

$$P_F = \{m \in M_{F,\mathbb{R}} : \langle m, v\rangle \geq -1 \text{ for every ray generator } v \text{ of } F\}.$$

For $k \geq 1$, choose each t_i to be zero or a primitive ray generator of F. The bundle $X = X_F(t_1, \dots, t_k)$ over $(\mathbb{P}^1)^k$ has rays $(v, 0)$, (t_i, e_i), and $(0, -e_i)$, with maximal cones obtained by choosing a maximal fibre cone and one ray from each base pair. Proposition G.2, applied successively, shows that X is smooth Fano. Put $n = f + k$, $[k] = \{1, \dots, k\}$, and

$$I(S) = \int_{P_F} \prod_{i\in S} \langle m, t_i\rangle \, dm, \qquad S \subseteq \{1, \dots, k\},$$

where dm gives the character lattice covolume one and an empty product is one.

Proposition G.3. *The anticanonical degree is*

$$D = n! \int_{P_F} \prod_{i=1}^{k} (2 + \langle m, t_i\rangle) \, dm = n! \sum_{S\subseteq[k]} 2^{k-|S|} I(S).$$

For each i, the two lifted base rays have equal divisor degree $(n-1)!J_i$, where

$$J_i = \int_{P_F} \prod_{j\neq i} (2 + \langle m, t_j\rangle) \, dm.$$

The relative tangent bundle satisfies

$$\mu(T_{X/(\mathbb{P}^1)^k}) \geq \mu(T_X) \quad \Longleftrightarrow \quad \sum_{S\subseteq[k]} |S| \, 2^{k-|S|} I(S) \geq 0. \tag{104}$$

The same equivalence holds with strict inequalities.

Proof. The anticanonical polytope is

$$\{(m, y) : m \in P_F, \ -1 - \langle m, t_i\rangle \leq y_i \leq 1 \text{ for } 1 \leq i \leq k\}.$$

All interval lengths are positive. Integration gives $D = n! \operatorname{vol}(P_X)$. Each base facet projects by a lattice isomorphism onto the region with the i-th interval omitted; this gives its degree $(n-1)!J_i$. The fibre subspace corresponds to the relative tangent bundle, which has rank f and degree $D - 2(n-1)! \sum_i J_i$. Comparing its slope with D/n is equivalent to $kD \geq 2n! \sum_i J_i$. A term $I(S)$ occurs in the last sum exactly $k - |S|$ times, with coefficient $2^{k-1-|S|}$ each time. Subtraction gives (104). □

For $k = 1$, write $b_F = \operatorname{vol}(P_F)^{-1} \int_{P_F} m \, dm$. The relative tangent slope is at least $\mu(T_X)$ exactly when $\langle b_F, t_1\rangle \geq 0$. Thus a Fano toric bundle of this form over $\mathbb{P}^1$ cannot have stable tangent bundle if $b_F = 0$.

For $k = 2$ and centrally symmetric P_F, the condition is

$$B(t_1, t_2) := \int_{P_F} \langle m, t_1\rangle \langle m, t_2\rangle \, dm \geq 0.$$

For $F = V_{2m}$, use its quotient-lattice presentation with rays $\pm e_0, \dots, \pm e_{2m}$. The form B is positive definite and S_{2m+1}-invariant. Irreducibility of the standard representation implies that it is a positive multiple of the form with entries $\delta_{ij} - 1/(2m+1)$. Hence, for nonzero

twists $t_1 = \varepsilon_1 e_i$, $t_2 = \varepsilon_2 e_j$, with $\varepsilon_1, \varepsilon_2 \in \{1, -1\}$, the relative tangent bundle has slope greater than $\mu(T_X)$ precisely when

$$t_2 = t_1 \quad \text{or} \quad i \neq j \ \text{ and } \ \varepsilon_1 \varepsilon_2 = -1.$$

For opposite twists with the same index, and for equal signs with different indices, the relative tangent bundle has smaller slope. These comparisons alone do not determine the stability of T_X.

G.4. **Projective-line bundles over a toric base.** Let Y be a smooth projective toric variety of dimension $n-1 \geq 1$, and let $X = \mathbb{P}_Y(\mathcal{O} \oplus L)$ be a toric projective-line bundle. Choose its fan presentation with fibre rays $(0, \pm 1)$ and lifted base rays (u_ρ, a_ρ). Set $L' = \sum_\rho a_\rho D_\rho$ on Y; the line bundle L corresponds to L' or $-L'$, according to the projectivization convention. Define

$$g(y) = (-K_Y + yL')^{n-1}.$$

Proposition G.4. *Assume X is Fano. Then $-K_Y \pm L'$ are ample, and*

$$D = n \int_{-1}^{1} g(y)\, dy, \qquad d_{(0,1)} = g(-1), \qquad d_{(0,-1)} = g(1).$$

Consequently

$$\begin{aligned} \mu(T_{X/Y}) \geq \mu(T_X) &\iff g(1) + g(-1) \geq \int_{-1}^{1} g(y)\, dy \\ &\iff \sum_{\substack{2 \leq j \leq n-1 \\ j \text{ even}}} \frac{j}{j+1} \binom{n-1}{j} (-K_Y)^{n-1-j} \cdot (L')^j \geq 0. \end{aligned}$$

For $n = 3$ this is $(L')^2 \geq 0$, and for $n = 4$ it is $(-K_Y) \cdot (L')^2 \geq 0$.

Proof. The two sections have normal bundles L and L^{-1}; adjunction therefore identifies the restrictions of $-K_X$ with $-K_Y + L'$ and $-K_Y - L'$. They are ample, hence in particular nef. Their convex combinations $-K_Y + yL'$ are ample for $-1 \leq y \leq 1$. At height y, the anticanonical polytope has slice

$$P_y = \{m : \langle m, u_\rho \rangle \geq -1 - a_\rho y \text{ for every } \rho\},$$

the polytope of $-K_Y + yL'$, of volume $g(y)/(n-1)!$. Integration gives D. The facets at $y = -1$ and $y = 1$ give the two divisor degrees. The relative tangent line bundle is $\mathcal{O}_X(D_{(0,1)} + D_{(0,-1)})$, so its degree is their sum. Finally, expand $g(y) = \sum_j c_j y^j$. The difference between the endpoint sum and the integral is $2 \sum_{j \geq 2,\ j \text{ even}} c_j j/(j+1)$, giving the assertion. □

Example G.5. The relative tangent bundle may have smaller slope even when the tangent bundle is unstable. Let $Y = \mathbb{P}^2 \times \mathbb{P}^1$ and $L' = 2H - F$, where H and F are the hyperplane classes of the two factors. Both $-K_Y \pm L'$ are ample; the strict support-function inequalities on the two sets of maximal bundle cones then show that X is Fano. Its rays may be written

$$-e_1,\ -e_4,\ e_2 - e_4,\ -e_2,\ e_4,\ -e_3,\ e_1 + e_3 + 2e_4.$$

This is the variety F.4D.0019 in the smooth Fano classification [Paf]. Here $g(y) = 3(3 + 2y)^2(2 - y)$, so

$$D = 400, \qquad \mu(T_X) = 100, \qquad \deg T_{X/Y} = g(1) + g(-1) = 84.$$

The divisor degrees in the displayed ray order are $(64, 75, 62, 62, 9, 64, 64)$. For example, the facet for $-e_1$ has degree $6\int_{-1}^{1}(3+2y)(2-y)\,dy = 64$, while the two facets for the $\mathbb{P}^1$ base directions have degree $3\int_{-1}^{1}(3+2y)^2\,dy = 62$. The plane $\operatorname{span}(e_2, e_4)$ contains four rays of total degree $75+62+62+9 = 208$. Its subsheaf has slope $104 > 100$, proving that T_X is unstable.

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